MODEL INSIGHT

The MSCI Macro-Finance Model

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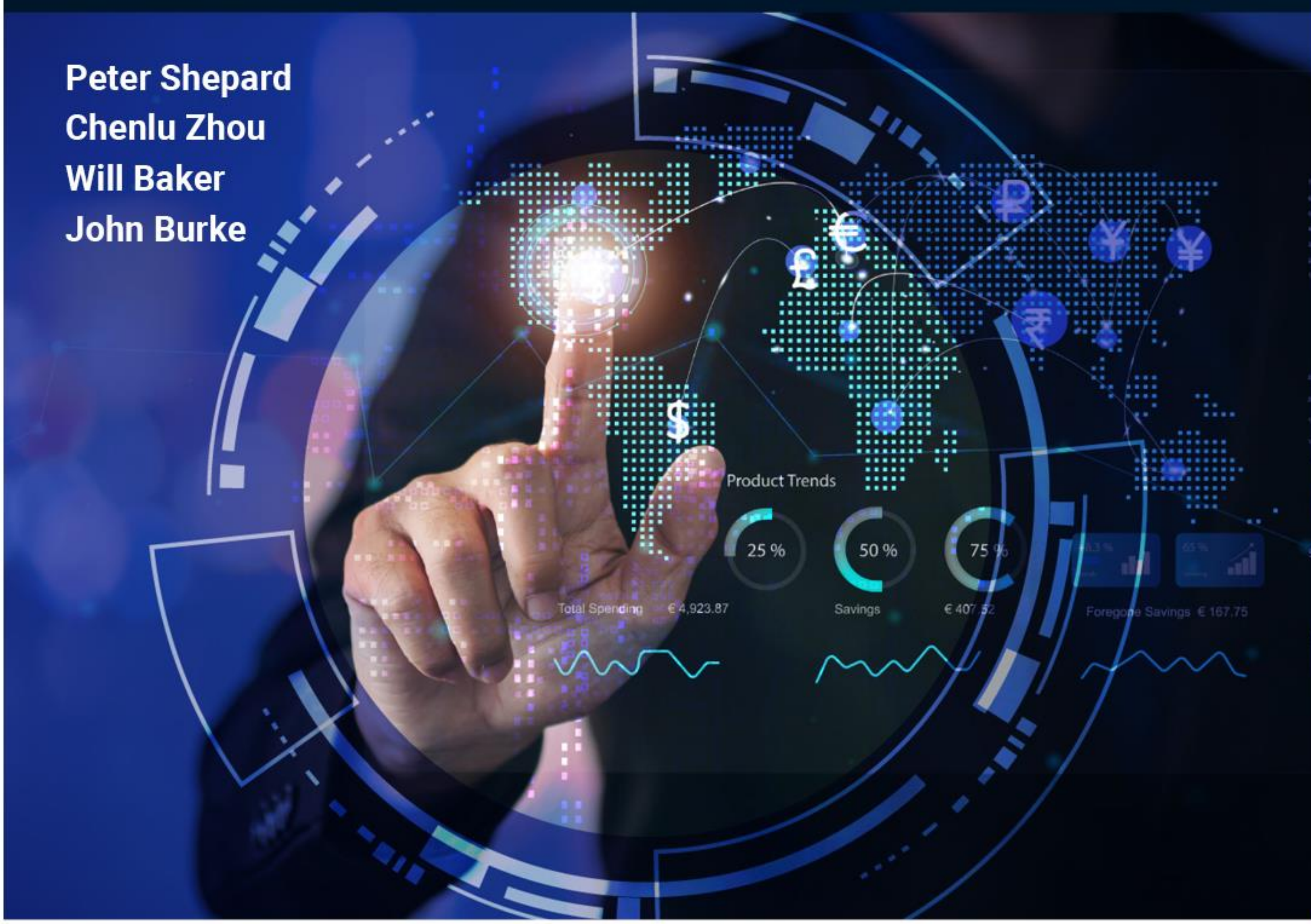

Executive summary

We introduce a framework to understand the financial markets through a macroeconomic lens. We relate cash flows and discount rates to common drivers across asset classes, horizons and regimes, in order to inform capital market assumptions, risk management, systematic strategies and asset-allocation decisions.

Despite the appeal of such a framework, establishing macro-market relationships has been notoriously difficult. It's obvious that economic growth is good for the equity markets, but the data can seem unaware of this link. Many papers have been written on the theoretical basis of macro-finance, but connecting theory to practice has lagged.

Our framework sees economic growth and inflation as fundamental drivers of asset returns, but their relationship to the markets requires understanding how both forward-looking expectations and the intervention of central bank policy play out over different horizons.

Macro risk is not macro volatility. Short-term growth shocks are painful, but the greater financial risk is of a long-term slowdown in trend growth. Linking macro to market requires understanding a full term structure of forward-looking growth expectations.

Similarly, inflation is a central driver of the markets, but the key factors are inflation pressure and central bank credibility, not realized inflation. Even when inflation is stable, central banks continuously act to channel inflation pressure into higher rates and lower real growth, which drive returns in all asset classes. And the effects of inflation differ widely whether it is driven by supply or demand, and depending on the credibility of the central bank.

The MSCI Macro-Finance Model provides a unified framework to understand the return and risk of multi-asset-class portfolios. This framework applies to four main use cases:

- Capital market assumptions based on market valuations and macro views
- Macro scenario analysis including multi-horizon causes and effects
- Long-horizon risk incorporating mean reversion and regime shifting
- Dynamic asset allocation and systematic strategies looking at market cycles and macro-sensitive liabilities

This note introduces the model and explores its applications.

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1 Overview

Many have sought a macro perspective to go beyond a short-term, backward-looking view of the markets, and to anchor expectations of risk and return into a forward-looking framework.

Projections of economic growth and inflation can inform the cash-flow and discount-rate expectations that drive asset returns, especially over a long horizon, and help investors understand how market and economic cycles are likely to evolve.

Economic uncertainties are also among the main sources of risk across asset classes. The impact of wide-ranging scenarios can be understood through their macroeconomic effects, and macro variables anchor sources of mean reversion and regime shifting that shape the landscape of risks at a long horizon.

But understanding the relationships between macroeconomics and markets has been elusive.[1] The apparent macro sensitivities of equity can seem completely backward, for example: Equity is often only weakly or even negatively correlated with growth, but positively correlated with interest rates and energy prices, and equity markets often react negatively to seemingly positive economic news like lower unemployment.

Understanding the connections between macro and market requires going beyond naive correlations to the underlying causes. Many practitioners employ a scientific approach to projecting the macroeconomy, but have lacked a robust framework for translating macro views to the markets.

We introduce a macro-finance model to provide this connection. The model aims to be "as simple as possible, but no simpler," in isolating the key macroeconomic drivers of returns across asset classes, time horizons and regimes.

The model starts with the dynamics of the macroeconomy, including key macro variables such as growth and inflation, as well as their interactions with central banks' interest rate policy. Crucially, the model goes beyond the short term to include major influences on the long-horizon behavior of the economy. Mean reversion significantly moderates some short-term risks, while other risks like slowing trend growth or a shift in regime become more and more important at longer horizons. Our goal is not to predict the path of the economy — many others attempt that — but to understand how various views and scenarios are likely to unfold in the markets.

To connect macro to markets, we model the dynamics of assets' cash flows and discount rates, and how they are sensitive to the macroeconomy. Understanding these sensitivities is a central challenge, and basic approaches like regressions can completely misunderstand the relationships. It's critical to understand:

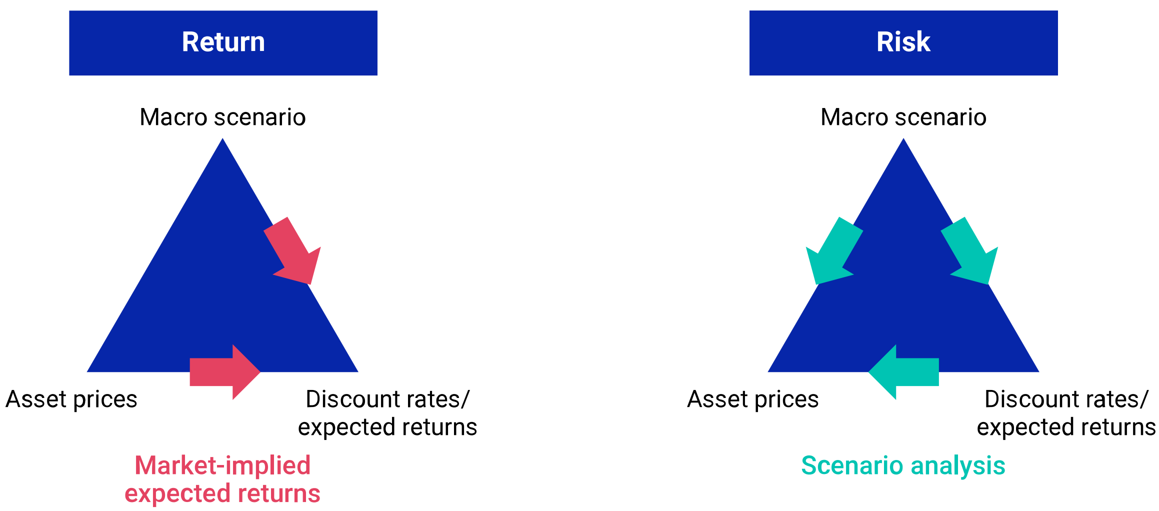
- *The roles of horizon, and forward-looking expectations*. Equity is growth-sensitive, despite low contemporaneous correlations with growth, but the important sensitivity is to sustained future growth, not the short-term, realized growth that is observed. Long-term growth is what matters at long horizons, and it eventually drives both the macroeconomy and equity markets together, despite following different short-term paths.

[1] Academics had to give a name — the equity premium puzzle — to the discrepancy between the large realized equity premium and what macro-based utility theory predicts. See for example, Cochrane (2005) and references therein.

- *Underlying fundamental, asymmetric causes*. Fluctuating demand has often moved equity, interest rates and energy prices up and down together, so they have often been positively correlated. But most stock markets would not celebrate either higher costs of capital or more expensive energy, even when they're positively correlated, and the macro sensitivities of equity to rates and energy are in the *opposite* direction as the correlations suggest.
- *The effects of central bank policy and its credibility*. The role of central bank policy is to moderate economic surprises by diverting their consequences: Lowering rates at signs of recession can shorten a slowdown and boost many asset classes; raising rates in response to inflationary pressures diverts that pressure away from inflation and into higher rates and lower real growth. A highly credible central bank is able to achieve the same benefits much faster, and with fewer side effects.

Getting past these core challenges of connecting macro to markets enables an intuitive framework to understand risk and return. In this framework, macro scenarios, market prices and expected[2] returns can be thought of as three parts of a triangle: Given two, the third is implied. A baseline macro scenario can be combined with market valuations to provide a term structure of implied expected returns (capital market assumptions); or, macro shock scenarios can be propagated through shocked discount rates and asset prices to understand a term structure of risk.[3]

Exhibit 1: A unified framework integrates macro scenarios with expected returns and risk analysis



In the MSCI Macro-Finance Model framework, macro scenarios, market prices and expected returns can be thought of as three parts of a triangle: Given two, the third is implied. This framework enables a consistent approach to risk and returns: A baseline macro scenario can be combined with market valuations to provide a term structure of implied expected returns; or, macro shock scenarios can be propagated through shocked discount rates and asset prices to understand a term structure of risk.

[2] Throughout this note, "expected" refers to the average value relative to a model-derived probability distribution, not a prediction. According to those same probability distributions, it is usually very unlikely that a specific "expected value" will actually occur. These "expectations" provide a summary statistic over a wide range of potential outcomes.

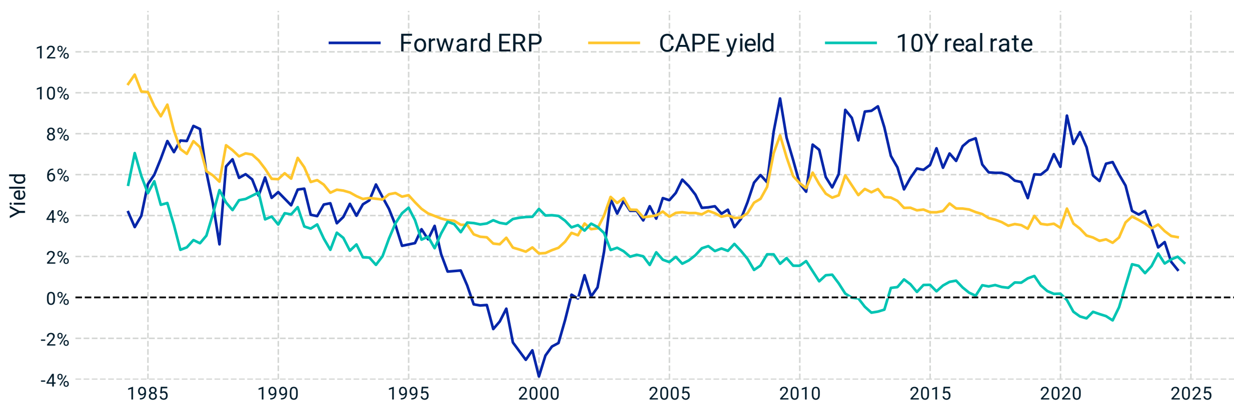
[3] The risk case can also just specify the macro scenario, using the model to propagate to both discount rates and asset prices. Macro shocks are hypothetical scenarios beyond the baseline expectations.

1.1 Capital market assumptions

Our model is no crystal ball to predict market returns, especially at short horizons, when markets can move far away from the fundamentals. But precisely because markets can move far from their fundamentals, investors have both a need and an opportunity to understand those fundamentals.

Exhibit 2 compares our model's view of equity valuations, the Forward equity risk premium (Forward ERP), to the commonly used cyclically adjusted price-to-earnings yield (CAPE yield). Like the yield of a bond, higher yields correspond to lower valuations. The Forward ERP reflects our model's estimate of the forward-looking return premium of stocks relative to risk-free assets, and provides an indicator of whether stock markets may be relatively over- or undervalued.

Exhibit 2: The Forward ERP provides an equity valuation signal



The Forward ERP incorporates both forward-looking growth expectations and risk-free rates, making it a more responsive and predictive signal compared to the CAPE yield. The Forward ERP is calculated using the MSCI Macro-Finance Model, with MSCI USA Index valuation; CAPE yield is calculated as the inverse of the Shiller CAPE ratio, a backward-looking measure; the 10-year real rate is calculated using U.S. Treasury yield and breakeven inflation.

At the time of Alan Greenspan's "irrational exuberance" call (though years before the markets corrected), both the CAPE yield and the Forward ERP had dipped well below their historical averages, indicating the markets were disconnecting from the fundamentals. Our Forward ERP provided a stronger signal that the markets were overpriced, projecting a *negative* return premium for the broad equity market for years before the dot-com crash.

The advantage of the Forward ERP relative to the CAPE yield became especially clear more recently. CAPE yields ventured toward their dot-com bubble lows during the COVID market recovery and rally, which suggested equity could be far overvalued, while the Forward ERP saw equity valuations as being in line with historical norms. The lower-real-rate environment was the most significant among various factors driving this difference. From this perspective, far lower real rates would have justified much higher equity prices soon after the 2008 global financial crisis (GFC), and the bull market that followed the GFC can be understood as a slow repricing of equity to reflect the new real-rate regime.

The CAPE and Forward ERP also signal different valuations at the time of this writing: The CAPE indicates valuations as moderately expensive, while the Forward ERP sees equity as well above any valuation seen over the past 20 years.

A macro view also provides insights into expected returns across other asset classes. Bond markets are often viewed with the "expectations hypothesis," which assumes future short-term rates to be the forward rate implied by the term structure of bond yields. This is often extended to assume that future inflation will be breakeven inflation (the difference between nominal and real risk-free rates), or that the rate of future credit losses is given by credit spreads.

These assumptions are a useful starting point, but have historically missed important return premia. Nominal forward rates have tended to overstate future interest rates, which is known as the term premium of long-dated bonds. And credit spreads have often been above the level of realized default losses, reflecting a risk premium awarded to investors in credit-risky assets.

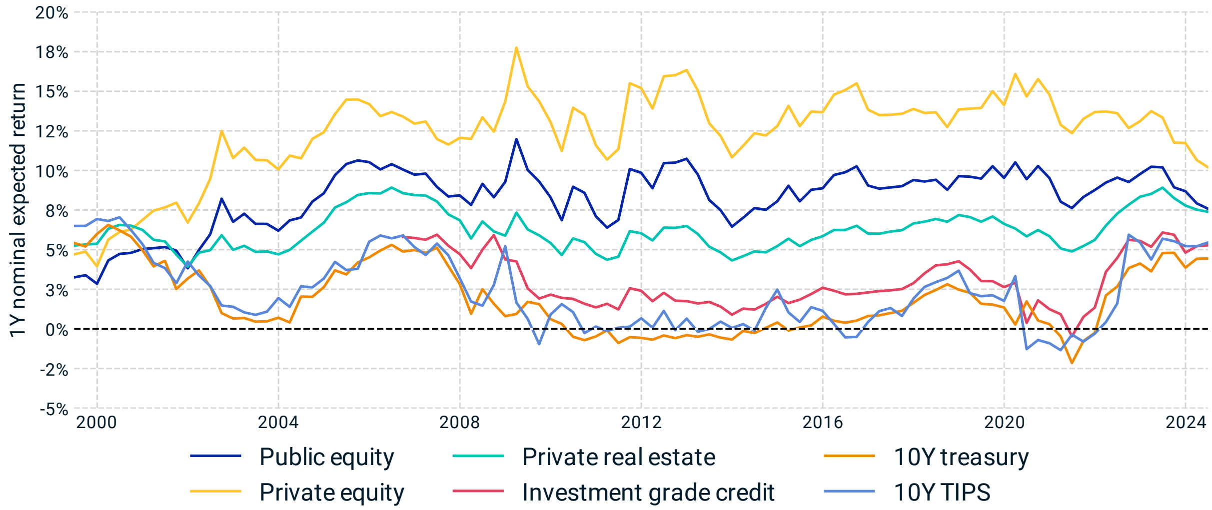
These premia are important for long-term asset-allocation decisions, but short-term fluctuations can be equally important for tactical asset-allocation decisions. These premia expand or contract whenever bond prices move out of sync with macro expectations. This would never happen if markets were fully efficient, but it has been common when liquidity, risk-hedging and other pressures have pushed bond markets away from their fundamentals, and also arises whenever an investor's macro views differ from what the market has priced in.

Private assets can also be understood with a forward-looking view of valuations. Investors in real estate have long looked at cap rates (the ratio of net operating income to property values) as a valuation indicator, similar to earnings or dividend yields for equity. As with equity and the Forward ERP, we can better understand cap rates by putting them in the context of the macro environment, to differentiate the effects of risk-free rates and cash-flow growth from a valuation premium. Many real-estate investors take a similar approach with discounted-cash-flow models, though they may not robustly capture the macro sensitivities in the cash-flow expectations, or be able to take a consistent view across asset classes.

Private equity is especially important to understand on a forward-looking basis. Private equity returns have significantly outperformed those of public equity historically, often attributed to an illiquidity premium and the potential advantages of private ownership.[4] But with so much capital now eagerly moving into private assets, can investors assume the same illiquidity premium will persist in the future? An ex-ante view of private-equity returns has been especially difficult because even basic valuation measures like earnings yield have been unavailable, due to the scarcity of data and transparency into private capital. MSCI's October 2023 acquisition of The Burgiss Group LLC provides significantly greater transparency, enabling us to look through to private-equity deal fundamentals to — perhaps for the first time — estimate broad market valuation ratios (EBITDA to enterprise value), upon which we can anchor forward-looking return estimates.

[4] There has been much debate about how much of the outperformance can be attributed to leverage, sector tilts or other factors. We don't claim to have a definitive answer, but our analysis suggests private equity has outperformed even after controlling for such factors.

Exhibit 3: Consistent, ex-ante and valuation-driven capital market assumptions



Model-estimated forward-looking expected returns (capital market assumptions) for an array of public and private assets. Private real estate exhibits significant growth exposure, closely tracking public equity movements. The expected returns (calculated on a nominal basis with a one-year return horizon) are based on U.S. assets and incorporate market valuation such as equity index prices, EBITDA to enterprise value ratio of private equity, the cap rate of private real estate, and yield curves.

1.2 Long-horizon risk

Risk management has often relied heavily on a backward-looking, statistical view of the markets: Assume returns come from some probability distribution …, look at historical data to calibrate the covariance matrix or value at risk … and then allocate assets and build portfolios to manage the risk according to that assumed distribution.

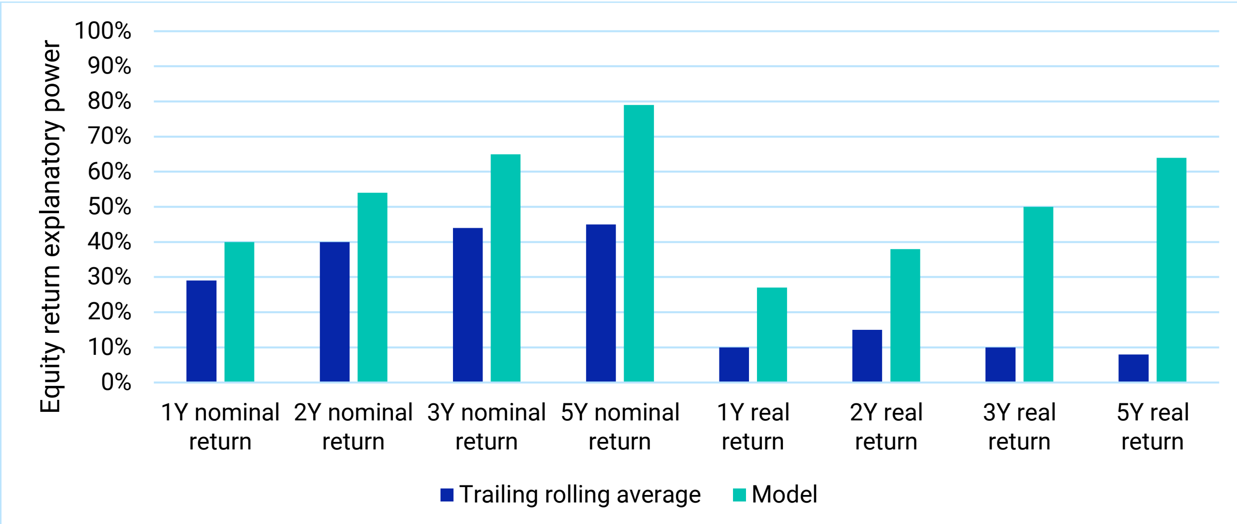
This approach has worked well over short horizons, and when the market regime has been relatively stable, but it has significant limitations:

- *Horizon matters*: Markets may look like a random walk at short horizons, but at longer horizons short-term noise becomes less important, mean reversion and the fundamentals dominate, and the risk-return landscape is reshaped.
- *Regime-shifts happen*: The important drivers of returns in the past may not be the most important in the future, and as we go to longer horizons regime shifting becomes increasingly likely. Correlation is not causation. The effects that followed from the primary causes in previous regimes may not reflect how future scenarios play out.

Mean reversion makes it easier to predict a long horizon, while regime shifting makes it much harder. Both become increasingly important as the horizon grows, and both can be understood with a macro view. Understanding these dynamics may help long-horizon investors harvest risk premia associated with some short-term risks, avoid paying premia for the short-term safety of others and understand the most important long-term risks.

Mean reversion is the tendency of variables to drift toward a long-run average or an evolving reference point, rather than take a random walk that can wander arbitrarily far away. It shapes the long-term dynamics of key macro factors, pulling short-run growth, interest rates, the corporate profit ratio and various risk premia toward their long-run trends, which moderates some short-term risks and introduces greater predictability.[5] This greater predictability can be seen in Exhibit 4, which shows the rising predictive power of the model's capital market assumptions as the investment horizon grows.

Exhibit 4: Equity return predictability improves with longer investment horizons



The model (teal bars) demonstrates stronger explanatory power for nominal and real U.S. equity returns, compared to using trailing rolling averages as forecasts (dark blue bars). As the investment horizon lengthens, the model's explanatory power increases thanks to the mean-reversion of macro drivers smoothing out short-term unpredictability. Trailing rolling averages are calculated using 10-year rolling returns. The explanatory power is computed as one minus the ratio of average squared return deviation (realized minus predicted return) over average squared return.

Mean reversion isn't the only way fundamentals temper the long-horizon return landscape. Equally important is the long-term divergence between the effects of cash flows and discount rates, closely related to the idea of "bad beta" versus "good beta" (Campbell and Vuolteenaho 2004). At a short horizon, return is return, but the effects of cash flows and discount rates are fundamentally different. Higher discount rates depress most asset prices in the short term, but (by definition) they also bring higher expected returns. The longer the investor's horizon, the greater the benefits from harvesting higher returns, and eventually higher discount rates are good, not bad.

But the long-horizon landscape is far from benign. The risks that emerge at a long horizon are harder to see, but no less severe. Many take the forms of regime shifting, or of changes in long-term trends. These risks are not captured by a traditional covariance matrix, but may be thought of as uncertainties in the distribution itself.[6]

[5] If a random walk (aka the drunkard's walk) describes someone equally likely to take the next step in any direction, then mean reversion describes someone who eventually stumbles home. Any one step is nearly as likely to be backward as forward, but over longer periods their direction of travel is easier to predict. Alternatively, consider weather forecasting. It's easier to predict how much it will rain tomorrow than to predict how much it will rain a year from tomorrow, but it's actually relatively easiest to predict how much it will rain all of next year, because it averages over the uncertainty among many days. In both cases, the short-term noise gives way to longer term predictability.

[6] See Shepard (2008) for a discussion of short-horizon risks arising from uncertainty in the probability distribution, and how we can begin to measure and manage such risks like more traditional risks. These risks are not the roulette-wheel-like randomness of known probabilities — they're sometimes more akin to the risk of the casino getting robbed

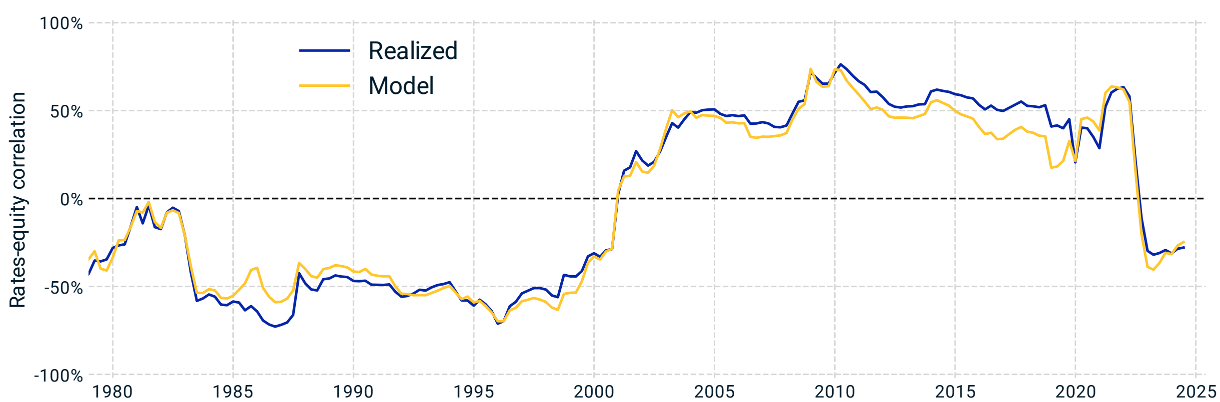
As we look to longer horizons, the likelihood of shifting to another regime increases, and the importance of subtle shifts in macro trends is amplified. At intermediate horizons, these risks may include risk-on/risk-off changes in volatilities, or cyclical changes in expected returns. At longer horizons there can be greater risks of structural changes, such as the relative importance of supply versus demand shocks, and the strength or credibility of central bank policy. One of the greatest risks facing many long-horizon investors is that of a persistent slowdown in trend growth.

Many risk managers have recognized the importance of regime shifting, and have increasingly incorporated stress testing to consider scenarios that may be much more likely in a different regime. The key challenges, at both long and short horizons, are to understand which scenarios are realistic, to understand what has already been priced in to the markets, and then how to propagate a scenario across markets and time horizons.

Take, for example, the relationship between interest rates and equity, which were positively correlated in most developed markets throughout the first two decades of this century. Starting with the crash of the dot-com bubble, and continuing through the GFC, the bull market years that followed, and even the first year of the COVID crisis, the rates-equity correlation effectively provided a hedge between bond and equity allocations, as bond prices tended to rise when equity fell, and vice versa.[7] Risk-parity strategies especially benefited by balancing the legs of the hedge. And then the correlation flipped sign ….

Although the rates-equity correlation held positive for two decades, it had been negative for long periods of previous history. When it flipped sign again in the second year of the COVID recovery, the protection offered by balanced bond and equity allocations disappeared. How can we understand the risk around this correlation?

Exhibit 5: The US rates and equity correlation sign flipped (again)



The U.S. rates-equity correlation was positive for most of the past two decades, serving as a hedge between bond and equity allocations. Before 2000, however, the correlation was negative, and it recently flipped sign again amidst entrenched inflation. The model attributes this correlation to more fundamental drivers, and aligns closely with realized correlation. The early 2000s flip can be linked to lower supply shock volatility and strengthened central bank credibility.

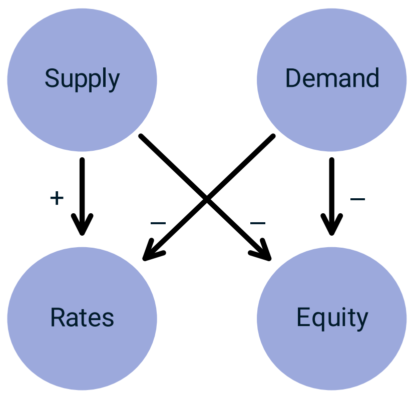
— but we can start to measure and manage them. Calling these risks "second-order risk" may have been a misnomer for the implication that these risks are of second-order importance, which they are not.

[7]A positive rates-equity correlation corresponds to a negative bond-equity correlation, since bond prices fall as rates rise.

Our macro framework understands the rates-equity correlation as a consequence of competing underlying causes, not a parameter calibrated from backward-looking data. The model-implied correlations in Exhibit 5 closely explain the historical correlation pattern based on changing supply- and demand-shock volatility and changing central bank credibility.

Demand shocks had been the primary driver throughout the era of moderate inflation and even into the initial surge of inflation that came in the first year of the COVID recovery. Demand shocks drove a positive rates-equity correlation when rising or falling growth expectations led both equity and interest rates up or down together, as in Exhibit 6.

Exhibit 6: Supply and demand are asymmetric drivers of the rates-equity correlation



The MSCI Macro-Finance Model sees through the correlation to its underlying, asymmetric drivers. In this simplified view, negative supply shocks drive rates up and equity down, while negative demand shocks drive both down. The rates-equity correlation reflects the combined effect of these competing forces. By identifying these underlying causes, investors can differentiate behavior across different scenarios, and understand how the rates-equity correlation might evolve in future scenarios or policy regimes.

As the COVID recovery put pressure on supply chains, and then geopolitical events strained them further, markets came to realize that the luxury of elastic supply could not be taken for granted. Central banks' credibility was also hurt when they were seen to be slow to tighten policy as inflation surged.

Both supply shocks and weaker central bank credibility drive a negative rates-equity correlation. The inflation pressure of a supply shock prevents cutting rates to mitigate the accompanying slower growth, and lower central bank credibility requires more painful cooling to bring inflation back under control.

With this view, we can both model the risks of a changing correlation structure, and explore stress tests at the level of more fundamental causes. Rather than manage risk assuming fixed correlations, or propagate stress tests using those same backward-looking correlations, we can understand which scenarios may be more likely than backward-looking data suggests, and propagate these scenarios with an understanding of their fundamentals.

The remainder of this section introduces the main pillars of the methodology, which are detailed further in the following sections.

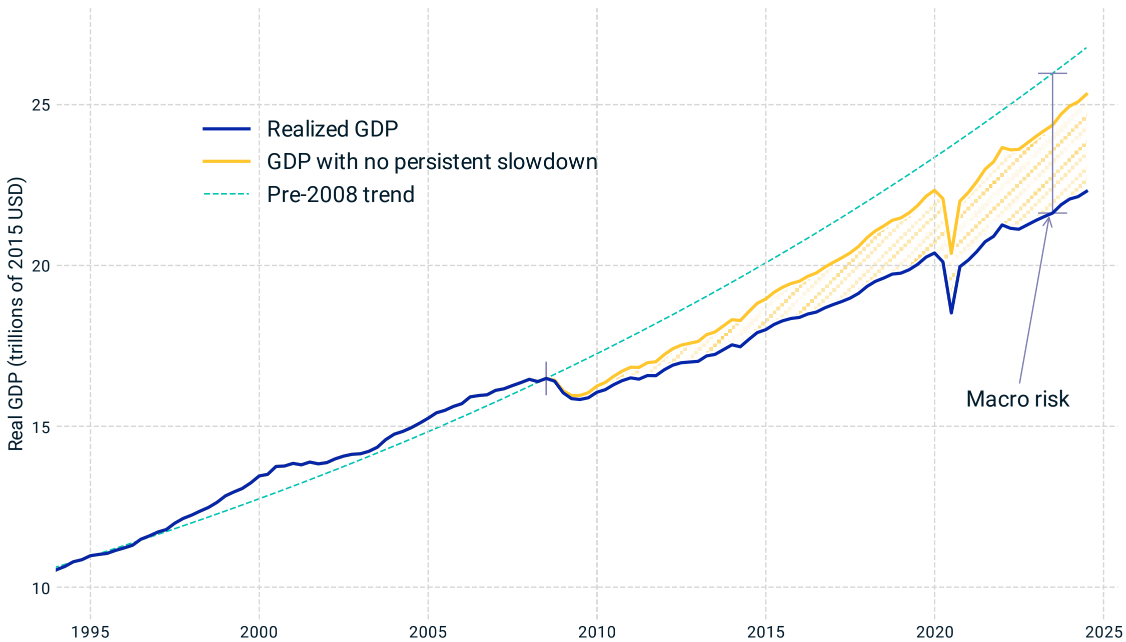
1.3 Real growth and equity

The first pillar of the model is understanding the relationship between growth and equity. This may be the most important macro relationship, and also the hardest to measure.

The growth-equity relationship is important because economic growth is the primary bet most investors make. Equity, and equity-like asset classes, are the main drivers of returns for many portfolios, for better or for worse. These asset classes likely do well in the long run if the economy continues to grow, and the primary risk investors in these assets face is that economic growth stalls.

This relationship is hard to measure because macro and market move out of sync. Macroeconomic data reflects what has already occurred, while markets look forward. There is modest correlation between changes in realized economic activity and rising or falling expectations, but that correlation is far from capturing the importance of the macroeconomy in driving long-term equity returns.

Exhibit 7: Macro risk is not macro volatility



Realized volatility of growth is low, but macro risk primarily lies in the uncertainty of long-run growth. U.S. GDP growth experienced a persistent slowdown of about 80 bps following the GFC. Had this slowdown not occurred and the pre-2008 trend continued (even with the same realized volatility and short-term drawdown), the U.S. economy would have grown about 13% larger over the intervening years through 2024, as seen in the gap between the yellow and blue curves.

The missing link is long-run expectations. Equity prices reflect the present value of cash flows stretching decades into the future, so changes in realized growth from quarter to quarter are much less important than changes in the long-term trajectory of the economy, as in Exhibit 7. Even a small change in the rate of long-run growth can be much more important than a short recession or growth spurt.

Whether the path of the economy is relatively smooth or punctuated by large bumps along the way, what matters most to investors is not volatility, but the trajectory: A sustained boost that lifts the economy onto a different path, or a persistent slowdown that reduces cash flows more and more, year after year.

The economic damage of the GFC/great recession was less from the contraction of 2008 and 2009, for example, but more from the small change to long-run growth that knocked the economy off its historical trajectory. Similarly, equity markets rose to record highs even as the economy contracted during the COVID shock of 2020. Market expectations for a V-shaped economic recovery and return to pre-crisis growth prevailed over the contemporaneous economic pain.

Going beyond a single growth factor, we introduce factors to capture a term structure of growth and growth expectations:[8]

- **Short-term growth** shocks, which are the primary source of growth volatility, but are only weakly persistent.
- **Cyclical growth** shocks, which persist over a one- to five-year horizon, but represent unsustainable growth spurts or temporary slumps.
- **Trend growth** shocks, which drive sustainable changes in growth over a 10- to 20-year horizon — driven, e.g., by a change in the rate of innovation.[9]

These macroeconomic dynamics are connected to equities with additional factors:

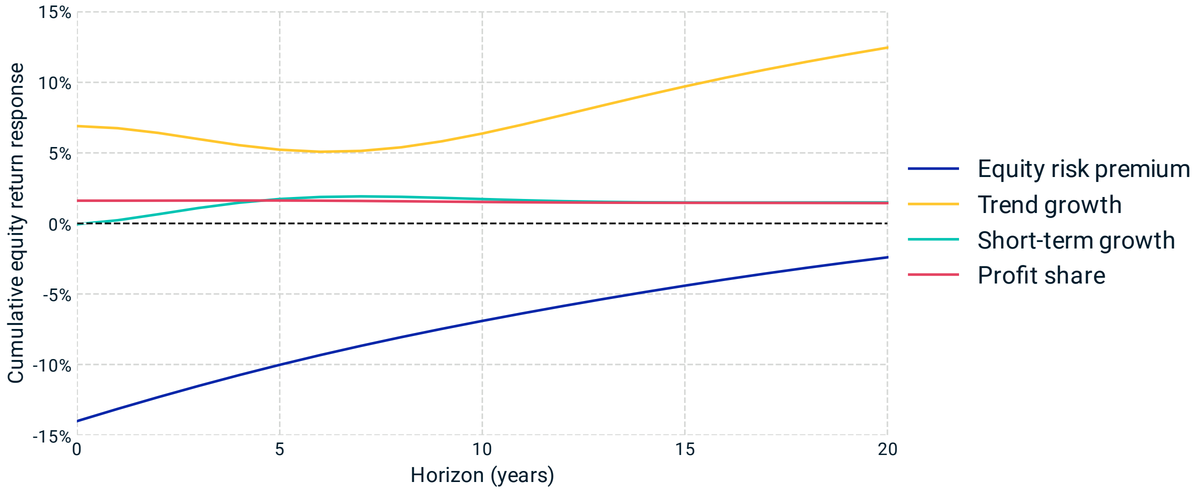
- **Equity risk premium**, which reflects the discount rate of equity relative to risk-free real yields. The risk premium is a key component of the expected return of equities, and it can fluctuate with equity valuations, tending to be higher (lower) when equity is relatively cheap (expensive) relative to the baseline macro assumptions.
- **Corporate profit share** relates overall GDP to the cash flows received by equity investors. What fraction of the total economic "pie" goes to equity investors? This can fluctuate significantly from year to year, including due to accounting artifacts, but the ratio has tended to mean revert over a three- to five-year horizon. This mean reversion makes equity cash flows much more closely connected to the macroeconomy than their short-term behavior suggests.

The influence of these factors on equity returns is shown in Exhibit 8. Short-term growth shocks are much less important for equity than changes in trend growth or changes in the equity risk premium, which is why the contemporaneous equity-growth correlation is low. Over a longer horizon, however, trend growth drives both the economy and equity.

[8] We have also explored understanding total growth as the separate effects of demographic growth and productivity growth and will likely incorporate this research into a future version of the model.

[9] We refer to the horizon up to about 20 or 30 years as "long-run." This horizon is set by the half-life of the mean-reversion speed of the long-run factors, which is approximately 13 years. Horizons longer than this are not dynamically modeled but treated as "long-long-run" parameters.

Exhibit 8: Key drivers of equity returns across different horizons



The impulse response function shows the impact on real equity returns from a 1% shock to various factors over different time horizons. In the short term, equity returns are dominated by changes in the equity risk premium, while short-term growth shocks have only a mild effect. This explains the low contemporaneous equity-growth correlation. Over longer horizons, trend growth becomes a key driver of both the economy and equity market. Equity returns are also influenced by factors driving the discounting of cash flows via real rates and equity risk premium, as discussed in Section 2.5.

1.4 Inflation, policy and interest rates

The second pillar of the model is central bank policy, which connects inflation and growth to interest rates, and feeds back into the path of the macroeconomy.

In a regime of strong and credible central bank policy, the role of inflation is subtle but powerful. Even when realized inflation is mild, many other macro dynamics revolve around the efforts to keep it that way.

A strong central bank managing inflation is a bit like a skilled acrobat walking a high rope: The likelihood of a fall (uncontrolled inflation) may be low, but only because of a constant dance of activity to respond to shocks and preserve the balance. We model central bank policy both to understand how inflation pressures get channeled into higher interest rates and lower growth, and to understand what happens if the “acrobat” were to lose their balance.

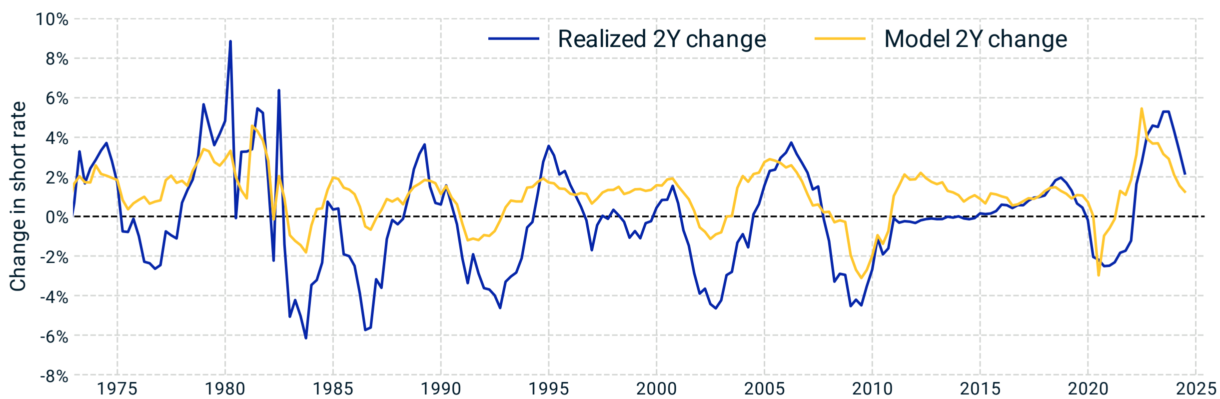
In real terms, moderate inflation and inflation shocks would not, on their own, have much effect on the economy. The economy can produce the same goods, and consumers are just as able to buy them, if sticker prices and wages have gone up. If inflation is predictable, people can enter long-term contracts and plan for the long run without much risk.

The danger of inflation is that it can spiral out of control if not kept in balance. Expectations of inflation are self-fulfilling, and once high inflation sets in the real consequences have historically been very painful. Since the Volcker era, central bank policy in most developed countries has been to aggressively quash high inflation by raising interest rates to push the economy toward recession. As unpleasant as that can be, weaker policy has been far worse, and periods of hyperinflation have often preceded social catastrophe.[10]

To avoid these possibilities and to maintain the benefits of strong credibility, modern central banks have been hawkish in keeping inflation under control. Rather than risk an inflationary spiral, most central banks quickly respond to inflationary pressures with offsetting monetary policy. As a result, inflation shocks play a key role in the economy, but much of their effect is seen in interest rates and growth, not just realized inflation itself.

As discussed in Section 3.1, we model the policy response to inflation and growth with a generalization of the Taylor rule (Taylor 1993), which prescribes interest rate policy to counterbalance inflation pressures and provide stimulus in a (demand-driven) recession. By raising rates when inflation and growth are high, and cutting rates when they are low, a central bank cools an overheating economy and provides stimulus in a downturn, aiming to balance the economy in a sweet spot of stable growth and inflation.

Exhibit 9: Policy rate model historically explained 40% of the US short rate changes



U.S. short rate changes over two-year periods as projected by the model (yellow) compared with the realized changes (blue). The model projection uses realized inflation and growth for each forecast period and accounts for interest rate mean-reversion. Over a four-decade test period, the model has a 40% R-squared explaining rate changes, and a 67% R-squared explaining the level of policy rates. The date labels correspond to the end of each two-year period.

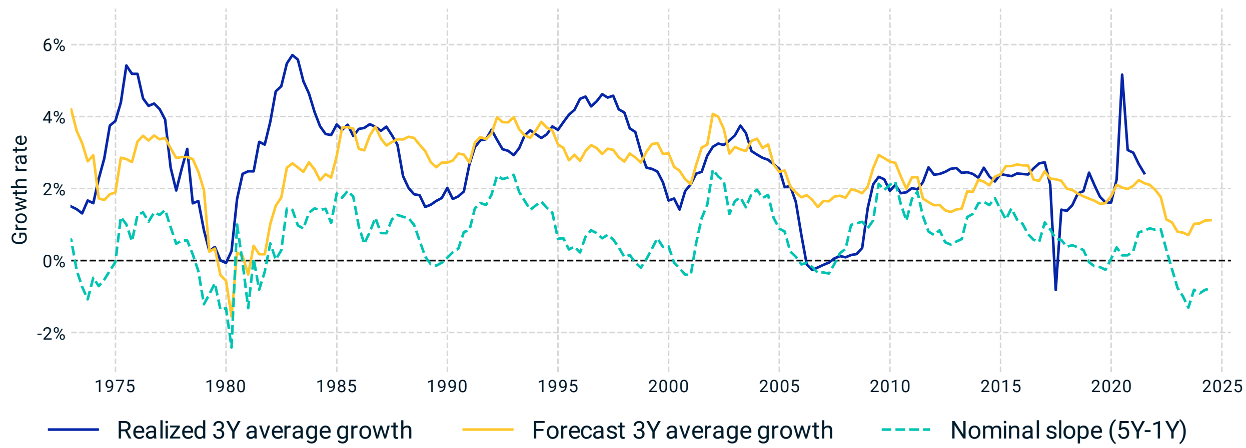
The link between macro variables and short-term interest rates also connects yield curves with expectations of future growth and inflation. For example, an inverted or downward-sloping yield curve is often interpreted as a sign of market pessimism. A downward slope suggests rates are expected to be lower in the future, which is associated with a recessionary environment.

The policy and growth models together make this intuition explicit. Exhibit 10 compares the model's growth projections, based on the yield-curve slope and other dynamics of the model. Inverted yield curves in 1980, 1999, 2006 and 2019 are associated with market expectations of an economic downturn, each of which correctly anticipated a recession.

[10] Low inflation and deflation can also be harmful, as consumers defer spending, the labor market stagnates, and central banks get stuck in a liquidity trap in which traditional monetary stimulus breaks down.

Altogether, the levels and shapes of yield curves, changes in yields over time and even the changing correlation structure across asset classes can be understood through the reactions of short-term interest rates to changes in the macro environment.

Exhibit 10: Reading growth expectations from bond market signals



Model-implied average growth forecasts, or "bond-market sentiment" (yellow) compared with realized average growth (blue) over a three-year horizon. Each data point represents the start of a three-year period, where the model's growth forecast incorporates market yield curve data available at that time, and it is aligned with the realized growth over the subsequent three years. The nominal yield curve slope (five-year minus one-year, teal dashed line) is overlaid to show its correlation with the bond-market implied growth sentiment. While not perfect predictors, these signals correctly anticipated many major economic shifts.

1.5 Private assets

As allocations have expanded, the need to understand private assets together with the total, multi-asset-class portfolio has grown. Rather than fall under a peripheral "alternatives" allocation assumed to be weakly correlated and highly diversifying, many institutional investors now see private assets as central to the core portfolio and investment mandate.

At the same time, private assets are increasingly understood not as a single asset class, but rather as offering a range of investment characteristics to play different roles in the portfolio, including income, growth and inflation protection.

Long holding periods and smooth, subjective valuations have made private assets much more difficult to understand, and contributed to the perception that they could be weakly correlated with other asset classes. But private assets live in the same economy as public assets, so they have many of the same fundamental exposures. Much like the relationship between public assets and the macroeconomy — there's *obviously* a close connection … until you look at the data — the interplay of expectations and horizons can make the relationship hard to see.

Take, for example, core real estate. As interest rates declined after the GFC, many institutions made up for a shortfall in income by shifting allocations from bonds into real estate, making private real estate part of an income portfolio. Even after some cap-rate compression, the historically stable cash flows made real estate attractive to many institutional investors, compared to yields on government or investment-grade bonds.

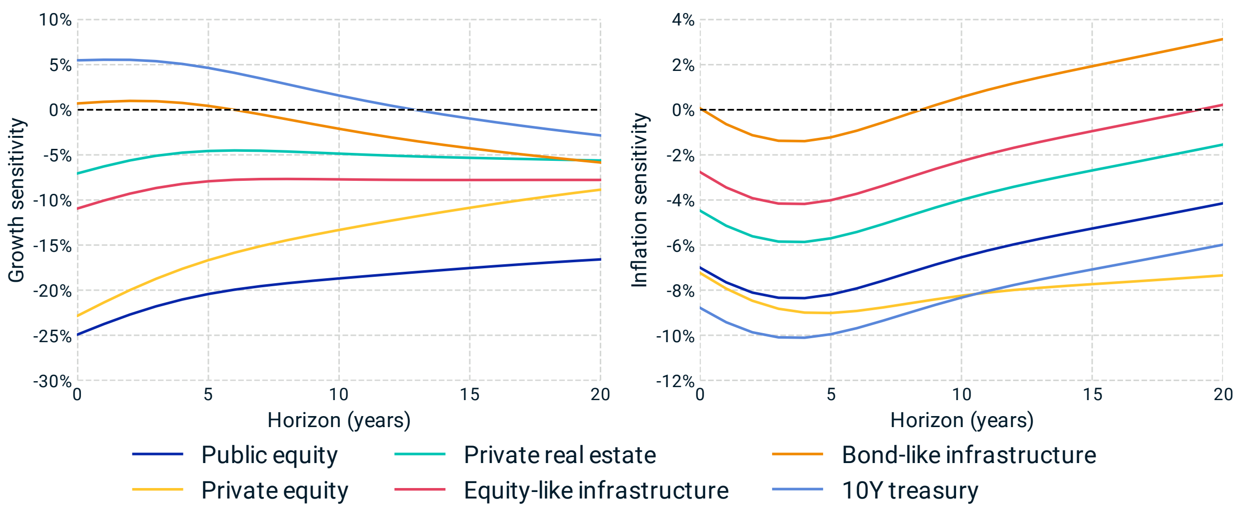
But private real estate was not a simple bond substitute. While the income stream of private real estate can be bond-like over the life of the lease, the subsequent lease and the underlying property value have large sensitivities to both growth and inflation. The COVID crisis laid bare how much growth-sensitivity was lurking in core real estate.[11]

These sensitivities make core real estate a hybrid asset, having an income stream like a bond combined with a significant dose of equity-like exposures in the long-run values. Real estate's growth and inflation sensitivities could be a benefit or a risk in the context of different portfolios and liabilities. In either case, the income streams of real estate and bonds come with markedly different exposures to macro conditions.

The sensitivities of future cash flows to growth and inflation shocks are critical to understanding private assets' role in the total portfolio. It is quite common to use discounted cash flow models (DCFs) to value private assets, but often only the systematic exposures in the discounting are considered. Ignoring cash flows' sensitivities can make any asset look bond-like, and give a wildly distorted view of the investment.

The range of macro sensitivities among private assets is as broad as public asset classes, ranging from highly growth-sensitive private equity to stable and inflation-hedging bond-like infrastructure, with everything in between found among real estate and more equity-like infrastructure. The role of private assets in the portfolio can be correspondingly diverse.

Exhibit 11: Private assets have diverse growth and inflation sensitivities



The response of U.S. asset valuations to growth and inflation shocks varies significantly by asset type and time horizon. Private equity demonstrates strong growth sensitivity, while bond-like infrastructure provides inflation protection. Growth sensitivity is calculated as the real-return response to 1% drops in short-term, cyclical and trend growth. Inflation sensitivity is calculated as the real-return response to 1% increases in short-term and trend inflation. For private assets, the model projection reflects the impact on fair value rather than valuation, so the immediate impact (t=0) is larger than what might be expected for smoothed valuations.

[11] It seemed somewhat radical at the time (Shepard 2015) to point out the growth sensitivity of real estate, but with the hindsight of the COVID crisis it now seems obvious.

1.6 Summary

We have introduced a new framework to explain a wide range of financial dynamics through the lens of a small number of macroeconomic variables. The dynamics of expected returns, market correlations, interest rate term structures, long-run risk and regime switching are explained by assets' underlying relationships to growth and inflation.

The key to these relationships is the interplay of expectations over multiple horizons. Contemporaneous growth and inflation explain only a small part of asset returns, but changing macro expectations explain more, and macro factors are the key driver in the long term.

The remainder of this note details the model framework and explores its applications to asset allocation, scenario analysis and long-horizon risk management.

2 Real growth, equity and credit

This section introduces the models relating the prices and returns of equity and equity-like assets to the cash flows, discount rates and expectations that drive them, and in turn to underlying macroeconomic factors that influence all asset classes.

Our starting point is one of the most basic relationships in finance,[12] the discounted cash flow model (DCF):

$$Price \ = \ Expected\ Present\ Value(Future\ Cash\ Flows).$$

To price equity, we consider the free cash-flow stream of companies, and model the sensitivities of these cash flows to macroeconomic expectations.

2.1 Warmup: A toy model

Start with the simple case of corporate income growing at a fixed rate, similar to the Gordon growth model (Gordon and Shapiro 1956).[13] In this case, the DCF reduces to:

$$Price \ = \frac{Income}{Discount\ Rate - Growth\ Rate}.$$

The constant growth and discounting assumptions are oversimplifications of the dynamics of cash flows and discounting, but this simple view provides key intuition.

First, the Gordon model gives a baseline for equity's sensitivities to three "factors": income, growth and discount rate.[14]

$$\frac{\Delta Price}{Price} \approx \frac{\Delta Income}{Income} + \frac{Price}{Income} \cdot \Delta Growth\ Rate - \frac{Price}{Income} \cdot \Delta Discount\ Rate$$

The coefficients show the relative importance of each factor. Income yields on equity are typically somewhere in the 3-10% range, so the $\frac{Price}{Income}$ coefficients represent large multipliers to growth and discount rates and a decades-long "duration of equity."[15]

[12] The DCF is arguably hundreds or thousands of years old.

[13] Our equity model is based on the present value of free cash flows, rather than the dividends of Gordon's model, because the discretionary and sticky nature of dividends muddies their sensitivities, and because the total payout to investors includes share buybacks in addition to dividends. For a company that balances its books, a DCF can be applied to either the total payout (dividends plus buybacks) or to the free cash flow stream. Price is *not* present value of future earnings/profits, but for simplicity, this toy model just considers "income" generically, and we refer to all of these constant growth variants as Gordon models.

[14] This derivation is useful for building intuition, but it is mathematically hand-waving, because it starts from a model assuming income, growth and discounting to be known, constant parameters, rather than dynamic, uncertain factors. Consistent factor exposures will be derived below in a model that includes the evolution and uncertainty of dynamical factors. The other mathematical handwaving of this equation is the linearization that considers a change in the left-hand side proportional to changes in each of the three variables on the right-hand side.

[15] Duration is defined as the negative of the relative price sensitivity to changes in the discount rate, and can be interpreted as the present-value-weighted average time to cash flows. In this Gordon model, duration is equal to the price-to-income ratio, expressed in years.

The sensitivity to the current cash flow is much milder. A 1% change in income — holding growth fixed — leads to a 1% change in price, while a 1% change in the growth or discount rates might lead to a 25% change in the price.

These sensitivities underscore the importance of long-run cash flows for equity. The details will change as the model incorporates more realistic dynamics, but the main conclusion remains: The value of equity is driven by discount rates and expectations for future growth much more than by short-term profits.

The constant-growth model also provides a useful baseline for thinking about expected returns:[16]

$$Expected\ Return = \frac{Income}{Price} + Expected\left(\frac{\Delta Price}{Price}\right) = \frac{Income}{Price} + Growth\ Rate = Discount\ Rate$$

These expectations for equity are closely analogous to the relationship for bonds, $Expected\ Return\ \approx\ Yield - \ Expected\ Default\ Loss\ Rate$, with a difference that equity cash-flow growth is usually positive, while credit losses are negative. This relationship is generalized with the Forward ERP, which provides market-implied expected returns for equity.

2.2 From macro to market

The factor sensitivities and expected returns in the simplified Gordon model provide intuition for some of the main goals of the Macro-Finance Model: Connect price levels, expected returns and price sensitivities to their underlying economic drivers, as in Exhibit 1. But we need to go beyond constant growth and discounting assumptions, and understand the dynamics of changing and uncertain cash flows, growth, discounting and market expectations.

We model the dynamics of cash flows and cash-flow expectations by relating aggregate corporate profits to overall economic output:

$$Profit\ = GDP\ \cdot\ Corporate\ Profit\ Ratio.$$

This relationship is on the one hand almost trivial — the corporate profit ratio is defined simply as the ratio of earnings to GDP, so it is true by definition. But it is also powerful, because it provides a key connection between equity and the macroeconomy. When combined with the DCF, it suggests a relationship between macro and market, as refined further in Section 2.4:

$$Price\ =\ Expected\ Present\ Value(Future\ GDP\ \cdot\ Corporate\ Profit\ Ratio).$$

Crucially, equity prices are sensitive to expectations about the economy stretching far into the future — the duration of equity is measured in decades — so to understand even short-term behavior of equity requires understanding long-run growth.

Previous macro-finance models took macro growth fluctuations to mean revert around a fixed, long-term trend growth rate. The original research on the equity premium puzzle (Shiller 1982; Mehra and Prescott 1985) took consumption growth to be log-normally distributed, which corresponds to immediate mean reversion of growth shocks.

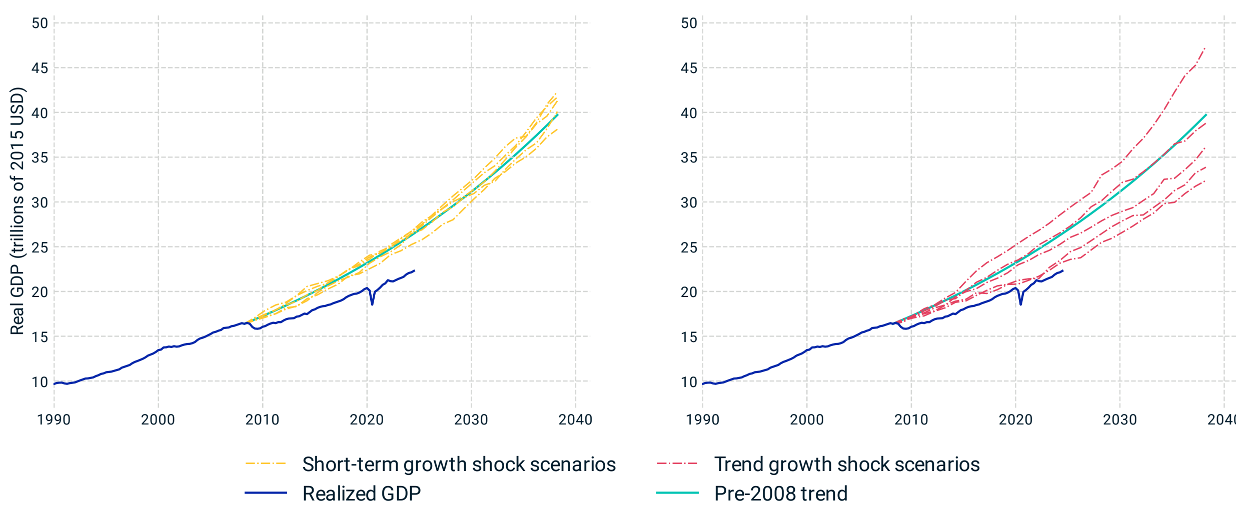
[16] These relationships use $Expected\left(\frac{\Delta Price}{Price}\right) = Expected\left(\frac{Price}{Income} \cdot \Delta Growth\ Rate - \frac{Price}{Income} \cdot \Delta Discount\ Rate + \frac{\Delta Income}{Income}\right) = Expected\left(\frac{\Delta Income}{Income}\right)$ because growth and discount rates have no predicted changes in this simple model, $\Delta Growth\ Rate = \Delta Discount\ Rate = 0$, and $Expected\left(\frac{\Delta Income}{Income}\right) = Growth\ Rate$, by definition of the growth rate.

If this were true, equity would be a very low-risk investment to long-term investors, offering a reliable stream of cash flows growing into the future, while even providing greater inflation protection than "risk-free" cash or nominal Treasurys. It's not very puzzling that assumptions of such stable long-term growth would suggest an unreasonably small risk premium.

Our model instead sees persistent shocks to trend growth, not macro volatility, as the primary source of risk. Short-term growth mean reverts, but to a moving target that can surge or lag for decades. As seen in Exhibit 12, for example, the U.S. economy had previously followed a smooth trajectory consistent with a stable trend growth rate, but the 2008 GFC and ensuing Great Recession brought on persistently lower growth for the following decade. If anything, the surprising aspect of the U.S. economy isn't the downturn in the financial crisis, but rather the remarkably stable run of growth that preceded it.

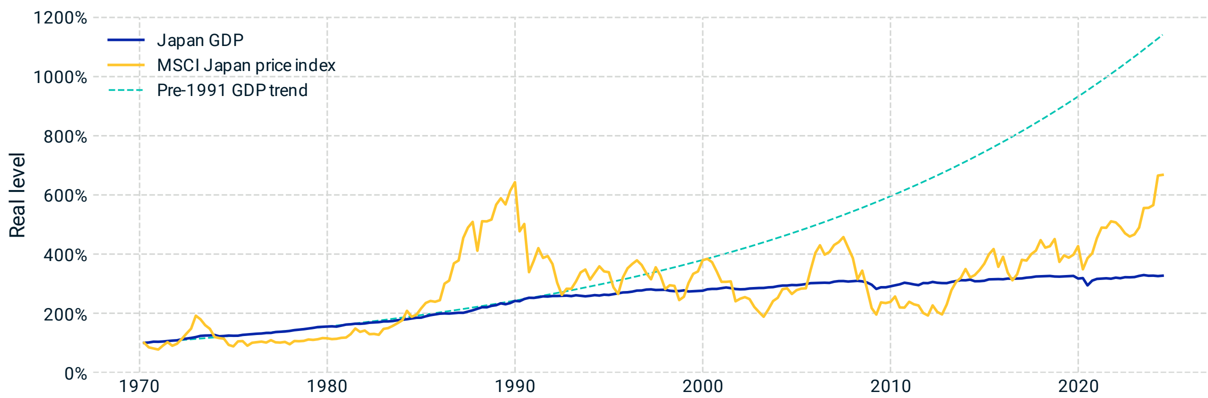
Japan has provided an example of what can happen when previously stable trend growth slows. After an extraordinary run of growth for the decades through the 1980s, Japan's growth sharply slowed in the early 1990's, as shown in Exhibit 13, bringing decades of slower growth and low returns to equity. Our model's trend growth factors reflect the risk of a similar slowdown in other economies.

Exhibit 12: The effects of short-term shocks are much less risky than trend shocks



Sample scenarios from a model with only short-term growth shocks (left panel) are similar to the pre-2008 behavior of the U.S. economy, showing limited deviation from the pre-2008 trend (teal curve), but are inconsistent with the slower post-GFC trend. In contrast, introducing shocks to trend growth (right panel) leads to much greater long-term uncertainty. While short-term volatility causes temporary fluctuations, small but persistent changes in trend growth accumulate over time, resulting in significant dispersion. The sample projections include trend shocks of a magnitude similar to the 2008 crisis. Since equity prices reflect the present value of all future cash flows, trend-growth shocks can have a much more significant impact than short-term volatility.

Exhibit 13: Trend-growth declines have a long-term impact on equity markets



The largest risk factor facing long-term equity investors is the potential for a sustained decline in trend growth, as seen in Japan's "Lost Decades" following the late 1980s asset bubble burst. The Japanese stock market performance reflects this long-term growth trend, offering a cautionary example for investors. The plot shows the growth of Japan's real GDP, its pre-1991 trend and inflation-adjusted MSCI Japan price index performance from 1970 to 2024, each normalized to their 1970 levels.

Trend- and cyclical-growth shocks fundamentally change the risk characteristics of equity. If we think of equity as a portfolio of future cash flows, persistent shocks amplify their correlations. This persistence of shocks is closely analogous in the time stream of cash flows to the role a market factor plays in the cross-section of securities. Both drive correlations that reduce diversification, and result in much higher risk relative to the level of stand-alone uncertainty.[17]

The risks of persistent growth shocks make equity quite risky, because small changes in trend growth correspond to large changes in present value. But trend-growth shocks do not usually announce themselves, and may be apparent only well after the fact. Investors are left to anticipate the future path of the economy and guess the long-term impact of news as it arrives. This leaves large uncertainty into the fair price of equity, and creates room for short-term volatility driven by market overreactions, both positive and negative, compounded by trend-following and forced selling. A second key driver of equity prices and risk, especially in the short term, is fluctuating equity risk premium, discussed in Section 2.5 below.

2.3 Real growth

Our objective in modeling macroeconomic growth is to describe it, not predict it. No one should look to our model as a particularly accurate forecast of GDP. Rather, our goal is to understand the range of possible macro scenarios and their dynamics as they play out and influence asset prices over the years ahead.

[17] This may explain the equity premium puzzle. The macro risk of equity is much larger than the short-term volatility/covariance of growth suggests. Scenarios of a persistent economic downturn could lead to large equity losses, whose pain is amplified by coinciding with the loss of present and future employment income.

A common starting point for macro-finance models (for example, Sims 1980; Stock and Watson 2001) is a first-order autoregressive model, in which short-term real growth mean reverts around a long-term trend:[18]

$$Growth(t) = b_g \cdot Growth(t-1) + \left(1 - b_g\right) \cdot Trend\ Growth + Growth\ Shock(t)$$

Growth partially carries over from one period to the next, while also mean reverting to the trend growth rate and responding to the arrival of new shocks. The parameter b_g, which determines the stickiness and the speed of mean-reversion of growth from one year to the next,[19] is observed to be small in most markets, around 0.2. Such a fast return to trend growth would lead to cash flows similar to the constant-growth case of Section 2.1. That would imply little risk in equity, unless growth volatility (or autocorrelation) were much higher than observed.

Short-term volatility of growth therefore cannot be the primary source of economic uncertainty, but we can do more to capture the term structure of growth. The critical addition is to promote the trend growth rate from a constant parameter to a dynamic variable subject to its own economic shocks:

$$Growth(t) = b_g \cdot Growth(t-1) + \left(1 - b_g\right) \cdot Trend\ Growth(\boldsymbol{t-1}) + Growth\ Shock(t)$$

$$Trend\ Growth(t) = \lambda_g \cdot Trend\ Growth(t-1) + \left(1 - \lambda_g\right) \cdot Longterm\ Growth$$
$$+ Trend\ Growth\ Shock(t)$$

The parameter λ_g (lambda) sets the timescale for persistence of trend-growth shocks, and their decades-long mean reversion toward the long-term growth rate.[20]

Comparing with the original form, the first line is nearly identical, except that the trend growth rate is now dynamical. Over short timescales, the two models look quite similar. However, as shown in Exhibit 12, trend-growth shocks change the longer-term dynamics. The model with only short-term shocks looks similar at short horizons, but behaves much more like the constant-growth Gordon

[18] Macro growth is defined as $Growth = \Delta \log(Real\ GDP) \approx \frac{\Delta Real\ GDP}{Real\ GDP}$.

[19] The half-life of mean reversion is $-1/\log_2(b_g)$. Small values of b_g correspond to fast mean reversion, while mean reversion is slow and shocks persistent as $b_g \to 1$.

Although persistent growth shocks could in principle arise by taking b_g close to 1, this parameter also determines the autocorrelation of growth (the degree to which higher than average growth one period is correlated with higher growth in the next period), which is observed to be less than 25% annually in most markets. So realistic parameter values in this model would lead to a fast return to trend growth after a shock.

[20] Trend-growth shocks persist with a half-life of $-1/\log_2(\lambda_g)$.

This model for growth has a close analogue in traditional interest rate term structure models, with short-term growth in the role normally played by short-term interest rates. In both cases, a full term structure is implied by a small number of factors, and the short end does not move in lockstep with the long end of the curve.

In theory, long-run expectations could be tightly coupled to short-run growth. Some theories assume that hyperrational consumers react to every changing macro expectation with changes in consumption, so short-run growth moves in sync with long-run expectations. In these worlds, changing economic expectations drive the *homo economicus* consumer to update their Hamilton-Jacobi-Bellman equations, determine their new optimal lifetime path of consumption and immediately adjust their spending patterns accordingly. A decline in expectations causes these consumers to quickly slash consumption in favor of saving, which in aggregate leads to a large, contemporaneous shock to short-run macro growth. Such behavior would be needed to justify short-run growth as the primary factor.

model at long horizons. Comparing with Exhibit 7, the persistent shocks capture the key to macro risk, which is much more about the long-term trajectory than short-term volatility.

The two-factor growth model above can be extended further with more factors to reflect a term structure of growth at additional timescales, and the influence of central bank policy through interest rates: when the short-term real rate is above its long-term equilibrium – corresponding to a positive real rate gap – it tends to put downward pressure on real growth. In between the short-term shocks and persistent trend growth, an intermediate-term, cyclical growth factor, explored further in Section 3.3, reflects an economic cycle:

$$\begin{aligned} Growth(t) = b_g \cdot Growth(t-1) + \left(1-b_g\right) \cdot \boldsymbol{Cyclical\ Growth(t-1)} \\ -\boldsymbol{\gamma_{r \to g}} \cdot \boldsymbol{Real\ Rate\ Gap(t-1)} + Growth\ Shock(t) \end{aligned}$$

$$\begin{aligned} Cyclical\ Growth(t) = \rho_g \cdot Cyclical\ Growth(t-1) + \left(1-\rho_g\right) \cdot Trend\ Growth(t-1) \\ +Cyclical\ Growth\ Shock(t) \end{aligned}$$

$$\begin{aligned} Trend\ Growth(t) = \lambda_g \cdot Trend\ Growth(t-1) + \left(1-\lambda_g\right) \cdot Longterm\ Growth \\ +Trend\ Growth\ Shock(t) \end{aligned}$$

The parameter ρ_g controls the timescale of the cycle, which is intermediate to those set by b_g and λ_g. When we later connect these growth factors to inflation and interest rate policy in Section 3, the trend will play the role of a sustainable, inflation-neutral level of growth, while fluctuations in cyclical growth reflect periods of out-of-equilibrium booms or recessions.

2.4 Equity cash flows

The connection between macroeconomic growth and equity starts with the corporate profit ratio, which is the fraction of total economic output going to corporate earnings:

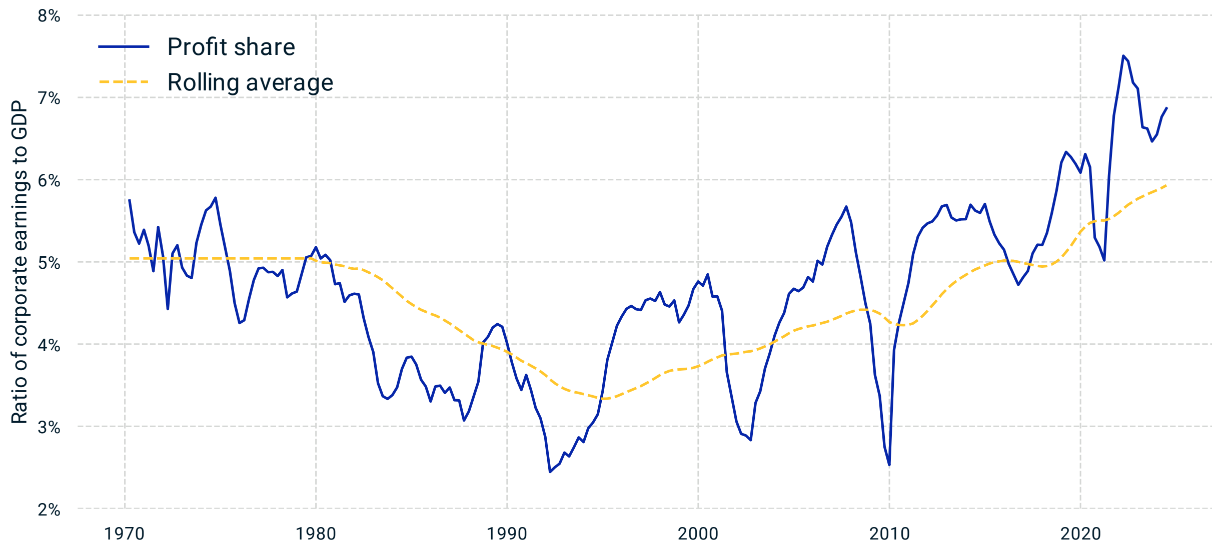
$$Corporate\ Profit\ Ratio = \frac{Earnings}{GDP}$$

The ratio roughly represents the share of the total economic "pie" that goes to equity holders. It fluctuates widely with earnings in the short term, but has tended to mean revert over a timescale of a few years, as shown in Exhibit 14.

Mean reversion of the corporate profit ratio is critical for the link between macro and market. If the ratio could drift away and maintain any value, then macro growth and equity could go their separate ways. But the tendency for corporate profits to be in the neighborhood of a stable fraction of the overall economy provides the macro-market link.

The mean reversion of the corporate profit ratio also plays an important role in equity valuation, as we explore in the next section. Equity prices reflect cash flows going decades into the future, while the corporate profit ratio tends to mean revert over timescales of just a few years. As a result, equity valuation is closely related to a cyclically adjusted earnings measure, similar to the Shiller CAPE ratio, which smooths out fluctuations in corporate profits to measure the part that is likely to be more persistent over the long term.

Exhibit 14: Mean reversion of the corporate profit ratio links cash flows to macro growth



The mean reversion of the corporate profit ratio is key to the macro-market link and equity valuation. While equity prices reflect long-term cash flows, profits tend to revert over shorter cycles, and the rolling average of corporate profit tends to remain relatively stable. The profit share ratio (blue) is calculated as the ratio of MSCI USA Index earnings to U.S. GDP, and the rolling average (yellow) is calculated using a 10-year rolling window.

The long data availability of the corporate profit ratio is important for establishing mean reversion, but price is present value of future free cash flow, not earnings.[21] Some of the accounting differences between the two don't materially change the assumptions for aggregate macro sensitivities, but some do. Most importantly, secular changes in corporate debt levels and interest rates change the burden of debt service, and change how the economic pie is split between holders of equity and corporate debt.

2.5 Discounting, the equity risk premium and valuation

Discount rates are the last piece needed to connect cash flow growth with equity, and they play two key roles: Discount rates determine the present value of future cash flows, and they determine expected returns. We use both, to project equity prices back from macro scenarios, and to project expected returns forward from market valuations.

Equity price levels are (fortunately) not mean reverting over a long horizon, but the equity risk premium (ERP) is, and that makes the ERP a useful valuation indicator. High ERP values (like those

[21] Price is not the present value of future dividends, nor future earnings. A DCF can be applied at the level of cash flows directly to the equity holder — i.e., dividends *plus* share buybacks; or at the level of the corporate balance sheet — i.e., free cash flow. For example, consider two companies with identical earnings and dividends streams, but the first needs large capital expenditure to sustain its growth, while the second is able to plow that cash into share buybacks: The latter is more valuable. Since dividends and buybacks are sticky and discretionary, their macro sensitivities are hard to calibrate, so we model free cash flow.

during the GFC) may go higher still, but they're more likely to moderate and bring price appreciation, while a low ERP (like during the dot-com bubble) is more likely to bring downward price correction.[22]

If this sounds similar to Shiller's interpretation of the CAPE, it is. The Shiller CAPE ratio compares equity prices to a rolling average of earnings. Shiller and Campbell (1988) found the CAPE to be mean reverting, and a predictor of future returns over an intermediate horizon, of five years or so.

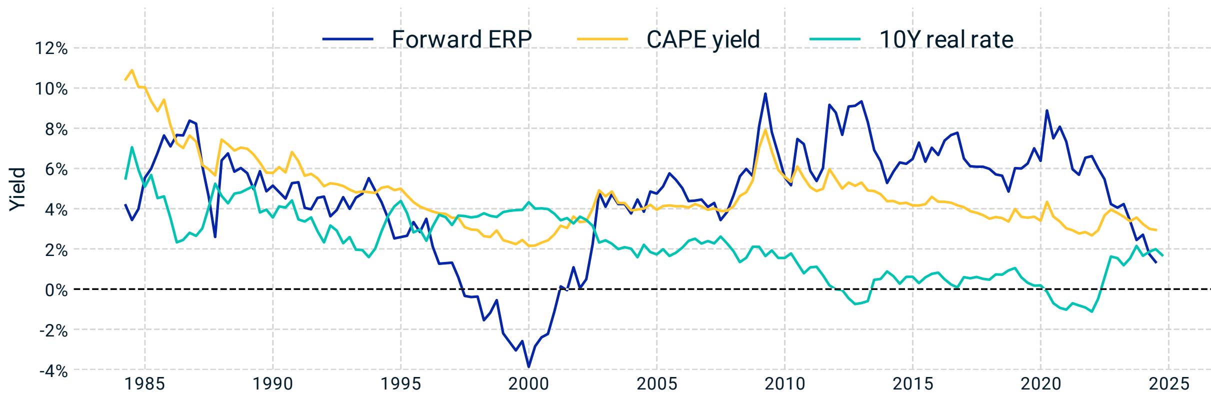
The CAPE can be understood as giving a reference price for equity relative to the fundamentals:

$$Price = \langle Earnings \rangle_{10y} \times Constant\ Earnings\ Multiple.$$

Market prices above this reference price are considered expensive, expected to lead to lower returns from slower price appreciation ("value") and lower yield ("carry"). Taking a rolling 10-year average, $\langle Earnings \rangle_{10y}$, smooths out the jumpiness in corporate earnings.

The CAPE provides a reasonable and intuitive place to start, but it can be distorted by secular shifts that change equilibrium valuations. CAPE ratios that look expensive or cheap relative to historical averages could be misleading, for example, because historical real rates were much higher.

Exhibit 15: The Forward ERP provides an equity valuation signal



Our modified valuation measure, the Forward ERP, uses a rolling average of not just earnings, as in the CAPE, but rather earnings as a fraction of the overall economy. If earnings are currently higher than their historical average, for example, they are less likely to mean revert down if they have risen with an expanding economy, as opposed to risen as a fraction of the economy. Forward ERP also takes into account real rates: A P/E ratio above its historical average is less likely to mean revert down in a low-rate environment. Reproduced from Exhibit 2.

By relating market prices to future cash-flow expectations and discount rates, the Forward ERP of our macro framework provides an alternative valuation measure to the CAPE. We model equity as

[22] Like a bond spread, the Forward ERP provides a view of future excess returns derived from expected cash flows and market prices. The ex-ante premium — the excess returns investors expect in the future — is not an extrapolation of the excess return equity investors have realized in the past. For example, if the ERP has compressed, then equity prices and the realized equity premium tend to be especially high, but the expected returns going forward are especially low.

real cash flows discounted with a dynamic ERP over the real yield curve, and define the Forward ERP as the premium that matches current market valuations to projected future cash flows.[23]

To understand the Forward ERP, start with cash flows from the simplified constant growth, Gordon-like model of Section 2.1. In this case the Forward ERP is closely related to the yield of equity:

$$Forward\ ERP \rightarrow \frac{Income}{Price} - Real\ Rate + Growth\ Rate$$

The three terms on the right — market yields, real rates, and growth expectations — remain the key influences even with more realistic cash-flow and discounting dynamics. We'll explore these in turn.

First, the income-to-price ratio reflects current market valuations. When cash flows are modeled as in Section 2.4 (rather than assuming constant growth), the role of the income yield is replaced by an adjusted ratio:

$$Macro\ Adjusted\ Yield \equiv \left\langle \frac{Free\ Cash\ Flow}{GDP} \right\rangle_{10y} \cdot \frac{GDP}{Price}$$

The macro-adjusted yield is much like the CAPE. (Squint your eyes and pretend free cash flow is the same as earnings, and that GDP can be pulled out of the average to cancel between the numerator and the denominator, and the macro-adjusted yield would equal the CAPE.) But the differences are economically and financially important.

The component $\left\langle \frac{Free\ Cash\ Flow}{GDP} \right\rangle_{10y}$ is a cyclically adjusted profit ratio, which arises as an estimate of the long-term share of the economic pie going to corporate free cash flows. As discussed in Section 2.4, the corporate profit ratio fluctuates widely over the short term, but tends to mean revert toward a long-term average. Because the duration of equity is long, this long-term average influences equity prices much more than the current cash flow. We can therefore understand the CAPE's cyclical adjustment as arising from mean reversion of the corporate profit ratio.[24]

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[23] The premium reflects the expected excess return to equity, including both rational risk premia — compensation for bearing systematic risks — and the effects of market mispricing relative to the fundamentals.

A lot of literature aims to model discount rates and the ERP arising from investor-utility functions, but we don't. Instead, we look to empirically measure the ERP and other discount rates from market prices. Various theories suggest what premium investors should demand, but we focus on the premium they actually require in the markets. The gap between theory and the markets has been especially wide during crises and bubbles, but theory also struggles to match everyday risk premia and risk-free rates.

While we have explored stochastic discount factor models (Cochrane 2005 and references therein), which model investors' optimal trade-offs between consumption today and in future states of the world, these models do not seem to robustly capture the real-world dynamics of asset prices. Specifically, they do not effectively account for equity price fluctuations during crises and bubbles, and they struggle to capture a realistic shape of the yield curve. This failure of theory may stem from the effects of central banks' asset purchases, market irrationality, and the limited ability of utility functions to represent rational investor behavior.

Note also that we model discount rates as evolving stochastically, but the discount rates relevant to prices today are known today. So price equals discounted expected cash flow, rather than the expected value of (discount factor x cash flow), with the discount rate varying by the source of the cash flow.

[24] The cyclically adjusted profit ratio is defined as a 10-year rolling average of free cash flow as a fraction of the total economy. The Forward ERP and profit ratio are based on the free cash flow, because price is the present value of

But while the CAPE averages earnings, our model implies that free cash flows as a fraction of the overall economy should be cyclically adjusted, rather than earnings. Profits that have risen with an expanding economy are more likely to persist, with no downward pull from a mean-reverting corporate profit ratio. A boost in trend growth would tend to push the CAPE to see equity as overvalued, for example, even when the equity risk premium is in line with historical averages.

The second main influence of the Forward ERP, real rates, are an important source of secular changes in market valuations.[25] A given level of the macro-adjusted yield or CAPE may look cheap or expensive compared to historical averages, but look very different relative to the real-rate environment.

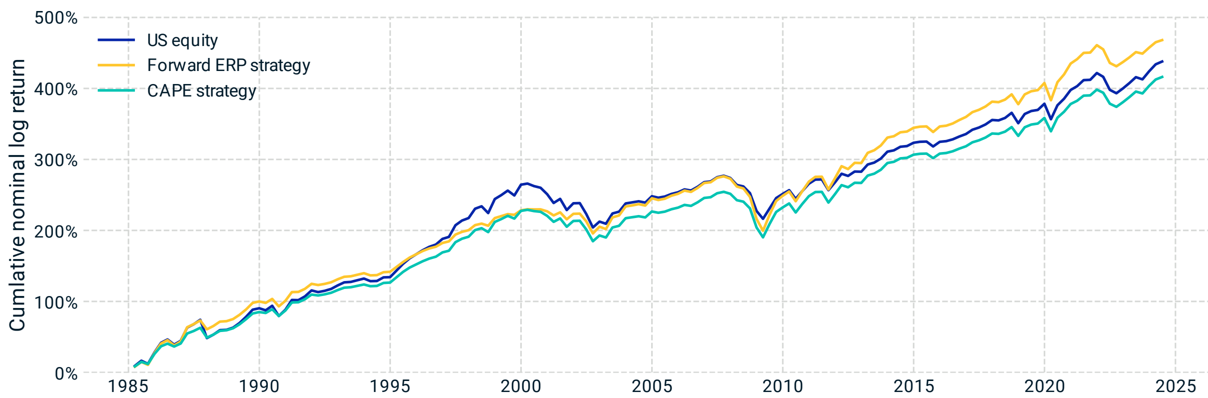
The CAPE suggested equity was overvalued throughout the late 2010s, and in the COVID recovery of 2022 the CAPE saw valuations as similar to the internet bubble. The Forward ERP saw it very differently, even after a long bull market. Rather than viewing the COVID recovery as akin to the dot-com bubble — during which sky-high valuations drove the Forward ERP into negative territory (meaning investors were paying a premium for the privilege of taking risk) — the Forward ERP saw equity valuations through the early 2020s as the rational (if long-delayed) response to a large decline in real rates after the GFC. As a result, a systematic timing strategy based on the Forward ERP overweighted equity during this bull market, while a CAPE-based strategy underweighted equity, as shown in Exhibit 16.

future free cash flows, not earnings. The distinction matters because, for example, an earnings stream sustained by a high degree of capital expenditure is less valuable.

Note that there are two forms of mean reversion at work with the Forward ERP (and the CAPE). Mean reversion of the corporate profit ratio is the reason for the cyclical adjustment. Mean reversion of the ERP is the reason to expect valuations to eventually revert — that is, the reason to associate these measures with “cheap” or “expensive” valuations.

[25] The relationship between the Forward ERP and real rates accounts for the term structure of rates and cash flows, so it is not the simple difference as in the Gordon model above. But the effect of real rates is qualitatively the same: Higher real rates lead to a lower ERP (or higher prices) for the same PE or profit ratio.

Exhibit 16: A Forward ERP-based systematic strategy has outperformed



Cumulative returns of three strategies over a 40-year period with quarterly rebalancing. "U.S. equity" reflects a static 100% equity allocation. The "Forward ERP strategy" and "CAPE strategy" dynamically adjust equity allocations based on Forward ERP and CAPE yield, respectively, using 10-year exponentially weighted moving averages. Higher Forward ERP or CAPE yield values translate to greater allocation to equity, funded by borrowing at short-term interest rates, while lower values correspond to reduced allocation to equity, with the remainder invested in short-term bonds. Over the 40-year analysis period, the "Forward ERP strategy" has outperformed, especially in the post-GFC period.

More recently, the combination of higher equity prices and real rates pushed the Forward ERP to see equity as much more expensive, similar to when Alan Greenspan called out irrational exuberance, but still far from the peaks of the dot-com bubble. This brings us to the third pillar of the Forward ERP: growth expectations.

Current valuations could be justified if AI-powered growth turns out to be as exceptional as markets seem to be expecting, but the Forward ERP implies a gap between the growth that markets have priced in and current OECD macro projections.[26]

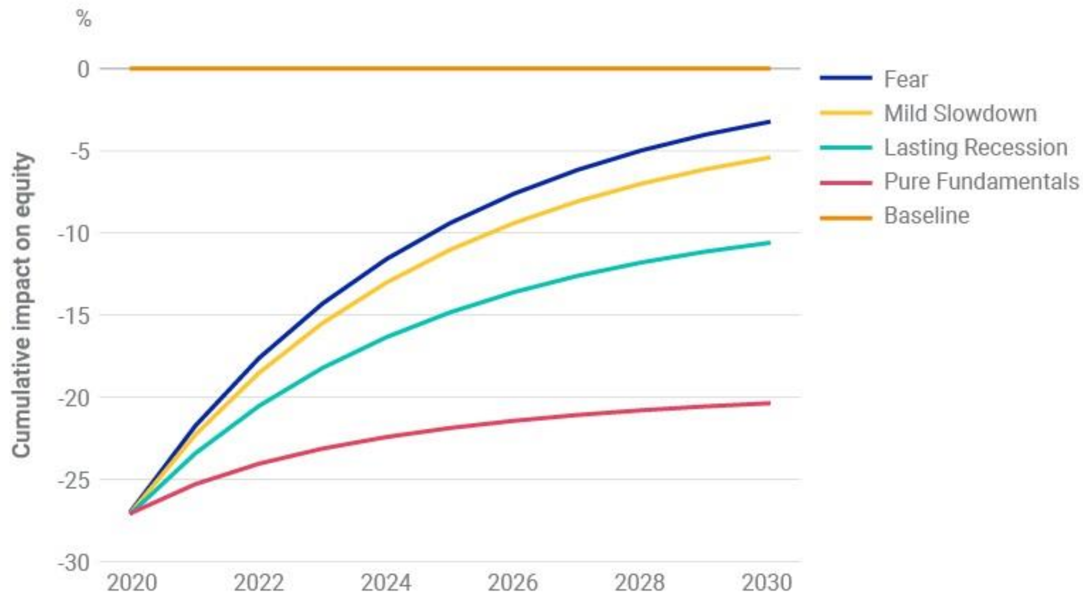
When valuations are high, does that reflect a low risk premium, or high growth expectations? If equity prices drop, is it because investors are more pessimistic about cash flow growth, or because they demand a higher premium for the same future cash flows? Our view is that equity returns and valuations are driven by both growth and ERP shocks. The broad cross-sectional dispersion of short-term stock returns can't be explained by changing systematic risk premia, as these would drive stocks up or down together along a small number of dimensions. But it is just as hard to attribute all price changes to changing growth expectations, especially during the heights of a bubble or the depths of a sell-off, at which times the valuations have required extreme growth expectations to fit with a stable ERP. A combination of changing growth expectations, changing systematic risk premia and nonsystematic premia is at work.

These differences can be seen in Exhibit 17, first published in the depths of the COVID crisis, on March 13, 2020 (Shepard, Amato and Zhou 2020), just after the markets had experienced a 25% drawdown. To assume such a sell-off was fully explained by a growth shock would require a decade-long economic contraction ("Pure Fundamentals"). However, a more plausible "Mild Slowdown" scenario would assume a combination of a shorter-term economic contraction alongside a spike in the ERP, reflecting that the markets had overreacted. These two scenarios would lead to

[26] "OECD Economic Outlook: Statistics and Projections," Organization for Economic Cooperation and Development, accessed Oct. 17, 2024.

significantly different equity market recovery paths. The latter view was largely validated by the market's recovery in the subsequent months.

Exhibit 17: Understanding return drivers can have long-term implications



As published on March 13, 2020 (Shepard, Amato and Zhou 2020), the model projected the effect of the unfolding COVID crisis on equity returns under a range of economic assumptions to explain the -25% shock the markets had then experienced. The chart illustrated four scenarios: The "pure fundamentals" scenario considered one end of the spectrum — what would be required to justify valuations based solely on economic fundamental changes — while the "fear" scenario looks at the possibility that the economy is barely disrupted and risk premium drives all of the market performance, with "lasting recession" and "mild slowdown" scenarios considering something in between.

2.6 Credit

Moving up the corporate capital structure, credit assets such as corporate bonds[27] offer equity-like exposure on top of the exposures of government bonds, but investors have found it difficult to understand that exposure, and credit allocations are often made as part of a broad allocation to bonds. A macro perspective can help investors unpack the hybrid nature of credit, and understand it alongside the drivers of returns across asset classes.

The macro model sees credit's equity-like exposures as arising both from growth exposures in the cash flows and from equity risk premium (ERP) exposures in the discount rates. But these exposures are not simply equivalent to an equity market beta. Credit's growth exposure is to nominal GDP instead of real GDP, and to the downside rather than upside — two distinctions that matter more at longer horizons. Credit's sensitivity to risk premia is also different because credit usually has a shorter duration than equity, so credit mispricings can't last as long as those for equity.

Discounted cash flow (DCF) models provide the standard view of credit, but the hard part is to understand the macro sensitivities of the cash flows and risk premia.

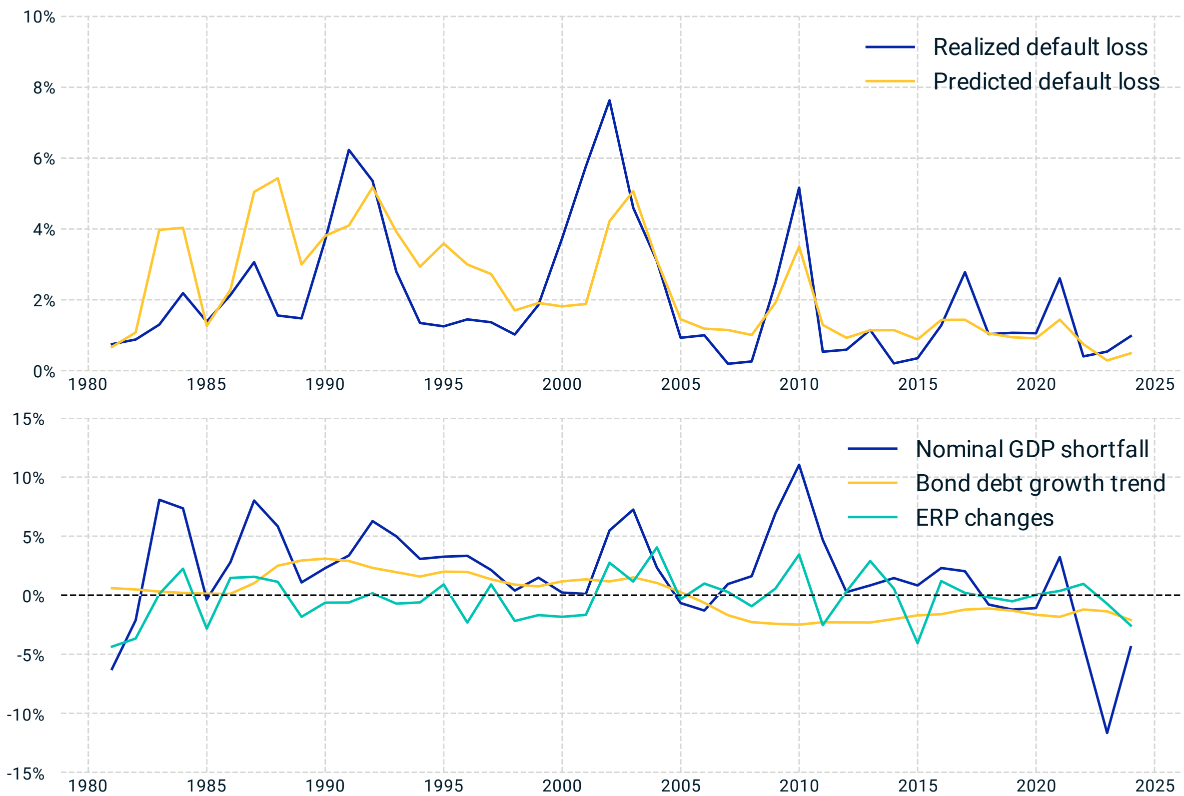
[27] This section focuses on publicly traded corporate bonds, interchangeably called "credit" or "debt."

Short-horizon investors often equate the discount rate with the yield of a bond — which is the discount rate assuming contractual cash flows with no defaults — and push expected default losses into the spread over the risk-free rate.[28] But the distinction between cash flows and discount rates matters: It is what determines expected returns. At the same spread, a bond with lower default expectations and a higher discount rate has higher expected return.

Our starting point for understanding credit-default losses is similar to structural models[29] of corporate credit, which look at a firm's enterprise value compared to its debt to model the conditions for default. Rather than look at the assets and income of individual borrowers, we model these at an aggregate level of overall corporate borrowers.

Three drivers — nominal GDP, the corporate debt level and the equity risk premium — connect macro to credit defaults, as seen in Exhibit 18.

Exhibit 18: Credit default losses correlate with economic cycles and equity risk premium



The top panel compares realized vs. predicted default losses for the MSCI USD High Yield Corporate Bond Index. The bottom panel tracks nominal GDP shortfall (defined as the difference between actual nominal GDP and the potential GDP as projected two-year prior), corporate bond debt growth trend and changes in the equity risk premium. Over the last four decades, default losses have shown a negative correlation with economic cycles, a positive correlation with the equity risk premium, and a lower trend as the debt level declined.

[28] How much of the market bond spread is credit risk premium (contributing to the true discount rate) versus expected default losses? Empirical evidence (Giesecke et al. 2011) indicates that about half of the spread is credit risk premium, and this ratio has been fairly consistent throughout history.

[29] See, for example, Merton (1974) and Geske (1977).

Nominal GDP loosely plays the role of enterprise value. It's *nominal* GDP (in contrast to equity cash flows' sensitivity to real GDP) because corporate debt is largely nominal. Inflation, by shrinking the real value of what is owed, at least makes debt easier to pay and default less likely. Nominal GDP is the strongest predictor of default losses.

Corporate debt levels influence defaults on a secular basis. Higher debt levels don't immediately trigger defaults, but a given GDP shock leads to greater defaults if corporations are more leveraged.

The equity risk premium may seem out of place as a predictor of cash-flow risks (we'll see it again below looking at discounting), but it's a good proxy for broad financial sentiment. When sentiment is high (ERP low) companies have an easier time riding out bumps in the road by, for example, securing more capital to sustain operations while they build toward a future opportunity. When sentiment dives (ERP high), companies can find themselves in a sudden cash crunch.

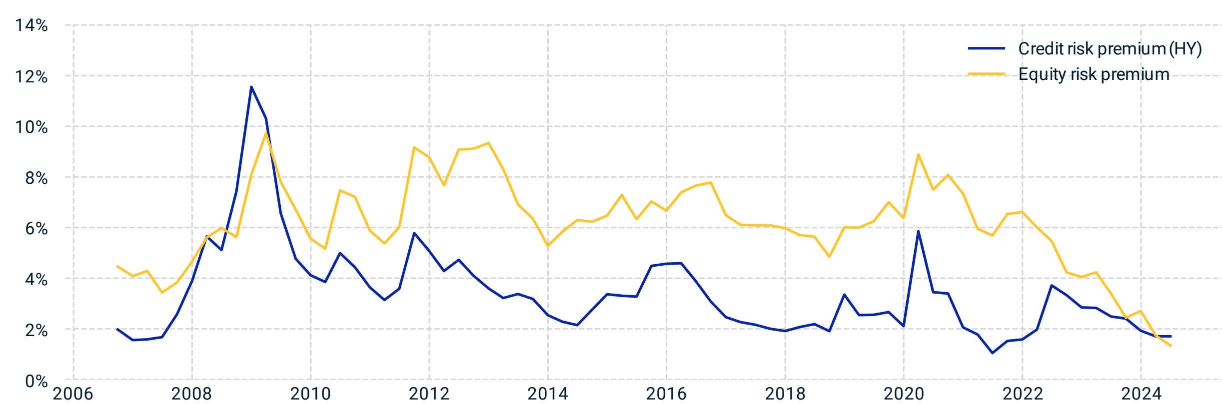
Understanding cash flows gets most of the way to understanding discount rates and expected returns. Given cash flow expectations and market prices, expected returns correspond to the discount rate fitting the market price. But projections into new scenarios require also understanding the dynamics of credit risk premia.

Like other premia, what we call the credit risk premium reflects both rational compensation for bearing systematic risks, and also return premia due to liquidity pressures or differences between investors' expectations and those priced in to markets.

There has been a close relationship between the credit risk premium — which is approximately credit spread minus expected default loss — and the equity risk premium (ERP), as seen in Exhibit 19. We model the sensitivity of credit spreads to macro scenarios through this relationship, projecting credit risk premia changes proportional to projected ERP changes. There is likely an additional component of credit liquidity premia, such as what drove the credit premium above the equity premium during the GFC, whose macro sensitivities we do not yet model.

By separating credit defaults from credit risk premia, and then connecting their dynamics to macro drivers, we arrive at a view of risk and return that consistently spans the traditional short-term market risk and long-term default risk points of view.

Exhibit 19: Credit risk premia have tended to rise in tandem with equity



Credit risk premium for the U.S. high-yield (HY) bond market and the equity risk premium from 2006 to 2024, estimated by the model. Credit risk premia have tended to rise alongside equity risk premia, particularly during period of market stress such as the GFC and the COVID pandemic.

3 Inflation, policy and interest rates

Inflation and interest rates play central roles in how the macroeconomic environment shapes asset returns. The most immediate link is between inflation pressure and short-term interest rates, but this connection propagates to influence all asset classes and horizons as rates influence growth and economic cycles.

This section introduces the interest rate model of monetary policy, the role of central bank credibility and how we relate interest rate term structure and inflation dynamics to expectations for economic growth. It also discusses how we can incorporate bond market valuations into expected returns, and the role energy prices and foreign exchange play in inflation dynamics.

3.1 Policy and the Taylor rule

Modern central banks typically have a dual mandate to maintain price stability and to promote economic growth. A standard starting point to understand central bank policy is the Taylor rule. The original form (Taylor 1993) prescribed the policy of nominal interest rates as a function of realized inflation and economic output compared with their “natural,” or equilibrium, rates. The Taylor policy raises rates when inflation or economic output is high, in order to slow the economy to bring inflation back toward the equilibrium rate, and conversely lowers rates to stimulate the economy in a recession or when low inflation allows it.

Our policy rate model is a central component of the overall Macro-Finance Model. We generalize the original Taylor rule to account for changing policy strength, to target the growth rate rather than the level of economic output, to incorporate forward-looking macro expectations, and incorporate changing long-term rates. We also account for stickiness of policy rate changes, to reflect that central banks often make changes in multiple steps rather than all at once.

First, the Taylor rule does not specify the overall strength of policy, and central banks can be more hawkish or dovish in how aggressively they raise rates to combat inflation pressure. A hawkish central bank would typically favor raising rates at the first sign of inflation, possibly to the detriment of economic growth. Section 3.2 explores how varying policy strength could cause a central bank to lose credibility and risk runaway inflation, as well as how a highly credible central bank can moderate inflation with little more than a slap on the wrist to cool the economy.

The original Taylor rule also referenced equilibrium levels of inflation and economic output. Many central banks have established the equilibrium rate of inflation to be around 2% per year, but there is much more ambiguity around the “natural” level of economic output: How much activity can an economy sustain without overheating and driving excess inflation? The original Taylor rule has been generalized (Orphanides and Williams 2002) to account for this ambiguity. In the absence of a clear natural level of output, studies (e.g., Orphanides 2003) have shown that a policy targeting the economic growth rate, rather than the absolute level of output, more accurately describes central banks’ behavior. In our model, long-term trend growth is the rate of growth an economy can sustain without driving higher inflation, and sets a regime-dependent target growth rate.

Another generalization of the original Taylor rule is to set policy rates based on forward-looking inflation and economic output instead of backward-looking values (see similar studies in Clarida et al. 2000). This important consideration reflects the ability of central banks to consider a broader array of macro variables in decision-making and their public statements.

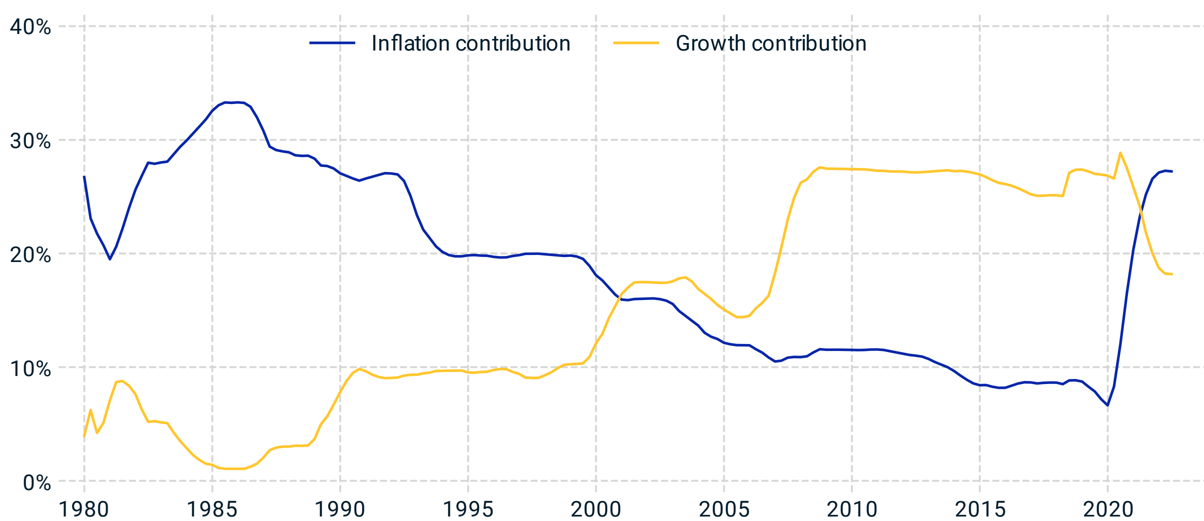
Lastly, we also generalize the Taylor rule to prescribe changes in rates, rather than their overall level, and to reflect a tendency of policymakers to smooth out changes in interest rates. Even if the optimal policy calls for a large change in rates, a central bank is unlikely to make a single, large adjustment, but instead tends to make a series of incremental steps. This smoothing naturally connects the Taylor rule with traditional term-structure models based on mean reversion of the short-term rate toward longer-term rates:

$$\begin{aligned} Policy\ Rate\ (t) &= b_r \cdot Policy\ Rate(t-1) + (1-b_r) \\ &\cdot \left(Longterm\ Real\ Rate(t-1) + Inflation(t) + \beta_\pi \cdot Inflation\ Gap(t) + \beta_g \cdot Growth\ Gap\ (t)\right) \\ &+ Policy\ Rate\ Shock(t) \end{aligned}$$

Here, the first term on the right-hand side models the stickiness of the policy rate with a (roughly) three-year half-life, and the second term accounts for the three components that form the “target” interest rate: the level of the long-term rate, and the gaps to both inflation and growth-rate targets, multiplied by policy strengths. We use regime-dependent β_π and β_g to reflect the changing regimes of central banks’ policies, as discussed further in Section 3.2.

The policy model’s projections closely tracked the actual policy rates in the U.S. over the past 40 years, as shown in Exhibit 9. Exhibit 20 below looks at the evolution of the influence of growth and inflation on policy, which is closely related to flips of the rates-equity correlation, discussed in the introduction and further in Section 4.2.

Exhibit 20: Inflation and growth have switched places as the main driver of policy rate



The 10-year rolling contribution of inflation and growth rates — two key components of the policy-rate model — to policy-rate volatilities. Before 2000, inflation was the dominant factor, but growth took over for the next two decades. Since 2022, we again saw a reversal of that trend. The lines represent the fraction of policy-rate volatility explained by these two pillars in the policy-rate model.

3.2 Inflation and central bank credibility

A second major lever of influence of central banks, in addition to setting interest rate policy, is in setting expectations. The widely accepted New Keynesian theory of inflation is, loosely, that inflation will be what people expect it to be. People adjust the terms of forward-looking agreements — such as wages in a labor contract and prices to deliver goods at a future date — based on expectations of inflation, and the terms of these agreements translate into future price levels.

If people have confidence in the central bank to hawkishly control inflation, then the central bank can keep modest inflation in check with only small, almost symbolic, enforcement of policy. When central banks have less credibility, the same amount of inflation pressure may come under control only after a much more severe and long-lasting cooling of the economy. When people lose all confidence in the central bank, expectations of higher inflation can spiral out of control into a cycle of hyperinflation that has historically ended in tears (or worse).

For these reasons, central bank credibility is critical:

- Because it's much less painful in the long run for central banks to strengthen credibility than to lose it, they often can seem to act as a strict chaperone "taking the punch bowl away just as the party was getting good" by counteracting higher growth with tighter interest rates.
- Because central bank credibility can be self-fulfilling, the level of credibility is key to determining how almost any shock plays out. Highly credible central banks can let the party go on, allowing growth to continue, or responding more modestly to fluctuations in inflation, while less credible central banks need to respond to surprises much more forcefully.
- And because a loss of credibility can spiral painfully downward, a regime shift in central bank credibility is a central long-term risk to understand and manage.

Our starting point to model the evolution of inflation is with a two-factor mean-reversion process:[30]

$$\begin{aligned} Inflation(t) = b_\pi(t-1)\cdot Inflation(t-1) + (1-b_\pi(t-1))\cdot Trend\ Inflation(t-1) \\ +\gamma_{g\to\pi}\cdot Growth\ Gap\ (t-1) + Inflation\ Shock(t) \end{aligned}$$

$$\begin{aligned} Trend\ Inflation(t) = \lambda_\pi\cdot Trend\ Inflation(t-1) \\ +(1-\lambda_\pi)\cdot Longterm\ Inflation + Trend\ Inflation\ Shock(t) \end{aligned}$$

Analogous to the real-growth model, the coefficients b_π and λ_π control the timescale of mean reversion of short-term inflation toward trend inflation, and trend inflation toward a very long-term average, respectively.[31] The growth gap term reflects how hotter- or colder-than-average growth can increase or decrease inflation.

However, mean reversion of inflation is not a given. If the coefficient b_π were a fixed parameter, the central bank could largely sit back and watch inflation take care of itself. Rather, the b_π coefficient is a dynamic variable that reflects the credibility of the central bank. We model b_π as a function of central bank credibility, as measured by inflation volatility and the gap to the inflation target, reflecting the fact that credibility is earned by the effectiveness of policy.

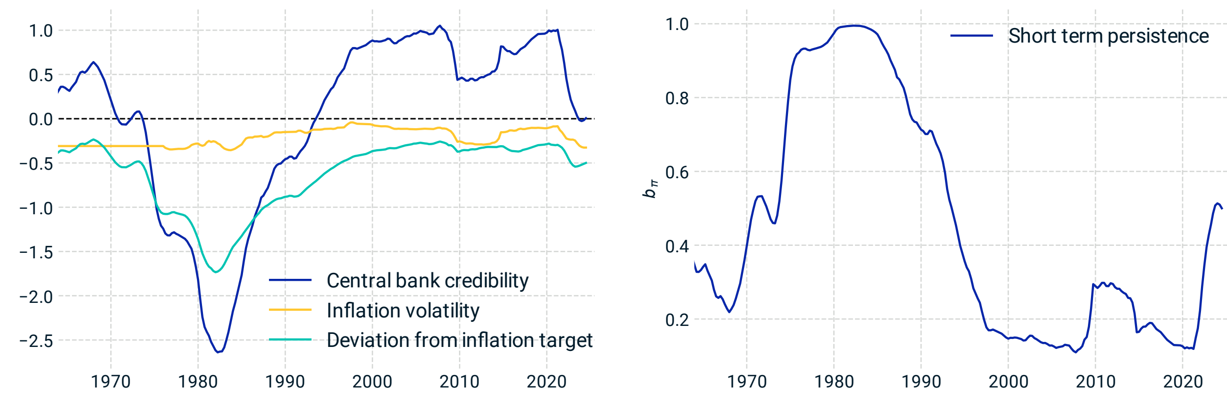
When credibility is high, people are confident that inflation will be managed, so expectations for future inflation put more weight on the inflation target, and less weight on the current fluctuation,

[30] This looks different from the standard New Keynesian relationship, $\pi_t = \beta\cdot E_t[\pi_{t+1}] + \lambda\cdot x_t + u_t$, using notation of Clarida et al. (2000), because we need to model the expectation $E_t[\pi_{t+1}]$ and output gap x_t in terms of state variables at time $t-1$.

[31] λ_π is set to 0.99, corresponding to a half-life of about 70 years. This reflects a strong persistence of trend inflation shocks, and helps fit the shape and dynamics of breakeven inflation yield curves.

corresponding to small values b_π. As seen in Exhibit 21, this was the prevailing regime in the U.S. starting in the late 1990's until the rise of inflation in the COVID crisis.

Exhibit 21: The evolution of central bank credibility



The U.S. central bank's credibility, highlighting key periods of volatility and deviation from the inflation target. During the early 1980s, credibility hit a low point but recovered after the Volcker era. The early 2020s, following the COVID recovery, presented new challenges as inflation surged. The left plot shows central bank credibility, calculated as a normalized linear combination of its components: inflation volatility and average deviation from the target (both components are plotted as their negative contribution to the credibility). Lower values reflect weaker credibility. The right plot depicts the evolution of the b_π parameter, which governs the short-term persistence of inflation, derived as a sigmoid function of the central bank credibility shown on the left. Higher b_π values correspond to weaker credibility and slower mean reversion.

When central banks are less credible, people expect fluctuations in inflation to persist, so b_π is close to 1. Rather than autocorrect, getting inflation under control in these regimes requires the full force of central bank policy of raising real rates and cooling real growth, until inflation is pushed back down. This was the case in the U.S. during the Volcker period, when austere monetary policy was needed to bring down inflation, at the expense of pushing a stagflationary economy further into recession.

A third category of regime, fortunately not seen in Exhibit 21, occurs when a central bank loses all credibility, described by $b_\pi > 1$. In these regimes of spiraling hyperinflation, rising inflation brings expectations of even greater inflation, so inflation grows exponentially.[32]

As explored in Section 4.2, in addition to its impact on how fast inflation shocks dissipate, central bank credibility is also a key influence on the correlation between equity and fixed income assets.

3.3 Term structure and premia

Interest rates are driven by growth and inflation, and yield curves reflect expectations for future interest rates, so there is a close link between yield curves and expectations for the macroeconomy. This link can be used to understand:

- What scenarios the bond markets are pricing in from the current shape of the yield curve,
- how macro scenarios could translate into shocks to the term structure and bond markets,

[32] In such regimes, inflation grows approximately as $\sim exp(t \cdot (b_\pi - 1))$.

- and what premia are available if investors' macro expectations differ from what the market has priced in.

For example, an inverted yield curve, characterized by downward-sloping rates, can be interpreted as an indication of lower expected interest rates, often associated with fears of a recession. Our model quantifies this association, to understand what bond markets expect for the macroeconomy, project hypothetical macro scenarios back to the bond markets, or understand what bonds may be cheap or expensive relative to an investor's macro views.

Traditional short-rate term structure models, such as those of Vasicek (1977) and Hull and White (1990) are foundational in finance and asset valuation. By incorporating factors such as long-term rate expectations and rate volatility, these models project future short rates based on a handful of current state variables.

The monetary policy model of Section 3.1 extends this framework by relating interest rates to the dynamics of growth and inflation. In this framework, factors driving the term structure of nominal and real yield curves are identified as the factors driving the term structure of growth and inflation expectations.

One important additional factor is needed to describe the dynamics of long-term real rates. Although there is quite a lot of theory saying what long-term real rates should be (for example, based on rational tradeoff of utility today against utility in the future), market rates bear little resemblance to theory. We adopt a view that is much more descriptive than prescriptive and model the dynamics of a trend real rate, which anchors the mean reversion of the short-term policy rate:

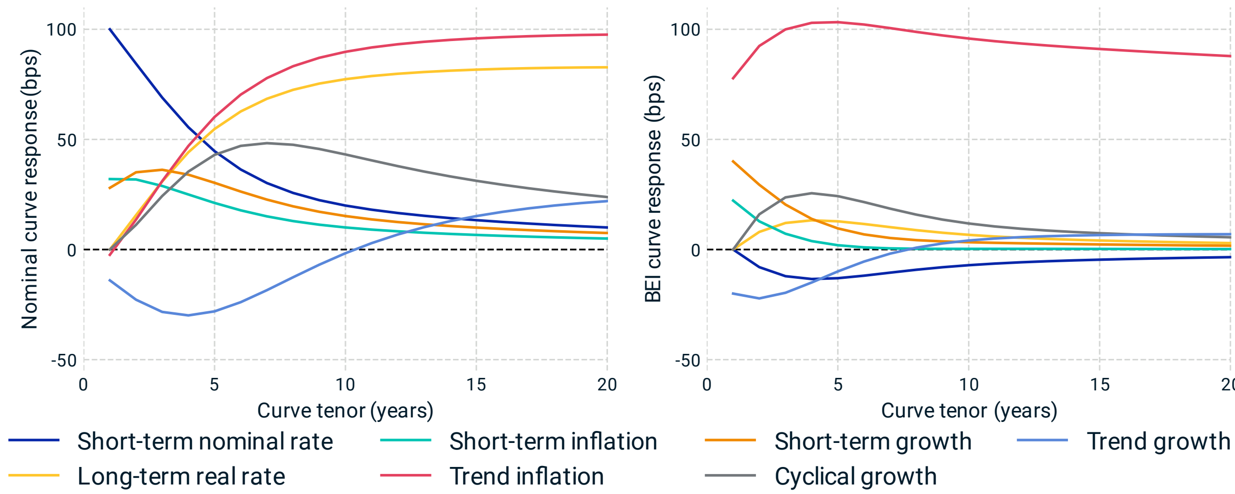
$$Longterm\ Real\ Rate(t) = \lambda_r \cdot Longterm\ Real\ Rate\ (t-1) + Longterm\ Real\ Rate\ Shock(t)$$

Importantly, mean reversion of real rates in the long term is very weak. Real rates have fluctuated up and down over short and intermediate horizons, but they have fallen significantly over horizons of decades and centuries, perhaps due to a secular increase in the availability of capital.[33] Uncertainty in long-term real rates is a major source of risk to long-term investors (though the direction of this risk may be opposite what is expected, due to the "good beta" effects discussed in Section 1.2).

If we assume the expectations hypothesis, which sets aside premia to equate forward rates with the expected value of future short rates, then there is a close link between yield curves and the macroeconomy.

[33] The parameter $\lambda_r = .99$ corresponds to a 70-year half-life of mean reversion of long-term real rates. A value slightly below 1 avoids technical problems of a unit root, while effectively turning off mean reversion over the horizons of interest to investors.

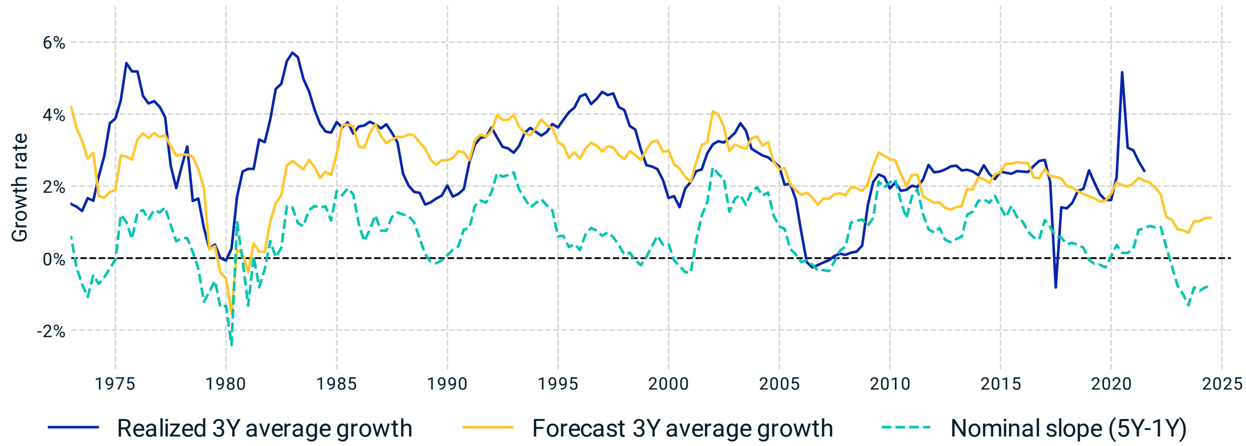
Exhibit 22: Model-implied nominal yield curve and breakeven inflation responses to macro shocks



The model-implied change in the U.S. Treasury nominal yield curve (left) and breakeven inflation curve (right) caused by macro factors. For example, a negative cyclical growth shock would tend to invert the yield curve. Each curve corresponds to a positive 1% shock to one macro factor, holding all others fixed. Realistic scenarios often include a combination of positive and negative shocks together.

As in Exhibit 22, macro factors are related to a wide range of fluctuations of the real and nominal yield curves. The links among these variables reflect the transmission mechanisms of monetary policy influencing and being influenced by the macroeconomy. For example, an inverted yield curve can now be interpreted not just directionally as a signal of economic pessimism, but directly related to a specific macro outlook, while controlling for other macro influences such as moderating inflation. Exhibit 23 shows that the medium-term, cyclical growth expectations implied by the market yield curve have historically preceded some recessionary periods, while avoiding some of the false alarms of looking at the slope alone.

Exhibit 23: Reading growth expectations from bond market signals



Model-implied average growth forecasts, or "bond-market sentiment" (yellow) compared with realized average growth (blue) over a three-year horizon. Each data point represents the start of a three-year period, where the model's growth forecast incorporates market yield curve data available at that time, and it is aligned with the realized growth over the subsequent three years. The nominal yield curve slope (five-year minus one-year, teal dashed line) is overlaid to show its correlation with the bond-market implied growth sentiment. While not perfect predictors, these signals correctly anticipated many major economic shifts. Reproduced from Exhibit 10.

The expectations hypothesis is a useful starting point for relating yield curves to future interest rates, but it neglects possible premia — both those due to rational compensation for systematic risks, and due to differences between investors' expectations and what markets have priced in.

A wide body of literature (e.g., Fama and Bliss 1987; Dai and Singleton 2002) has studied the bond term premium — the excess return of long-term risk-free bonds over short-term bonds, beyond what can be explained by the expectations hypothesis. Yield curves have tended to slope upward, offering higher yields for longer-duration bonds, which has been explained by a term premium required by investors to compensate for risks associated with holding longer-term bonds, such as interest rate risk and inflation uncertainty.

However, there is little consensus on a reliable methodology to estimate bond premia, due to their variability over time, potential macro-regime dependencies and unclear rationale.[34] There have also been periods when liquidity pressures have driven bond prices — particularly TIPS — far from what seemed like reasonable expectations. Kim and Orphanides (2012) demonstrated that a "term premium correction" can be derived using survey forecasts to anchor inflation and interest rate expectations, providing a dynamic, forward-looking view of what the market yield curves might have priced in.

[34] The tendency for nominal yield curves to be upward sloping might be explained as compensation for inflation risk, but then why has the real yield curve also tended to be upward sloping? Perhaps the real curve's slope has been a premium for interest rate risk? But long-horizon investors bear *less* interest-rate risk by locking in long-duration assets, and until recently interest rate risk was negatively correlated with equity risk, making bonds a hedge for short-term market risk. Those could suggest the real curve would be downward-sloping, corresponding to an insurance premium, rather than an upward-sloping risk premium. Could the upward slope of the real curve be a case of the expectations hypothesis with incorrect expectations, rather than a risk premium? The decades-long secular decline of real rates, which only recently reversed to bring rates back above zero, came as a surprise to many investors.

We adopt a similar approach: Our model incorporates expected inflation and growth trajectories to derive expectations for future interest rates and inflation, based on consensus economic forecasts, or a user's macroeconomic views. These inform the nominal- and real-rate term structures, and imply time-varying, regime-dependent premia in the valuation of risk-free bonds.

The estimated premia based on baseline macro projections have been small and inconsistent, with exceptions such as when TIPs experienced a liquidity squeeze during the GFC, but the model can project much larger premia when investors have macro views that differ more strongly from the baseline.

3.4 Energy supply and demand

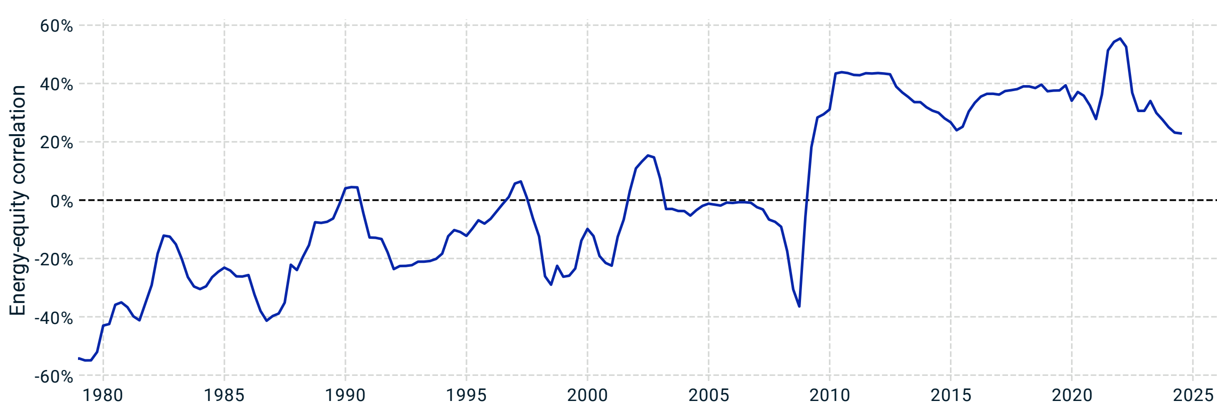
Oil and energy shocks have played a large role in shaping macroeconomic dynamics, but not all energy shocks are alike. The effects of supply shocks, often related to geopolitical tension, tend to be very different from those of demand shocks driven by the rise and fall of economic cycles.

Energy supply shocks like the 1973 oil embargo and the 2022 start of the Russia-Ukraine war typically lead to decreasing energy supplies, higher prices of both energy and other goods and services, and a headwind to global growth. The surge in U.S. fracking in 2015 and 2016 worked in the opposite direction, as excess supply cooled inflation while stimulating global growth. In these examples, energy supply was a root cause, and the broader macroeconomy followed.

Demand shocks typically see energy prices reacting to the economic cycle, rather than drive it. The sharp fluctuations of oil prices around the 2008 global financial crisis and the unprecedented negative price of crude-oil futures at the onset of COVID pandemic were stark examples of demand-driven shocks. Less dramatic events are more common, such as an everyday rise or fall in economic optimism driving growth expectations and energy prices up or down together.

Understanding the difference between supply and demand shocks is critical to understanding both their macroeconomic and financial implications. Much like the asymmetric rates-equity relationship shown in Exhibit 6, energy prices typically have a positive correlation with equity prices under demand shocks, but a negative correlation during supply shocks, when increased energy costs cut into corporate profitability, as shown in Exhibit 24.

Exhibit 24: Correlation of energy and equity prices varies with macro regimes

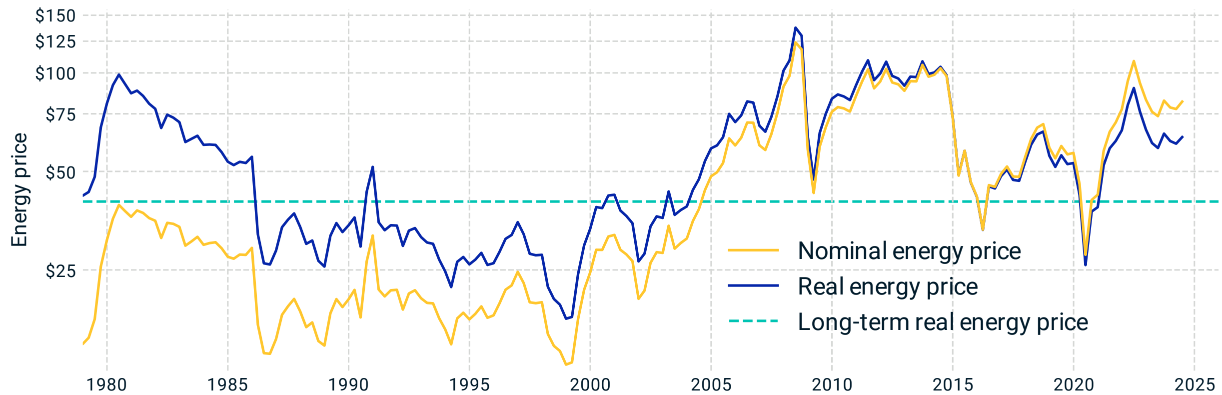


Energy and equity correlation tends to be more negative during supply shocks and more positive during demand shocks. This exhibit shows the exponentially weighted moving-average correlation, with a three-year half-life, between the annual price returns of U.S. equity and West Texas Intermediate crude oil, over the period of 1978 and 2024.

To understand these relationships, let's first understand the standalone dynamics of energy. Over a long horizon, nominal energy prices are not mean reverting, but real prices have been, as seen in Exhibit 25. So we adopt real energy prices and shocks as macro variables, with mean reversion at about a 2-year half-life, modeled with an AR(1) process:[35]

$$\begin{aligned} Real\ Energy\ Price(t) &= b_p \cdot Real\ Energy\ Price(t-1) \\ &\quad + (1-b_p) \cdot Longterm\ Real\ Energy\ Price + Real\ Energy\ Price\ Shock(t) \end{aligned}$$

Exhibit 25: Real energy prices mean revert



West Texas Intermediate crude-oil prices, adjusted for CPI inflation, display a tendency to mean revert to a long-term equilibrium. The real energy price is the logarithm of the CPI-adjusted nominal energy price, expressed as the price in 2015 USD.

Real energy prices have exhibited close interaction with the macroeconomy. First, as seen in Exhibit 26, there is a strong relationship between energy price shocks and the contemporaneous "inflation surprise" (defined as the change in inflation net of expected mean reversion and growth impact). This relationship has a mechanical connection — energy prices are a direct component of inflation[36] — but it also reflects the impact of higher energy prices feeding into higher costs of production of many goods, and the influence into inflation expectations.

This motivates the energy price shock as an additional component driving inflation, modifying the inflation dynamics:

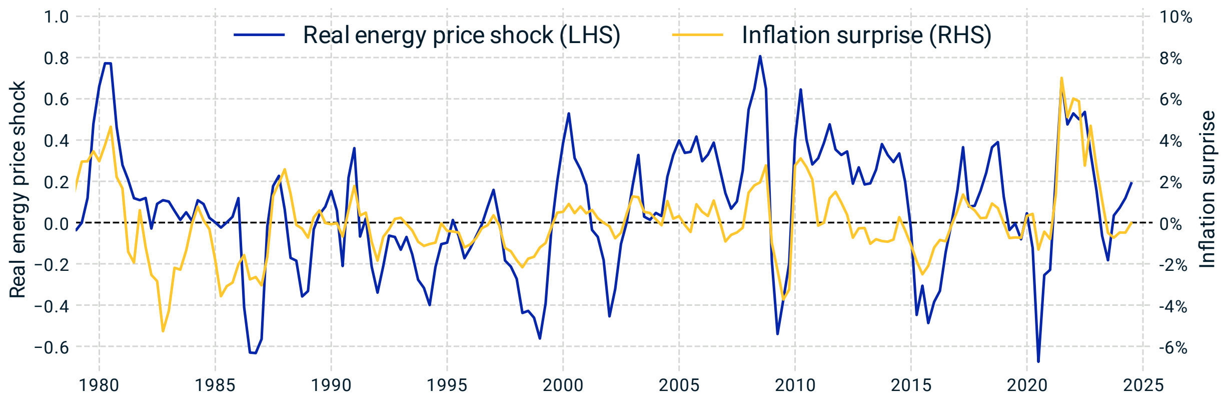
$$\begin{aligned} Inflation(t) &= b_\pi \cdot Inflation(t-1) + (1-b_\pi) \cdot Trend\ Inflation(t-1) \\ &\quad + \gamma_{g\to\pi} \cdot Growth\ Gap(t-1) + \boldsymbol{\Lambda_{p\to\pi}} \cdot \boldsymbol{Real\ Energy\ Price\ Shock\ (t)} + Inflation\ Shock(t) \end{aligned}$$

Note the "Inflation Shock" term now represents the non-energy contribution of inflation surprises.

[35] The "Real Energy Price" variable is defined as the natural logarithm of the CPI-adjusted nominal energy price. We refer to this log real price as "Real Energy Price" for simplicity in this section unless noted otherwise. In this analysis, "Energy" is represented as the price of crude oil, which is typically priced in USD in global markets, and the real price is adjusted for inflation using the U.S. consumer price index (CPI), since oil is traded globally and predominantly denominated in USD.

[36] Most countries have multiple versions of the consumer price index (CPI) to measure inflation. For example, in the U.S., the two most used indexes are the headline CPI and core CPI. The headline CPI includes all consumer goods and services including food and energy, while core CPI excludes volatile food and energy prices to provide a measure of long-term inflation trends. Our model uses headline CPI as an input to capture the dynamics of overall inflation.

Exhibit 26: Energy is a large factor driving inflation



Energy is a major component in inflation, both directly and indirectly. The correlation between real energy price shocks (left axis) and inflation surprises (right axis) is about 60%. Energy price shocks are defined as the change in the real energy price net of mean reversion and inflation surprises as the change net of mean reversion and growth impact.

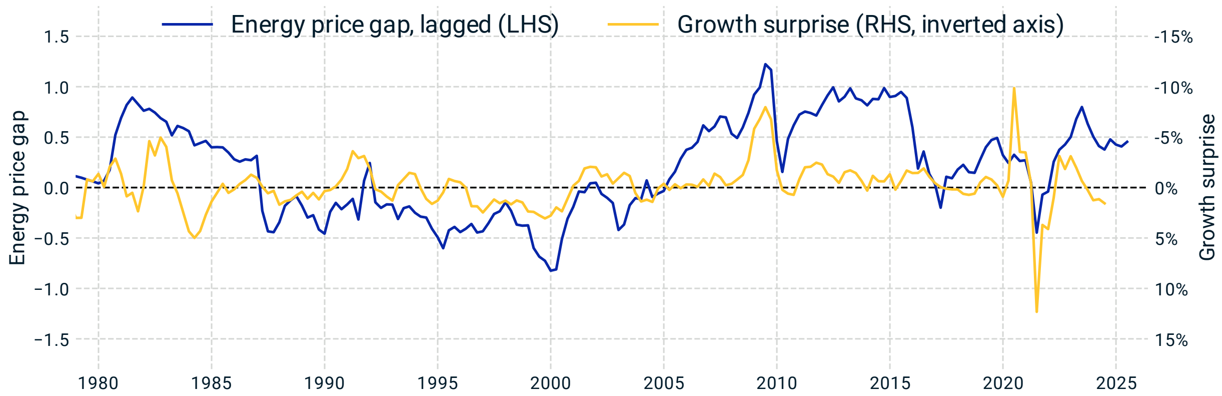
The relationship between growth and energy is less direct. Changes in energy prices have surprisingly little contemporaneous correlation with growth, but as seen in Exhibit 27 there is a strong longer-term influence. Real energy prices above or below the long-run average act as a drag or stimulant to subsequent growth, respectively. Growth surprises (defined as the change in growth net of mean reversion and the expected influence of real rates) have had a significant negative correlation with the relative price of energy.

This motivates a modification to the dynamics of growth to include the energy price gap, defined as the deviation of real energy price from its long-term average:

$$\begin{aligned} Growth(t) = b_g \cdot Growth(t-1) + \left(1 - b_g\right) \cdot Cyclical\ Growth(t-1) \\ -\gamma_{r \to g} \cdot Real\ Rate\ Gap(t-1) - \boldsymbol{\gamma_{p \to g}} \cdot \boldsymbol{Energy\ Price\ Gap(t-1)} + Growth\ Shock(t) \end{aligned}$$

The introduction of energy into growth and inflation dynamics adds an important channel for the propagation of growth and inflation, somewhat mirroring the role of real rates in shaping the economic cycle. The role of supply and demand shocks is explored further in Section 4.2.

Exhibit 27: Expensive energy slows growth



High energy prices have tended to act as a drag on growth, and vice versa. The energy price gap (left axis, defined as the real energy price relative to its long-run average) is a logarithmic variable. A value of 1 corresponds to real energy price of about 2.7 times the historical average, and -1 to a value about 70% below the average. Growth surprises (right axis, defined as the change in growth net of mean reversion and the expected influence of real rates) tend to be negative after periods when the energy price gap is positive, and positive after periods when the gap is negative. The energy price gap is shifted by one year so the lagged relationship is visible. The right axis is inverted to align the fluctuations in the curves.

3.5 Foreign exchange

The relationships between currencies and the macroeconomy are important for both: Macro variables drive currencies, and currency fluctuations change the relative costs of foreign and domestic goods and labor, feeding back into domestic inflation and growth.

A common view of foreign exchange rates (FX) has been based on the idea of purchasing power parity (PPP): If the same basket of goods costs more in one country than another — converting currencies at the market exchange rate — PPP predicts a depreciation of the currency in the more expensive country until equilibrium is reached.

The Economist's Big Mac Index is a somewhat lighthearted PPP model, based on a simple, easily comparable basket of goods: a McDonalds burger. As of this writing, their Burgernomics[37] lists the Swiss franc as the most "overvalued" currency, and the Taiwanese dollar the most "undervalued," based on the relative costs in the two countries.

The idea of purchasing power parity is intuitive, but it also happens to be wrong: Large gaps in purchasing power among currencies have persisted for decades (and for reasons unrelated to hamburgers). In the absence of transportation and sales costs, tariffs and other trade frictions, something like PPP could hold, or else there would be an arbitrage opportunity when the same good had a different price in different markets.

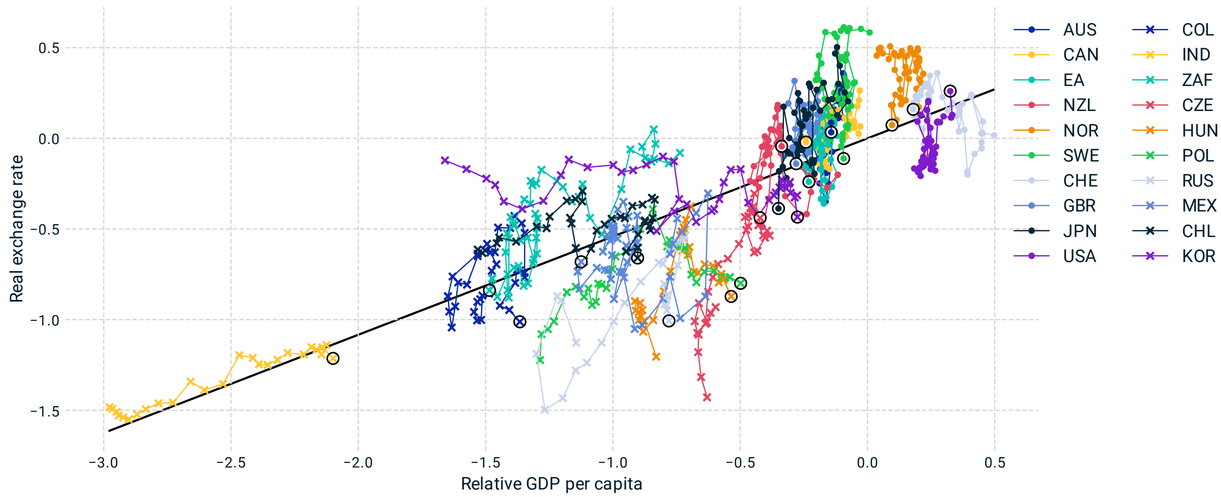
Even without tariffs and trade frictions, the costs of goods include local costs: Local staff and real estate, local logistics for delivery, local marketing and media, etc. So even if wholesale goods cost

[37] "Our Big Mac index shows how burger prices differ across borders," *Economist*, July 31, 2024.

the same anywhere, the retail cost is likely to be higher in richer countries, where shoppers can also tolerate a higher price.[38]

The resulting relationship between exchange rates and productivity is known as the Balassa-Samuelson effect (Balassa 1964; Samuelson 1964). Exhibit 28 shows our version of the Balassa-Samuelson relationship, comparing the log real exchange rate on the vertical axis to the log relative GDP per capita on the horizontal axis.

Exhibit 28: Real exchange rates tend to rise with GDP per capita



Countries with lower relative GDP per capita (left side of the plot) tend to have lower real exchange rates, while wealthier countries (right side) usually have higher real exchange rates. Poorer contries like India (shown by the yellow crosses at far left) that have experienced rapid increases in wealth also see their exchange rates moving toward the global average. At upper right, the grey dots represent the Swiss franc, which has moved up and left over time. The vertical axis is the log of the real exchange rate, and the horizontal axis is the log of GDP per capita relative to the global average, with the trendline fitted to the panel data. The data spans 1975 to 2022, though some countries have shorter history. "EA" refers to the euro area. For each country, the most recent data point is marked with a black circle, and the lines connecting the dots connect subsequent years.

The real exchange rate (RER) for a country is defined as the price of a basket of goods in that country compared with a global average[39] price for the same goods at market exchange rates. A real exchange rate above 1 (or above 0 in log terms, as in Exhibit 28) corresponds to more-expensive goods in that country. A basket of goods that costs 120 francs in Switzerland costs the equivalent of about 100 francs elsewhere, so the real exchange rate for Switzerland is about 1.2, and the log RER is about 20%. So far, this is quite similar to the Burgernomics view: Goods in Switzerland are expensive.

Exhibit 28 compares these real exchange rates to a reference line based on per-capita income. The log relative GDP per capita measures the average income in each country compared with the global

[38] There can also be offsetting effects in poorer countries due to weaker infrastructure, inefficient supply chains and scarcity of goods, which can drive prices higher.

[39] We consider a weighted average that puts more weight on wealthier countries, so the median country is below average.

average at each time. Positive values correspond to countries with higher-than-average income per person, and vice versa.[40]

The simple relationship — spanning frontier, emerging and developed markets over five decades — is remarkable. Norway today could hardly be more different from India in the 1990s,[41] but the real exchange rates of both hew to the same straight-line relationship. As India's GDP per capita grew, its currency also steadily climbed in value.

Purchasing power parity would imply a flat line, but our model recognizes that goods in Switzerland will likely remain more expensive as long as the Swiss economy is strong. In fact, the Swiss franc is almost exactly neutral today, according to our model, having previously been *undervalued* relative to income.

We incorporate this relationship as a source of mean reversion for currencies and inflation, implying a gradual pull of the real exchange rate back toward the line. The slope of the line is less than 1: An increase in a country's real income tends to be offset by higher prices, but only partially.

If a real exchange rate is low — that is, if local prices are inexpensive even after adjusting for relative income — our model implies upward pressure for the real exchange rate. This pressure could be released either through the nominal exchange rate (NER), or through inflation boosting local prices. Empirically, this pressure is usually released though currency appreciation if the central bank is strong and the currency is freely floating, and otherwise leads to inflation pressure (including in eurozone member countries).

The effects of mean reversion of the real exchange rate can be modeled with the following nominal exchange rate[42] and modified inflation dynamics:

$$\begin{aligned} NER(t) = NER(t-1) - b_{RER} \cdot RER\ Deviation(t-1) - Inflation\ Differential(t-1) \\ - Inflation\ Shock\ Differential(t) + NER\ Shock(t) \end{aligned}$$

$$\begin{aligned} Inflation(t) = b_{\pi} \cdot Inflation(t-1) + (1-b_{\pi}) \cdot Trend\ Inflation(t-1) \\ + \gamma_{g\to\pi} \cdot Growth\ Gap(t-1) - \boldsymbol{\gamma_{rer\to\pi}} \cdot \boldsymbol{RER\ Deviation(t-1)} \\ + \Lambda_{p\to\pi} \cdot Real\ Energy\ Price\ Shock\ (t) + Inflation\ Shock(t) + \boldsymbol{\Lambda_{ner\to\pi}} \cdot \boldsymbol{NER\ Shock\ (t)} \end{aligned}$$

Here RER deviation refers to the difference between the real exchange rate and the equilibrium value given by the straight line in Exhibit 28. The coefficient b_{RER} reflects both the speed of RER mean reversion and how much is channeled through NER. This term reflects a generalization of purchasing power parity to include the Balassa-Samuelson effect.

The coefficient $\gamma_{rer\to\pi}$ reflects the strength of RER mean reversion through the price-level channel, and varies by country and regime. In countries with independent and robust monetary policies, the impact of exchange rates on inflation is typically small, while real exchange-rate imbalances have tended to have a larger influence on inflation in countries with weaker central banks, particularly in

[40] The logarithms convert from an absolute to a relative scale: A given fractional change (e.g. doubling) corresponds to the same increment, regardless of the starting point.

[41] The charted exchange-rate data of the Indian rupee starts in 1997.

[42] Like RER, the sign convention of NER is such that higher values correspond to a more expensive currency, compared to a basket of foreign currencies.

emerging markets. The NER shock term reflects the concurrent impact of RER mean reversion from currency shocks.

The inflation differential term implies that currencies with above-average inflation tend to depreciate. The differential is the gap between a country's inflation and the global average inflation excluding that country. Its effect on NER arises from RER mean reversion, as the FX change is needed to offset changing price levels.[43] The "inflation shock differential" term approximately reflects the component of this effect due to inflation in the current period.

Because higher inflation is associated with higher interest rates, the inflation differential effect is closely related to two other common ideas about expected returns of currencies: uncovered-interest parity (UIP) and currency-carry strategies, which can be thought of as competing interpretations of the effects of interest-rate differences between countries.

Uncovered-interest parity assumes expected returns should be the same across currencies, so if short-term interest rates are higher in one country, it predicts an offsetting currency depreciation. This amounts to assuming that investors require no currency risk premia, and maybe more importantly (and more incorrectly) that it's supply and demand among international investors that determine FX fluctuations, rather than trade imbalances or central bank currency intervention.

A currency carry strategy, on the other hand, takes advantage of interest rate differentials among countries — borrowing in a low-interest-rate currency and investing in a high-interest-rate currency — and bets that the UIP relationship will *not* hold — i.e., that high interest rates can be earned without an offsetting currency depreciation. The strategy has been referred to as "picking up nickels in front of a steamroller" to highlight the inherent tail risk: The carry trade has often earned small but steady returns, but then a market stress can lead many investors to unwind the trade at the same time, bringing a sudden, painful currency move.

Empirically, UIP generally has not held for developed-market currencies, and the predictive power of the interest rate differential on exchange rate changes has generally been weak, and country- and regime-dependent.

Carry trading can even lead to the opposite effect predicted by UIP, as carry-trade investors put selling pressure on low-rate currencies and buying pressure on high-rate currencies. For example, Japan's prolonged period of ultralow interest rates starting in the 1990s made the yen a popular funding currency for carry trades. Contrary to the prediction or UIP, the yen did not appreciate consistently in the subsequent decades. Rather the large short positions often caused the yen to depreciate, further helping the carry trade, though this was punctuated by appreciation of the yen during flight-to-quality events such as the GFC, when investors buying the safety of Japanese bonds triggered unwinding of the carry trade.

When a UIP-like relationship has held, it has often been in cases of high inflation, high rates and currency depreciation that accompany periods of central banks struggling to maintain control,

[43] The levels of log nominal exchange rate (NER) and log real exchange rate (RER) are related through price levels: $NER = RER - p_c + p_f$, where p_c and p_f are the log price levels of a country and a global-average-ex-that-country, from which it follows that changes in NER and RER relate to an inflation differential: $\Delta NER = \Delta RER - \pi_c + \pi_f$, where π_c and π_f are the rates of inflation of a country and the global-ex-country foreign counterpart. If RER mean reverts exactly according to our modified Balassa-Samuelson rule, then $\Delta RER = -b_{RER} \cdot RER\ Deviation + shock$, so $\Delta NER = -b_{RER} \cdot RER\ Deviation - (\pi_c - \pi_f) + shock$.

especially in emerging markets. Various currency crises, such as Mexico in 1994-1995, Russia in 1998 and Turkey in 2021, saw high inflation and high interest rates as precursors to drastic currency depreciations.

To account for this range of influences of higher inflation and higher rates, we generalize the above NER dynamics to include separate influences of inflation and nominal interest rate dynamics:

$$
\begin{aligned}
NER(t) = NER(t-1) - b_{RER} \cdot RER\ Deviation(t-1) \\
-\boldsymbol{\gamma_{\pi\to NER}} \cdot Inflation\ Differential(t-1) + \boldsymbol{\gamma_{i\to NER}} \cdot \boldsymbol{Rate\ Differential(t-1)} \\
-\boldsymbol{\Lambda_{\pi\to NER}} \cdot Inflation\ Shock\ Differential(t) + \boldsymbol{\Lambda_{i\to NER}} \cdot \boldsymbol{Rate\ Shock\ Differential(t)} \\
+NER\ Shock(t)
\end{aligned}
$$

The currency-specific coefficients $\gamma_{\pi\to NER}$, $\gamma_{i\to NER}$, $\Lambda_{\pi\to NER}$, $\Lambda_{i\to NER}$ distinguish between the effects of UIP and the carry trade. In most cases, currencies are expected to depreciate when domestic inflation exceeds global inflation (positive inflation differential), with this effect being more pronounced in emerging markets than in developed markets. For currencies that are heavily involved in carry trades, where rate differentials could drive an outcome opposite to UIP prediction, they tend to appreciate if the domestic interest rate is higher than those of other countries (positive rate differential). Similarly, the inflation shock differential and rate shock differential terms reflect the concurrent impact from imbalanced shocks in domestic and foreign economies.

4 Total portfolio management

For many investors, managing a multi-asset-class portfolio requires much more than a mean-variance approach. Investors have objectives, liabilities, constraints and investment views spanning horizons from days to decades. Private assets have made it more challenging, obscuring sources of commonality across the portfolio, and bringing cash-flow management and liquidity risk to the fore. And as introduced in Section 1.2, managing long-horizon risk requires understanding regime shifting, especially in the correlation between interest rates and equity.

Whether it's a fully integrated total portfolio approach (TPA), or more traditional asset allocation, investors need a common language cutting across public and private asset classes, and across market regimes. This section explores how private assets are connected to macro drivers, and how supply and demand shocks, and central bank credibility, drive the rates-equity correlation. See Shepard et al. (2022) for a look at how such a framework is put into practice at a large sovereign wealth fund.

4.1 Private assets

As private assets have moved from the periphery toward the core of many allocations, and as the opportunity set for deals has become more competitive, it has become increasingly important to understand what drives returns. Equally, "private assets" are an asset class about as much as "public assets" are an asset class, which is to say … not really. Private assets bring a broad range of characteristics and exposures which can play different roles in meeting investors' objectives, and contribute to the same systematic risks as the rest of the portfolio.

There is also a lot of misunderstanding about private assets. Their smooth valuations can give the impression that they're much less risky than public counterparts, and the impression that they're weakly correlated with systematic risk factors. In the short run, private asset valuations are not volatile or strongly correlated. In the long run, the smooth valuations mean revert to a fair value, which reflects the obvious fact that private assets operate in the same economy as everything else. Valuation data is critical for understanding private assets, and our model relies on MSCI's broad set of private asset data (formerly from Burgiss, Real Capital Analytics and IPD), but interpreting that data requires seeing past the smoothness to understand the underlying sensitivities.

Confusion around private assets can also arise as a common side effect of the use of discounted cash flow models (DCFs) to model their return and risk. We also use DCFs for private assets (as for public assets) to benefit from their ability to fold together current market valuations with forward-looking views, and to understand income and price return together. But a common problem arises because it is much easier to understand the macro sensitivities of discount rates than cash flows … and if you only model discount-rate risk then everything looks like a credit-risk-free bond.[44]

Missing the macro sensitivity of cash flows can totally distort the view of private assets. Taking a public company private — while pouring on leverage — does not turn equity into gold. Private real estate, as many investors learned during the COVID downturn, brings equity-like growth sensitivity and other cash-flow risks.

[44] We once read a paper (not cited) about a very detailed DCF of private real estate that quoted predictions to four or five decimal places … and then came to the conclusion that more than 98% of the risk of an apartment building in Bucharest, Romania, could be captured with U.S. Treasury bonds.

Aware of these distortions, many investors look to public proxies to model private assets, which often give a completely different view. Smooth valuations and DCFs can give the impression that private assets have almost no connection to public equity, but using public proxies goes to the other extreme that private and public are identical.

A final source of confusion about private assets comes from using backward-looking returns to inform forward-looking expectations. As capital has poured into private assets, the competition for deals risen, and the supply of "fixer-upper" assets diminished, it may be dangerous to simply extrapolate past performance.

Our approach aims to combine the benefits of each of the various approaches above, while avoiding their pitfalls. We start with a DCF for each asset class, and aim to model the fair value of private assets, not their smoothed valuations. Our focus is understanding not just the expected cash flows and discount rates, but also their macro sensitivities, in order to understand how private assets may behave in different macro scenarios.

To calibrate these sensitivities, we draw on a wide range of data, including private asset valuations and cash flows. We fit sensitivities using a variety of econometric techniques, so the most economically important outputs of the model — such as long-term beta to equity, and long-term cash flow growth — are consistent with what we can observe empirically. For each private asset class, both the cash flows and discount rates have components driven by the macroeconomy, and idiosyncratic, "pure private" components.

The result is a framework that models the expected behavior of a wide range of private assets, over a wide range of scenarios and horizons. The macro sensitivities capture the key sources of systematic risk and commonality with public assets, while the profiles of cash flows and discount rates also capture how private assets are different. Very roughly, about two-thirds of the returns of diversified private asset portfolios are attributable to macro drivers, while the other third is specific to the asset class.

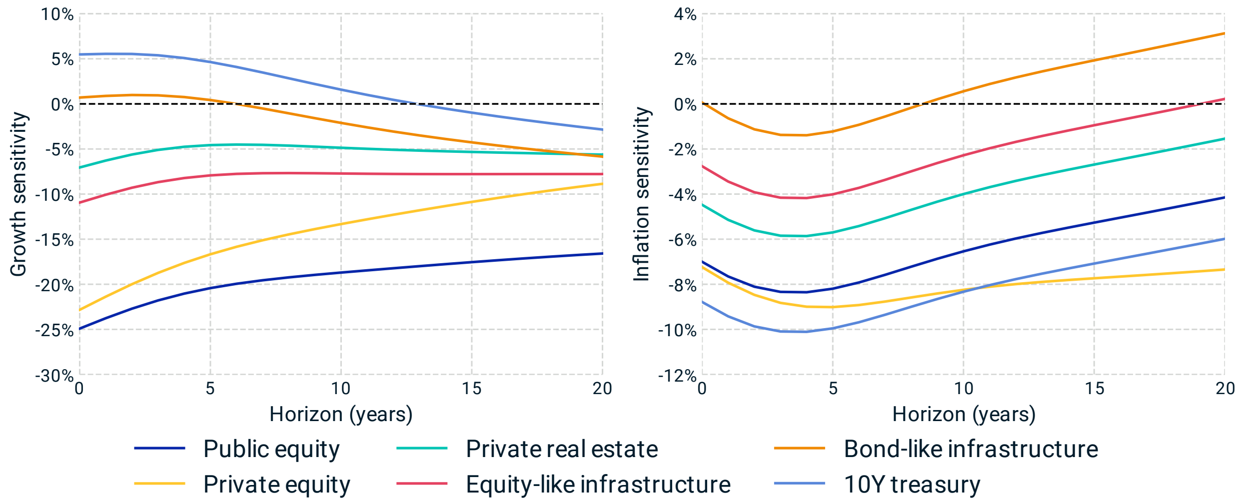
The range of behavior of private asset classes is demonstrated in Exhibit 29, which shows the projected real-return consequences of persistent growth and inflation shock scenarios, along with public equity and Treasurys, for horizons out to 20 years.

In the growth scenario, assets fall on a spectrum: Private equity suffers nearly as much as public equity, followed by equity-like infrastructure, real estate and bond-like infrastructure. In this scenario, Treasurys benefit from lower real rates and lower inflation. The responses to this growth shock are similar to the range of equity betas, though not exactly.

The effects of an inflation scenario are distinctly different from those of the growth scenario, and they make clear that risk is not one-dimensional. All asset classes suffer in this scenario, but to different degrees and through different channels. Treasurys, far from a safe haven, fare the worst in the inflation scenario. Higher rates hurt in the short term, while persistent inflation steadily erodes the real value of the cash flows.

But real assets are not automatically safe from the effects of inflation … because of their growth sensitivity. Central banks respond to higher inflation with higher real rates to cool the economy enough to bring inflation under control. Although the cash flows of real assets don't inflate away like the nominal cash flows of bonds, their growth sensitivity is painful.

Exhibit 29: Private assets have diverse growth and inflation sensitivities



The response of U.S. asset valuations to growth and inflation shocks varies significantly by asset type and time horizon. Private equity demonstrates strong growth sensitivity, while bond-like infrastructure provides inflation protection. Growth sensitivity is calculated as the real-return response to 1% drops in short-term, cyclical and trend growth. Inflation sensitivity is calculated as the real-return response to 1% increases in short-term and trend inflation. For private assets, the model projection reflects the impact on fair value rather than valuation, so the immediate impact (t=0) is larger than what might be expected for smoothed valuations. Reproduced from Exhibit 11.

Understanding private assets' connection to macro drivers is equally important for understanding expected returns. Many private assets have historically outperformed public counterparts by hundreds of basis points. If that return premium could always be assumed to hold, then many allocations might abandon public assets for anything other than a role providing for unexpected liquidity needs.[45]

But such a premium cannot hold if the capital allocated to private assets grows larger than the investment opportunity set. The logic of an illiquidity premium is that investors require extra return to be enticed into holding undesirable assets, but if investors are instead eager to hold these assets, then the premium could diminish. Extrapolation of historical returns is especially problematic if something is overvalued: Compression of discount rates boosts past returns at the expense of future returns. So as allocations to private assets rise, it becomes critical to have a forward-looking view of private asset returns.

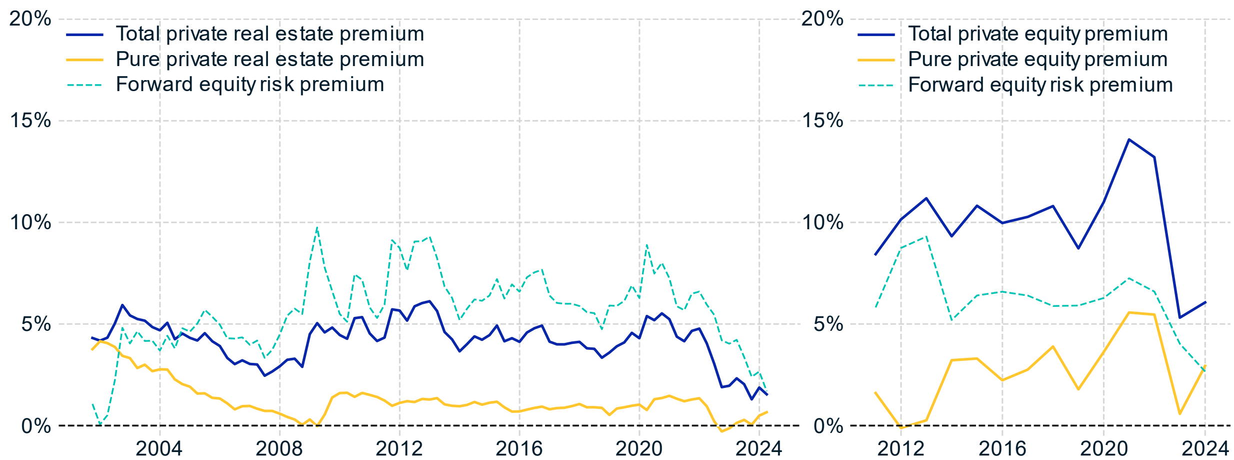
Looking back to Exhibit 1, connecting private asset cash flows to views of the macroeconomy is one important pillar of understanding the second pillar of ex-ante expected returns/discount rates, but the stool tips over without the third leg of valuation data. Such data has been more widely available in some private asset classes — private real estate cap rates have long been visible — but we are only beginning to see the equivalent of a P/E ratio for private equity.

Exhibit 30 shows the model's ex-ante view of U.S. private equity and real estate premia, using baseline cashflow assumptions. This view combines a baseline macro outlook with MSCI's deal-

[45] The risk-return trade-off of private assets looks even more attractive if smooth valuations are assumed to imply low risk.

level valuation multiple and real estate cap rate data. High valuations and rising real rates have squeezed these premia well below the level of outperformance in the past decade of high growth and low interest rates. These baseline scenarios can be refined to reflect investors' macro views or cash-flow growth expectations.

Exhibit 30: Forward-looking private asset premia



Ex-ante private asset premia combine baseline macro scenarios with MSCI's deal-level valuation multiples and real estate cap rate data. The total premium reflects the model-implied forward-looking excess return, which has recently been compressed by high valuation multiples and high real rates. The pure private components reflect the excess return above the beta-adjusted ERP.

4.2 Correlation regimes

The dynamics of cross-asset correlations are central to asset allocation and risk management, as introduced in Section 1.2. Portfolios constructed to hedge risks with offsetting correlations have significant exposure to risk in the correlation structure itself.[46]

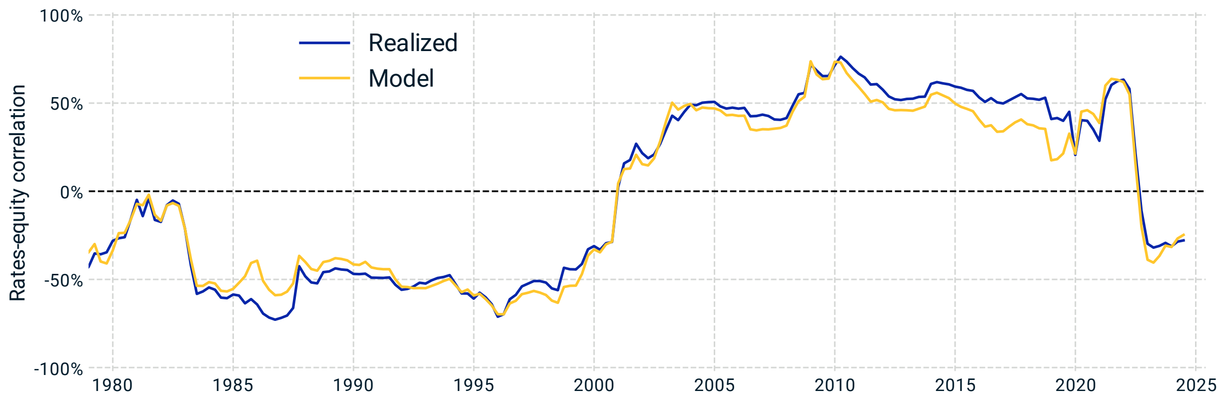
Stress testing can play a large role in managing the risks of a regime shift by considering scenarios representing different regimes. The challenges are to understand which scenarios are realistic, and how to propagate a scenario across markets and time horizons: Both have often relied on … correlations.

The adage that correlation is not causation is right. We need to see through statistical, backward-looking correlations to understand the structure of underlying causes. Correlations can be thought of as an average over various causes (weighted by variance or frequency). A simple model of the rates-equity correlation, as in Exhibit 32, views this correlation as arising out of the balance between the competing effects of supply and demand shocks: Demand shocks drive a positive rates-equity correlation, while supply shocks push it negative. The resulting correlation is a result of the balance of causes, whose importance varies with regimes.

The reversal of the rates-equity correlation when supply shocks kicked in during the COVID recovery was an example of such a regime shift, which hurt balanced and risk-parity allocations that had benefited from offsetting volatility between bonds and equities.

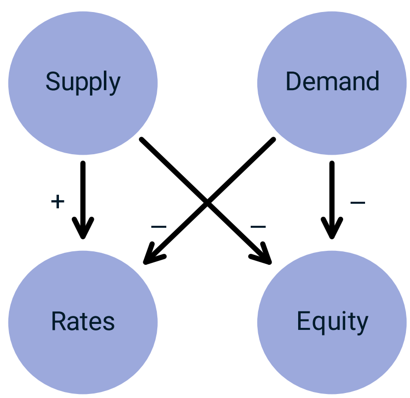
[46] Risks due to uncertainty in the covariance matrix were explored in Shepard (2008). See also footnote 6.

Exhibit 31: The US rates and equity correlation sign flipped (again)



The U.S. rates-equity correlation was positive for most of the past two decades, serving as a hedge between bond and equity allocations. Before 2000, however, the correlation was negative, and it recently flipped sign again amidst entrenched inflation. The model attributes this correlation to more fundamental drivers, and aligns closely with realized correlation. The early 2000s flip can be linked to lower supply shock volatility and strengthened central bank credibility. Reproduced from Exhibit 5.

Exhibit 32: Supply and demand are asymmetric drivers of the rates-equity correlation

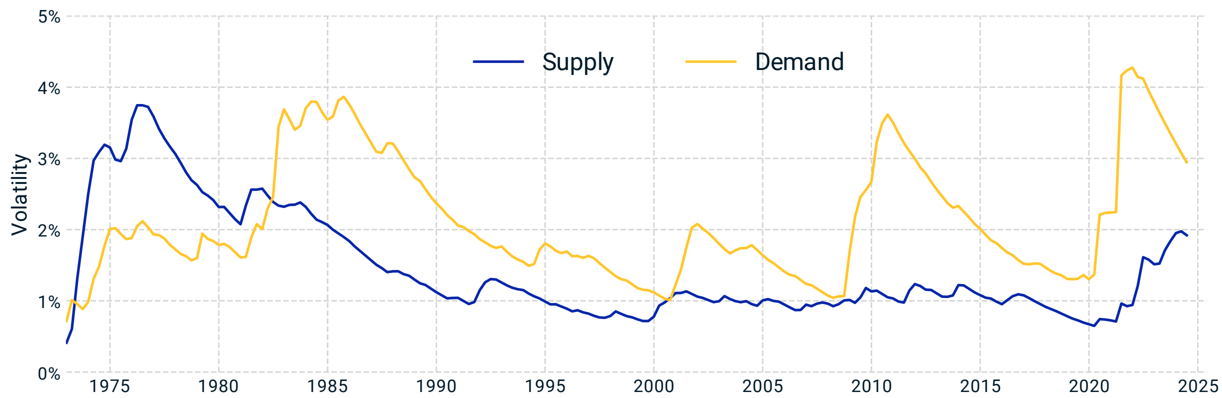


The MSCI Macro-Finance Model sees through the correlation to its underlying, asymmetric drivers. In this simplified view, negative supply shocks drive rates up and equity down, while negative demand shocks drive both down. The rates-equity correlation reflects the combined effect of these competing forces. By identifying these underlying causes, investors can differentiate behavior across different scenarios and understand how the rates-equity correlation might evolve in future scenarios or policy regimes. Reproduced from Exhibit 6.

The rates-equity correlation is not quite as simple as the supply/demand model of Exhibit 32. The influence of supply shocks in the U.S. and elsewhere waned at least a decade before rates-equity correlations flipped in the early 2000s, as seen in Exhibit 33. A second key influence was a regime shift related to central bank credibility.

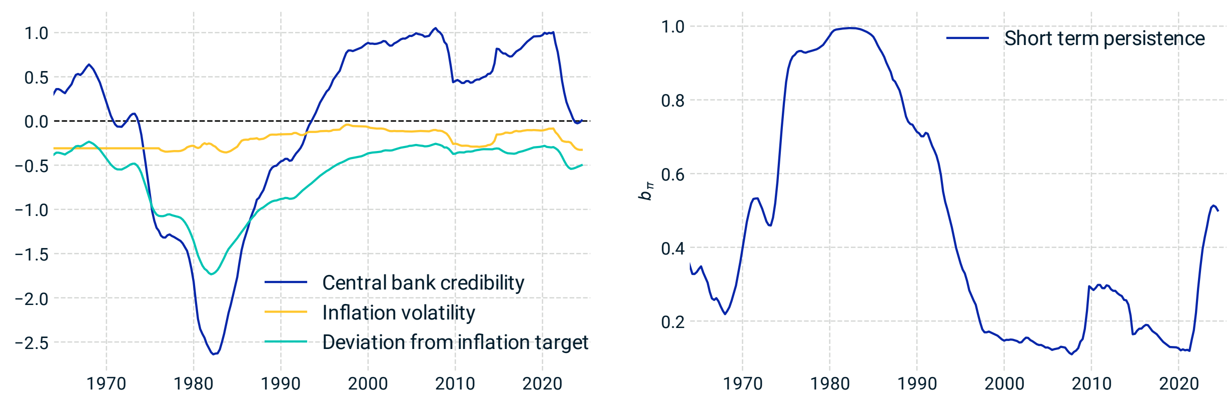
After the stagflation of the 1970s was brought under control with the tight monetary policy of the Volcker era in the 1980s, inflation and supply-shock volatility were stable, but central bank credibility was not. People largely expected inflation shocks to persist, and so they did, as seen in Exhibit 34 and discussed in Section 3.2. During this period, controlling higher inflation required a strong central bank response to raise interest rates and cool growth.

Exhibit 33: Historical volatility of supply and demand factors



The historical volatility of supply and demand factors estimated from 1973 to 2024. Supply shocks dominated in the late 1970s and early 1980s, while demand shocks took over since the mid-1980s. The 2022 surge in supply volatility came as demand volatility began to decline from a 2021 peak, which contributed to the rates-equity correlation sign change. Roughly speaking, a 1% negative supply shock translates to about 1% inflation and -0.5% growth, and a 1% negative demand shock corresponds to about -1% growth and -0.5% inflation. Over the analysis period, the supply and demand factors together have explained about 80% of macro-variable variance.

Exhibit 34: The evolution of central bank credibility



The U.S. central bank's credibility, highlighting key periods of volatility and deviation from the inflation target. During the early 1980s, credibility hit a low point but recovered after the Volcker era. The early 2020s, following the COVID recovery, presented new challenges as inflation surged. The left plot shows central bank credibility, calculated as a normalized linear combination of its components: inflation volatility and average deviation from the target (both components are plotted here as their negative contribution to the credibility). Lower values reflect weaker credibility. The right plot depicts the evolution of the b_π parameter, which governs the short-term persistence of inflation, derived as a sigmoid function of the central bank credibility shown on the left. Higher b_π values correspond to weaker credibility and slower mean reversion. Reproduced from Exhibit 21.

Although supply shocks influence equity consistently through both cash flows and discount rates, the net effects of demand shocks are less clear-cut. Demand shocks drive rates and growth together, which have competing influences on equity. A positive rates-equity correlation (the norm in the first two decades of this century) occurs if the growth effect is stronger than the rates effect. But when central bank credibility is weaker, a macro surprise requires a more forceful policy response, and growth faces headwinds. In these regimes, higher demand-driven inflation would require more aggressive interest rate hikes to keep it in check, cutting short the economic growth. As a result, rates tend to go up more, while equity gains are more limited.

As markets move through different regimes, investors must anticipate — rather than react to — changes in the correlation structure of the markets. By understanding the underlying drivers that could shape future correlations, such as the balance between supply and demand forces, and how they shift with policy regimes, investors can better navigate this long-horizon challenge.

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Appendix

The MSCI Macro-Finance Model sees asset returns as driven by fundamental macroeconomic causes. This framework enables a consistent approach to risk and returns: A baseline macro scenario can be combined with market valuations to provide a term structure of implied expected returns; or, macro shock scenarios can be propagated through shocked discount rates and asset prices to understand a term structure of risk. The model consists of two components: integrated macro dynamics and asset valuation models.

The integrated macro dynamics capture the evolution of macroeconomic variables, using a vector autoregressive (VAR) model:

$$\theta_t = M\theta_{t-1} + (I - M)\bar{\theta} + \Lambda\eta_t$$

θ_t represents a vector of macro variable time series and the matrix M governs both the mean reversion of individual variables to their long-term mean ($\bar{\theta}$) and the causal interactions among variables, such as the tendency for higher growth to drive up inflation. [47] The matrix Λ models how independent, unexpected shocks (η_t) propagate through the system contemporaneously, such as how unexpected inflation shocks could lead to a central bank's response of raising the interest rate in the same period.

The macro dynamics use quarterly frequency data and annual step size — i.e., a quarterly time series of θ_t is estimated, with parameters M and Λ reflecting the movement over a year-long horizon, from θ_{t-1} to θ_t.

The asset valuation models connect the macro dynamics with multi-asset-class valuations using a discounted cash flow framework: [48]

$$Asset\ Price_t = \sum_{n>0} Discount\ Factor_{t,n} \cdot E_t[Cash\ Flow_{t+n}]$$

This approach ensures consistent and differentiated macro exposures across various asset classes. For example, both public and private equity are significantly exposed to economic growth, with private equity typically having higher growth exposure due to greater leverage taken on. Similarly, interest rates factor into the discount rates for almost all asset classes, making the sensitivity of equity assets to rates more explicit than in traditional equity factor models.

In this appendix, Sections A and B cover the details of the macro dynamics and asset valuation models for each asset class, followed by input data and estimation methodology in Section C and validation results in Section D.

A. Macro dynamics

The core pillars of the model are real growth, inflation and interest rates. These factors — spanning a long horizon — interact with each other as well as with other macro variables, such as the equity risk

[47] For variables without a long-term mean (e.g., nominal exchange rate), the dynamics are determined by the rest of the autoregressive variables, without the corresponding $\bar{\theta}$.

[48] The $Discount\ Factor_{t,n}$ is derived from the n-year node on the discount curve, as of time t, which is used to discount $Cash\ Flow_{t+n}$, the cash flow n years from the analysis date t. The bi-temporality of the discount factor is important for analyzing asset prices at a future date.

premium, exchange rates and energy prices. Below, we summarize the model dynamics, followed by their mathematical representations.

1. Model dynamics

Short-term real growth (g_t) is influenced by several factors:

- **Short-term growth shocks (u_t^g):** These shocks are the primary source of growth volatility, though they only modestly contribute to macro risk for most growth-sensitive assets due to a short half-life.
- **Short-term growth persistence:** Higher-than-average growth in one year tends to be followed by somewhat higher growth the following year.
- **Medium-term cyclical growth (c_t):** Cycles of above- or below-trend growth become evident over a two-to-three-year horizon. Market expectations for the medium-term economic cycle are reflected in the shape of the yield curve, whose short end has tended to exhibit a downward slope in anticipation of a recession or a steep upward slope in anticipation of a boom.
- **Trend growth (a_t):** The largest source of long-term macro risk for equity and other growth-sensitive assets is a persistent slowdown in growth, which can last many years or even decades.
- **Short-term real interest rate (r_t):** Low short-term real rates (relative to the long-term real rates ϕ_t) tend to stimulate economic activity, while higher real rates tend to cool the economy. Adjusting interest rates is the primary lever used by the central bank to control inflation by managing economic activity.
- **Real energy prices (p_t):** Higher energy prices tend to have a slowdown effect on real growth in the near future. The energy price itself, in log term, follows a mean reversion to a long-term average.

Short-term inflation (π_t) is influenced by the interplay between inflationary pressure, central bank response and how central bank credibility and inflation expectations shape realized inflation:

- **Short-term inflation shocks (u_t^π):** These unexpected inflation shocks are the primary source of inflation volatility, though their persistent impact on inflation risk is highly dependent on the central bank's credibility.
- **Central bank credibility (b_π) and short-term inflation persistence:** When a central bank is highly credible (corresponding to a small b_π), short-term inflation surprises tend to reverse even before a strong response from the central bank, as expectations of stable inflation tend to be self-fulfilling. Conversely, if a central bank is less credible (corresponding to higher b_π), short-term inflation shocks may persist and accumulate, until brought under control by much more aggressive economic policies. The dynamic mean-reversion parameter b_π captures the shifting credibility.[49]
- **Long-term inflation (ψ_t):** The target or equilibrium inflation rate has evolved over time, hovering around 2% in many developed markets in recent decades, compared to higher levels in earlier periods.
- **Short-term real growth (g_t):** Higher real growth is often inflationary, leading to rate hikes by the central bank to cool the economy.

[49] To calculate b_π, we first calculate "credibility" as a linear combination of inflation volatility and EWMA of absolute inflation deviation from equilibrium, then apply a sigmoid function on this credibility measure. The higher the credibility, the lower the b_π, and vice versa.

- **Real energy price shock (u_t^p):** Energy prices can directly influence inflation as a key component, and its effect is usually felt concurrently.
- **Real exchange rate (rer_t):** If domestic prices of a basket of goods are lower than global prices (after adjusting for per-capita GDP), domestic prices tend to rise to align with global levels.

Short-term nominal rate, or policy rate (i_t) is modeled with a generalized Taylor rule, reflecting how central bank policy responds to the economic environment:

- **Short-term inflation (π_t):** Central banks typically raise rates when inflation is above its long-term target π_t^*, and vice versa.
- **Short-term growth (g_t):** Central banks also tend to raise rates when growth is above the long-term growth target g_t^*, and vice versa.
- **Policy strength (β_π, β_g):** Central banks' responses to inflation and growth conditions can vary depending on their policy stances. Paul Volcker notably initiated a regime of hawkish policy, in which the Federal Reserve and many other central banks aggressively raised rates to fight inflationary pressure. A higher β_π indicates a more hawkish central bank facing inflation pressure, and a higher β_g reflects a larger rates action in response to a growth shock. Both parameters are country- and regime-dependent.
- **Short-term rate persistence:** Central banks tend to adjust rates incrementally, with a series of smaller changes anchored to the previous rate, even when economic conditions alone may justify larger changes.
- **Long-term real rate (ϕ_t):** The baseline for setting the policy rate is influenced by the long-term real interest rate, which evolves alongside the long-term real rate and inflation targets.
- **Short-term nominal rate shocks (u_t^i):** Central banks can surprise the market by being more or less aggressive in their policy actions than what a Taylor-like rule would suggest.

Nominal exchange rate (ner_t) is driven by a combination of factors, including real exchange rate imbalances, inflation differential, and interest rate differential:

- **Real exchange rate (rer_t):** Real exchange rate tends to mean revert to its equilibrium level determined by relative GDP-per-capita (RW_t), as suggested by the Balassa-Samuelson effect. This channels through NER mechanically.
- **Inflation differential ($\pi_t - \pi_t^{foreign}$):** Currency tends to depreciate when it experiences higher inflation compared to its foreign counterpart, as it loses real value more quickly.
- **Interest rate differential ($i_t - i_t^{foreign}$):** Currency tends to appreciate when it has a higher nominal interest rate compared to its foreign counterpart, as higher rates make it more attractive for carry trades. Though this effect is highly country- and regime-dependent.
- **Inflation differential shocks ($u_t^\pi - u_t^{\pi_f}$)** and **rate differential shocks ($u_t^i - u_t^{i_f}$):** Reflect the impact of inflation and rate differentials on the exchange rate in the current period.

Term structure of equity risk premia (erp_t) is modeled with a mean-reverting short-rate model.

Real energy price (p_t) over a long horizon tends to mean revert, and is modeled as an AR(1) process.

2. Mathematical representation

The following equations describe the model dynamics.

Generally, the b_v, λ_v, ρ_v parameters control the mean reversion of macro variable v; $\gamma_{v_1 \to v_2}$ and $\Lambda_{v_1 \to v_2}$ typically govern how the level of or shock to variable v_1 influences variable v_2. Additionally, β_π and β_g

capture the strength of central bank response when inflation and growth are out of balance. Unless otherwise noted, all these parameters are positive, therefore the plus or minus sign associated with each term can provide a directional guidance on the impact of each channel.

Growth pillar:

- Short-term growth: $g_t = b_g g_{t-1} + (1 - b_g)c_{t-1} - \gamma_{r \to g}(r_{t-1} - \phi_{t-1}) - \gamma_{p \to g}(p_{t-1} - \bar{p}) + u_t^g$
- Cyclical growth: $c_t = \rho_g c_{t-1} + (1 - \rho_g)a_{t-1} + u_t^c$
- Trend growth: $a_t = \lambda_g a_{t-1} + (1 - \lambda_g)\bar{a} + u_t^a$

Parameters b_g, ρ_g, λ_g govern the mean reversion of the growth rate at different horizons, together forming a "term structure" of growth rates.

Parameters $\gamma_{r \to g}$ and $\gamma_{p \to g}$ capture how the real interest rate and real energy price channels influence short-term growth.

Parameters $\bar{a}$ and $\bar{p}$ are the long-term averages of growth rate and real energy prices, respectively. Stochastic variables u_t^g, u_t^c, u_t^a represent the unexpected shock of short-term, cyclical and long-term trend growth. The short-term real rate r_{t-1} is defined as the short-term nominal rate minus the expected short-term inflation, i.e., $r_{t-1} = i_{t-1} - E[\pi_t]$.

Inflation pillar:

- Short-term inflation: $\pi_t = b_{\pi,t-1}\pi_{t-1} + (1 - b_{\pi,t-1})\psi_{t-1} + \gamma_{g \to \pi}(g_{t-1} - g_{t-1}^*)$
$$-\gamma_{rer \to \pi}(rer_{t-1} - rer_{t-1}^*) + \Lambda_{p \to \pi}u_t^p + \Lambda_{ner \to \pi}u_t^{ner} + u_t^\pi$$
- Long-term inflation: $\psi_t = \lambda_\pi \psi_{t-1} + (1 - \lambda_\pi)\bar{\pi} + u_t^\psi$
- Growth target: $g_t^* = w_a a_t + (1 - w_a)\bar{a}$

Parameters $b_{\pi,t-1}$ and λ_π govern the mean reversion of inflation at different horizons, forming the basis of the inflation trajectory. $b_{\pi,t-1}$ is time-dependent, hence the additional t-1 subscript.

Parameters $\gamma_{g \to \pi}$ and $\gamma_{rer \to \pi}$ capture the lagged inflationary effects of growth and real exchange rate mean reversion, while parameters $\Lambda_{p \to \pi}$ and $\Lambda_{ner \to \pi}$ capture the contemporaneous effects on inflation through energy prices and FX. $\gamma_{rer \to \pi}$ and $\Lambda_{ner \to \pi}$ are generally small for countries with an independent and credible central bank, and larger for emerging markets or countries without strong monetary policies.

Parameter $\bar{\pi}$ is the long-term average of inflation rate. Stochastic variables u_t^π, u_t^ψ, u_t^p and u_t^{ner} represent the unexpected shock of short-term and long-term inflation, real energy prices and nominal exchange rate, respectively.

$g_{t-1} - g_{t-1}^*$ represents the "growth gap": the distance of growth rate from its target, where the target g^* is a weighted-average of the long-run averages $\bar{a}$ and current long-term values a_t.

$rer_{t-1} - rer_{t-1}^*$ represents the RER deviation. The equilibrium real exchange rate rer_t^* is a linear function of the relative GDP per capita, reflecting the Balassa-Samuelson effect.

Monetary policy and interest rates pillar:

- Short-term nominal rate: $i_t = b_r i_{t-1} + (1 - b_r)\left(\phi_{t-1} + \pi_t + \beta_\pi(\pi_t - \pi_t^*) + \beta_g(g_t - g_t^*)\right) + u_t^i$
- Long-term real rate: $\phi_t = \lambda_r \phi_{t-1} + (1 - \lambda_r)\bar{\phi} + u_t^\phi$
- Inflation target: $\pi_t^* = w_\pi \psi_t + (1 - w_\pi)\bar{\pi}$
- Growth target: $g_t^* = w_a a_t + (1 - w_a)\bar{a}$

Parameters b_r and λ_r govern the mean reversion of interest rates at different horizons, forming the basis of the yield curve term structure.

Parameters β_π and β_g capture the inflation and growth pillars of the monetary policy and are country- and regime-dependent.[50]

Parameters $\bar{\phi}$, $\bar{\pi}$ and $\bar{a}$ are the long-term averages that real rates, inflation and growth are assumed to mean revert to, respectively. Stochastic variables u_t^i and u_t^ϕ each represents unexpected shock of the short-term nominal rate and long-term real rate.

The inflation target π_t^* and growth target g_t^* used in formulating monetary policy targets are a blend of long-run averages $(\bar{\pi}, \bar{a})$ and current long-term values (ψ_t, a_t). This approach strikes a balance between stability and responsiveness to changing conditions. Essentially, the central bank sets its policies with reference to a very long-run anchor while also assessing whether current inflation and growth are deviating from recent trends. Parameters w_π and w_a determine how much weight the targets put on each component.

Energy price process:

- Real price of energy: $p_t = b_p p_{t-1} + \left(1 - b_p\right)\bar{p} + u_t^p$

Parameter b_p governs the mean reversion of real price of energy (in log term) to its long-term average $\bar{p}$, while u_t^p is the unexpected shock.

Exchange rate process:

- Nominal exchange rate: $ner_t = ner_{t-1} - b_{rer}(rer_{t-1} - rer_{t-1}^*) - \gamma_{\pi\to ner}\left(\pi_{t-1} - \pi_{t-1}^{foreign}\right)$
  $+\gamma_{i\to ner}\left(i_{t-1} - i_{t-1}^{foreign}\right) - \Lambda_{\pi\to ner}\left(u_t^\pi - u_t^{\pi_f}\right) + \Lambda_{i\to ner}\left(u_t^i - u_t^{i_f}\right) + u_t^{ner}$

Parameters b_{rer} governs the mean-reversion of RER. $\gamma_{\pi\to ner}$ and $\gamma_{i\to ner}$ govern how inflation and rate differential channels influence the change in nominal exchange rate. In addition, $\Lambda_{\pi\to ner}$ and $\Lambda_{i\to ner}$ parameters capture the concurrent effects. The effects of $\gamma_{i\to ner}$ and $\Lambda_{i\to ner}$ are related to carry trade and are significant only in some countries.

Risk premium process:

- Equity risk premium (ERP): $erp_t = b_{erp} erp_{t-1} + \left(1 - b_{erp}\right)\overline{erp} + u_t^{erp}$

[50] For the U.S. model, for example, we model four different policy regimes: 1970-1978 “Pre-Volcker”, 1978-1982 “Volcker shock”, 1982-2008 “Great moderation” and 2008-now “Post GFC”. Policy regimes can continue to evolve, but it happens infrequently.

Even though the notation for β_π and β_g does not include a subscript t (to distinguish them from the dynamic parameters such as $b_{\pi,t}$), they have different values for each of the policy regime above.

The parameter b_{erp} governs the mean-reversion process of erp_t, and the parameter $\overline{erp}$ is the long-term average. u_t^{erp} is the unexpected ERP shock.

Foreign variables and dynamics:

In addition to modeling the dynamics of the domestic macroeconomic variables for a target country ("domestic variables"), as described above, the model also incorporates parallel dynamics for the same set of macro variables (excluding energy price and exchange rate[51]) for the global market, referred to as the "foreign variables". These foreign variables represent the global economy excluding the target country.[52]

While the parameters governing the foreign macro dynamics have the same conceptual meanings as those for the domestic economy, some of their values differ, driven by empirical data. For example, the long-term growth rate assumptions for Japan and Germany are lower than that for the U.S. As a result, in the U.S. model, the foreign component reflects a lower overall long-term growth rate compared to the domestic growth rate.

Global markets are interconnected through trade, capital flows and income convergence. These factors create a system where economic events in one region can quickly influence the economic conditions of others, creating both opportunities and vulnerabilities on a global scale. The model captures this effect by incorporating channels for cross-country shock propagation. For example, a short-term growth shock in the U.S. can influence the global economy, while a long-term inflation shock originating outside the U.S. can have ripple effects on U.S. inflation as well.

B. Asset valuation models

The model uses a discounted cash flow (DCF) framework for asset valuation, offering several key benefits:

- Forward-looking: The cash flow component reflects future macro projections, and the discount rate serves as an estimate for future expected returns by making connection with current asset valuation.
- Multi-horizon: The model provides a common language for evaluating risk and returns over different investment horizons.
- Systematic: This framework enables understanding the price changes as results of fundamental drivers such as risk-free rate, risk premium and growth expectations. Investors can aggregate risk and returns for total portfolios through consistent macro factors.
- Intuitive: The DCF approach is straightforward and easy to understand.

[51] The energy price and exchange rate variables do not have a "foreign" counterpart, because crude oil is globally traded, with prices impacting markets worldwide, and a country's exchange rate is inherently a bilateral variable, with foreign variables (such as inflation and interest rates) influencing its dynamic.

[52] As of this writing, the global market includes Australia, Canada, France, Germany, Italy, Japan, the U.K. and the U.S., weighted by their respective equity market cap according to the MSCI World Index. The inclusion criteria depend on the country's weight in the capital markets, as well as the availability of its macro data (such as breakeven-inflation data), which makes robust estimations of the long-term variables possible. These countries collectively account for 91% of the market cap of the MSCI World Index, and 81% of the market cap of the MSCI ACWI Index, as of Sept. 30, 2024.

This section elaborates the DCF valuation model for each asset class. For each asset class, we focus on the models and assumptions for the stream of expected cash flows $E_t[Cash\ Flow_{t+n}]$ and the term structure of discount rate, $Discount\ rate_{t,n}$, the two building blocks of the valuation model:

$$Asset\ Price_t = \sum_{n>0} E_t[Cash\ Flow_{t+n}] \cdot Discount\ Factor_{t,n} = \sum_{n>0} E_t[Cash\ Flow_{t+n}] \cdot e^{-n \cdot Discount\ Rate_{t,n}}$$

Equity

As discussed in Section 2, the equity valuation model ties economic growth and corporate profit to future equity cash flows, and connects the risk-free rate and equity risk premium to the discount rate.[53] The cash flows that public equity investors are entitled to are known as free cash flows to equity (FCFE), representing the earnings available for distribution to shareholders after covering all expenses and reinvestments.[54]

We model the expected future cash flows of public equity n years from the analysis date t as follows:

$$\begin{aligned} E_t[Free\ Cash\ Flow\ to\ Equity_{t+n}] = E_t[Real\ GDP_{t+n} \times Corporate\ Profit\ Ratio_{t+n}] \times k \\ -Principal\ Payment_{t+n} - Tax\ Adjusted\ Interest\ Expense_{t+n} \end{aligned}$$

The starting point for this calculation is free cash flows to firm (FCFF), which are cash flows available to both equity and debt holders. The key difference between FCFE and FCFF is changes in debt and tax-adjusted interest rate expense:

$$FCFE\ =\ FCFF\ +\ \Delta Debt - Tax\ Adjusted\ Interest\ Expense$$

Because the level of debt and interest expenses can fluctuate significantly with secular trends and rates environment, FCFF (rather than FCFE) is expected to grow proportionally with the overall economy. The total FCFF of public companies is driven by the size of the economy and the corporate profit ratio[55]. In addition, a constant ratio k translates corporate profit to FCFF, a ratio that has remained historically stable.

The relationship between debt and cash flows is crucial for equity holders. When a company's debt level increases (i.e., when $\Delta Debt$ is positive), more cash becomes available to equity holders. Conversely, an increase in interest rate can burden equity holders with higher interest expenses and result in lower cash flows.

[53] The equity valuation model treats the broad equity market as the model asset. Further research is required to provide more granularity on different segments and tilts of the equity market, such as sector and style segmentation.

[54] Payout ratio is another popular cash flow measurement for DCF analysis. FCFE and payout ratio are essentially two sides of the same coin. FCFE is the amount of cash available for distribution after covering capital expenditure, debt servicing, and other expenses, while the payout ratio – including dividend yield and buyback yield – reflects what is actually paid out to shareholders. In the long run, assuming a stable cash reserve retained by the company, they tend to be equivalent.

[55] Corporate profit ratio is modeled as an AR(1) process, reflecting its fluctuation around a long-term average, as discussed in Section 2.4.

For simplicity, the model assumes that companies, in aggregate, will pay down their current outstanding debt over a 15-year horizon[56], in addition to covering the interest expense on remaining outstanding balance each year. Furthermore, the model reflects that most publicly traded companies are financed with fixed coupon debt[57].

The discount rate term structure for equity is modeled as the sum of the real risk-free rate and the equity risk premium:

$$Discount\ Rate_{t,n} = Real\ Riskfree\ Rate_{t,n} + Equity\ Risk\ Premium_{t,n}$$

If we only needed to price an asset today, we would simply use the current market discount curve. However, this discount curve framework has two key advantages: it allows us to estimate asset values over a long horizon, factoring in macro expectations, and it enables the application of the valuation model in stress testing. In both cases, macro shocks can adjust the risk-free rate as well as the equity risk premium term structure, leading to updated discount curves.

As shown in Exhibit 1, we can use the valuation model in two directions. Given current market prices and baseline assumptions of future macroeconomic paths, we can estimate the market-implied Forward equity risk premium. Conversely, we can propagate macro shocks through the DCF to estimate the price impact.

Treasurys and inflation-linked bonds

The cash flows of Treasury bonds are straightforward, as they are generally considered risk-free.[58] The discount rate component determines the expected returns of bonds.

As discussed in Section 3.3, the model estimates a term structure of yield premium ($d_{t,n}$) by comparing model-derived expected future short (one-year) rates ($E_t^{mdl}[i_{t+n}]$), which are based on expected inflation and growth trajectories, and market-implied future interest rate, also known as the market forward curve ($f_{t,n}^{mkt}$):

$$d_{t,n} = f_{t,n}^{mkt} - E_t^{mdl}[i_{t+n}], \qquad n = 0, 1, 2, \ldots N$$

By construction, the zeroth node of the 1-year forward curve equals the one-year spot rate:

$$f_{t,0}^{mkt} = i_t \ \rightarrow \ d_{t,0} = f_{t,0}^{mkt} - E_t^{mdl}[i_t] = f_{t,0}^{mkt} - i_t = 0$$

We assume that these yield premia roll down over time:

[56] It's important to note that companies typically do not pay down outstanding debt without also simultaneously issuing new debt. Empirically, the aggregate corporate debt-to-equity ratio remains relatively stable over time. However, these assumptions (whether debt is paid down in a fixed time period or is effectively "perpetual") are equivalent for the purpose of asset valuation, as equity value is equal to enterprise value minus debt. Our model errs on the conservative side for equity valuation by front-loading debt servicing: Under tight credit conditions, for example, companies may struggle to refinance or issue new debt, while simultaneously facing higher interest payments, which can severely reduce equity cash flows. If the model assumed perpetual debt rollover, the impact of credit tightening would be understated. By modeling debt payment over a fixed horizon, the model ensures the valuation will respond adequately to such stress scenarios.

[57] This assumption may be modified if empirical evidence for different segments of the equity market suggests a different ratio between fixed coupon vs. floating coupon bonds.

[58] In the case of a currency bloc such as eurozone, the Treasury bond cash flows may not be risk-free. Additional modeling is required to capture the sovereign default risk.

$$E_t^{mdl}[d_{t+h,n}] = \begin{cases}(1-b^n)E_t^{mdl}[d_{t+h-1,n+1}], & h \geq 1, n = 0,1,2,\ldots N-1 \\ (1-b^n)E_t^{mdl}[d_{t+h-1,N}], & h \geq 1, n = N\end{cases}$$

Here, $E_t^{mdl}[d_{t+h,n}]$ represents the expected value for the market-model differences at a future time $t+h$. We set $b = 0.05$, based on the autocorrelation of the forward rate. It also ensures that yield premia dissipate slowly, resulting in stable expected returns.

Combining above, we have a model projection for expected forward curves at time $t+h$:

$$E_t[f_{t+h,n}^{mdl}] = E_t^{mdl}[i_{t+h+n}] + E_t^{mdl}[d_{t+h,n}], \;\; n = 0,1,2,\ldots N$$

The expected discount curve used for bond pricing at a future time $t+h$ is a mechanical conversion from the expected forward curve via expectation hypothesis:

$$E_t[Discount\ Rate_{t+h,n}] = \sum_{i=0}^{n-1} E_t[f_{t+h,i}^{mdl}], \qquad n = 1,2,\ldots N$$

The same methodology applies to inflation-linked bonds, with nominal yield curve replaced with real yield curves.

Corporate bonds

The corporate bond valuation model links macroeconomic fundamentals to credit default losses and cash flows, and connects risk-free rate and credit risk premium with the discount rate.

Cash flows of corporate credit are modeled as scheduled payments adjusted for expected default losses:

$$E_t[Cash\ Fows_{t+n}] = Scheduled\ Cash\ Flows_{t+n} \times e^{-nE_t[Default\ Loss\ Rate_{t+n}]}$$

The default loss rate is the cumulative result of the default loss *hazard* rate, which incorporates the impact of nominal GDP growth, debt levels and changes in the equity risk premium, modeled using a proportional hazard framework:

$$Default\ Loss\ Hazard\ Rate = e^{(b_0 + b_1 \times Detrended\ nominal\ GDP\ growth + b_2 \times Debt\ level\ growth + b_3 \times Change\ in\ ERP)}$$

$$Default\ Loss\ Rate_{t+n} = \frac{1}{n} \cdot \sum_{i=1}^{n} Default\ Loss\ Hazard\ Rate_{t+i}$$

The discount rate for corporate credit is calculated as:

$$Discount\ Rate_{t,n} = Riskfree\ Rate_{t,n} + Credit\ Risk\ Premium_{t,n}$$

Here, the credit risk premium is a linear function of credit spread, expected default losses, and changes to the equity risk premium:

$$\begin{aligned} Credit\ Risk\ Premium_{t,n} = {} & Credit\ Spread_{t,n} - E_t[Default\ Loss\ Rate_{t+n}] \\ & + \beta(E_t[Equity\ Risk\ Premium_{t,n}] - Equity\ Risk\ Premium_t) \end{aligned}$$

Private equity

Private equity is representative of an evolving portfolio of leveraged buyouts (LBOs). Instead of relying on the reported *valuations,* which often lag due to infrequent reporting, this model focuses on the intrinsic or "fair" *value* of the portfolio, derived from expected cash flows and discount rates.

Leveraged buyouts share similarities with public equity: The cash flows of both asset classes are sensitive to economic growth, while their discount rates are sensitive to interest rates and equity risk premium. However, LBOs also have distinct features:

- Earnings growth: The earnings of LBO companies typically grow faster during the initial years following the buyout.
- Higher leverage: Cash-flow growth is typically more leveraged than that of public equity, although the gap has narrowed over the past decade.
- Debt structure: LBOs are typically financed with floating-rate debt, which often has shorter maturities and higher credit spreads compared to the debt of publicly traded companies.
- Returns to investors: Investment returns to the limited partners (LPs) are net of both management and performance fees paid to the general partners (GPs).

Both the expected cash flows and the discount rates share similar equity-like exposures, and are adjusted to account for the unique features of private equity:

$$E_t[Free\ Cash\ Flow\ to\ Private\ Equity_{t+n}] = Growth\ Adjustment_n \times B^{PE} \times E_t[Real\ GDP_{t+n}] \\ -Principal\ Payment_{t+n} - Tax\ Adjusted\ Interest\ Expense_{t+n}$$

$$Discount\ rate_{t,n} = Real\ Riskfree\ Rate_{t,n} + \beta_n^{PE} \times Equity\ Risk\ Premium_{t,n} + \alpha_n^{PE}$$

There are two key modifications for cash flows:

1. Cash flow growth: Free cash to firm (FCFF) for private equity companies is modeled as proportional to the size of the economy (through the coefficient B^{PE}). An additional growth adjustment term captures the faster-than-public growth rate during the buyout period, which the model takes to be seven years.
2. Higher leverage and debt structure: Principal and interest payments are modeled separately for two types of debt: one similar to that carried by publicly traded companies, and another taken on during the buyout. These two types of debt differ in size, time-to-maturity and coupon rate.

For discount rates, the model assumes an affine relationship to public equity's discount rates, where β_n^{PE} represents the higher systematic risk (return beta) of private equity, and α_n^{PE} captures the pure private alpha (excess return). The forward rate of the discount curve beyond seven years is set to the same as for public equity, reflecting the assumption that after the buyout period, the underlying companies in the private equity funds behave like public companies. β_n^{PE} is calibrated to the return beta in the MSCI Private Equity Model (PEQ2), and α_n^{PE} is calibrated to the enterprise value to EBITDA ratios in the MSCI private-capital data universe.

The model also accounts for fees that are typical for private equity funds, which reduce the raw expected returns. Assuming a 2% management fee and 20% performance fees, we adopt an affine model for fee adjustment:[59]

$$E_t[After\ Fee\ Return] = 0.81 \cdot E_t[Before\ Fee\ Return] - 1.9\%$$

[59] We use historical private equity return and fee data to estimate the affine parameters for fee adjustment. The results closely match the well-know "two and twenty" fee structure.

Private real estate

Private real estate is modeled as a portfolio of core real estate investments. Like private equity, we model the intrinsic value of these assets, which reacts more quickly to economic shocks, in contrast to reported valuations that often lag due to infrequent transactions and lack of mark-to-market pricing.

The model captures key characteristics of core real estate investments:

- Growth sensitivity: Real estate is not just an income-generating asset; it has significant sensitivity to economic growth.
- Slower cash-flow growth: Cash flow from real estate tends to grow more slowly compared to public equity.
- Lower beta: Unlevered real estate cash flows generally have a beta of less than one, relative to public equity cash flows.

The model assumes an affine relationship between real estate's expected cash flows to real GDP, and relates discount rates to the public equity risk premium:

$$E_t[Real\ Estate\ Cash\ Flow_{t+n}] = Growth\ Adjustment_n \times (A^{PRE} + B^{PRE} \times E_t[Real\ GDP_{t+n}])$$

$$Discount\ Rate_{t,n} = Real\ Riskfree\ Rate_{t,n} + \beta^{PRE} \times Equity\ Risk\ Premium_{t,n} + \alpha^{PRE}$$

The affine term $A^{PRE} + B^{PRE} \times E_t[Real\ GDP_{t+n}]$ captures both the growth sensitivity (through the coefficients B^{PRE}) and the lower cash flow beta (through a positive A^{PRE} term[60]). Further, a growth adjustment is added on top to model the slower growth rate based on empirical observations. [61]

The discount rate formula is in the same form as that for private equity, with the difference being both the β^{PRE} and α^{PRE} parameters are constants. The public ERP beta β^{PRE} is consistent with the MSCI Private Real Estate Model (PRE2), and the pure private real estate premium α^{PRE} is calibrated to the cap rate of core real estate index.

Private infrastructure

Private infrastructure is the most difficult to model due to its lack of cash-flow and valuation data. We adopt a simple approach with a structure similar to that of the private real estate model, providing the same set of macro sensitivities:

$$E_t[Infrastructure\ Cash\ Flow_{t+n}] = A^{Infra} + B^{Infra} \times E_t[Real\ GDP_{t+n}]$$

$$Discount\ Rate_{t,n} = Real\ Riskfree\ Rate_{t,n} + \beta^{Infra} \times Equity\ Risk\ Premium_{t,n} + \alpha^{Infra}$$

The parameters A^{Infra}, B^{Infra} and β^{Infra} are calibrated using the MSCI Private Infrastructure Model (PIN1), distinguishing between equity-like and bond-like infrastructure. The pure private infrastructure premium α^{Infra} is estimated using historical premia.

[60] The A^{PRE} term embeds bond-like behavior that contributes to real estate's low beta to public equity, and has the same scale as $B^{PRE} \times E_t[Real\ GDP_{t+n}]$.

[61] By comparing the historical growth rate of net operating income (NOI), and nominal GDP, we found that real estate cash flow growth is roughly 1.4 percentage points slower than that of GDP.

C. Model inputs and estimation

The model utilizes various sources of input data, drawing from both public sources and proprietary MSCI datasets:

1. **Macroeconomic data:** Sources include the OECD (GDP, inflation, purchasing power parity), Federal Reserve Bank of St. Louis FRED (historical yield curves, energy prices, corporate bond liabilities) and the World Bank (population). These data are essential inputs for the vector autoregressive (VAR) model that drives the macro dynamics.
2. **Equity:** MSCI Equity Index data provides equity index earnings, which are used to model equity cash flows and equity prices, which in turn help calibrate the model-implied Forward equity risk premium (Forward ERP).
3. **Yield curves:** MSCI derived curves, including both nominal and breakeven inflation (BEI) curves, are used to estimate hidden variables such as long-term real rates and long-term inflation. This is done by aligning the yield curves generated by the Macro-Finance Model with the MSCI curves derived with market data.
4. **Private assets**: MSCI private-capital data universe provides valuation data for private asset classes. Aggregate enterprise value and EBITDA are used to inform private equity valuation and estimate ex-ante risk premium. Similarly, private real estate cap rates are used as valuation inputs.

The model output supports various use cases:

1. **Capital market assumptions:** Estimate forward-looking expected returns of each asset class implied by its market valuation, under either the baseline or a macro stress scenario.
2. **Macro scenario analysis:** Propagate a macro scenario consisting of a small set of shocks into the broader set through the estimated macro shock covariance matrix, and further into asset prices through the asset valuation functions. Simulate the future trajectories of macro variables and asset prices.
3. **Long-horizon risks:** Understand the long-horizon macro sensitivities of asset classes, and asset correlations informed by potential regime changes.
4. **Dynamic asset allocation:** Build a macro-resilient strategic asset allocation given capital market assumptions under a range of scenarios, and respond to market volatilities with tactical adjustments.

Three main challenges of model estimation are: selecting appropriate parameter values, calibrating hidden variables, and estimating the macro covariance matrix used for long-horizon scenario analysis.

- **Parameter selection:** We calibrate model parameter values using a variety of techniques. First, we ground the identification of the structural shocks in economic theory. This determines which parameters are non-zero in the autoregressive matrix M, and generally sets a sign constraint. Then we use a combination of regressions and impulse response functions (IRFs) to narrow in on parameter values, ensuring they align with both theoretical expectations and empirical data.

- **Hidden variable calibration:** Some variables in the macro dynamics, namely, cyclical and trend growth, long-term real rate and inflation, are not directly observable.
    - For trend growth, we use a Kalman filter to estimate its value, updated as new observations of short-term growth come in.
    - For cyclical growth, long-term real rate and inflation, we estimate their values from nominal and BEI curves, which embed market expectations for interest rates, inflation, as well as growth. By calibrating hidden variables to these curves, we "kill two birds with one stone": Extract the long-term expectations embedded in market data, and ensure the macro model's yield curves directly reflect macroeconomic expectations.
- **Long-horizon scenario macro covariance matrix estimation**: We estimate a macro shock covariance matrix using the full history of estimated model shock variables. This covariance matrix is used in translating long-horizon macroeconomic scenarios, expressed through movements in observed macro and market variables, into projected time series of their underlying macro drivers (or "shocks"). It plays a role in distinguishing, for example, whether rising inflation stems from supply-side or demand-side shocks, given other inputs such as growth and interest rates. As a result, when applying such scenarios to asset valuation, the outcome relies less on the recent asset covariance (which may be suitable only for a particular regime), but more on the fundamental causes, and enables the detection of a potential regime shift.

D. Model validation

Validating a macro-finance model presents significant challenges, primarily due to limited data availability and shifting regimes.

Our approach balances empirical grounding and economic intuition. Exhibit 35 below outlines the model's key objectives, and the corresponding metrics used for validation. The validation framework includes a range of both quantitative and qualitative metrics, recognizing the importance of each in assessing the model's robustness and intuitiveness.

Exhibit 35: Validation metrics of the Macro-Finance Model

Objective		Metric	Metric type
Integrated macro dynamics	Model causal relationships between macro variables and long-horizon effects of macro shocks	Intuitive macro variables impulse response to macro shocks	**Qualitative**
	Model the monetary policy response to inflation and growth shocks	Explanatory power of interest rate changes given growth and inflation foresights	**Quantitative**
	Estimate market-implied macro expectations across horizons	Differences between model-estimated yield curves and market yield curves	**Quantitative**
	Generate reasonable macro projections given point-in-time levels	Unconditional forecasting power of short-term macro variables	**Quantitative**
Asset valuation model	Systematically estimate macro-driven and market-implied capital market assumptions	Forecasting power of historical CMA estimations	**Quantitative**
		Model CMAs are in the same ballpark as industry publications	**Qualitative**
	Capture asset correlation changes under different macro regimes	Model captures asset correlation regime shifts	**Qualitative**
	Evaluate long-horizon impact of stress scenarios	Intuitive asset impulse response to macro shocks	**Qualitative**

The rest of this section presents the validation results of the model, evaluating each of the objectives and metrics above.

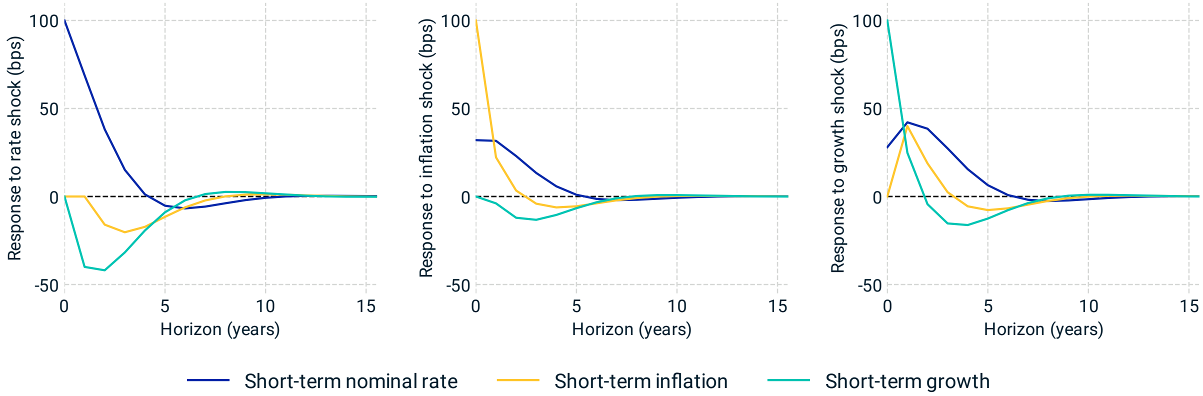
Objective #1: Model causal relationships between macro variables and long-horizon effects of macro shocks.

Exhibit 36 below illustrates the impulse response functions for the three most important macro shocks: growth, inflation and interest rates.

These IRFs capture the following intuitive relationships:

- Growth shock: A positive growth shock leads to higher inflation as increased demand drives prices upward. In response, central banks typically raise interest rate to balance growth and inflation.
- Inflation shock: A positive inflation shock prompts interest rate hikes as central banks aim to contain inflation, which in turn dampens economic growth.
- Interest rate shock: A surprise increase in short-term interest rates impacts both inflation and growth negatively, as higher borrowing costs slow down economic activity and reduce inflationary pressures.

Exhibit 36: Impulse responses to policy rate, inflation and growth shocks



Impulse response functions (IRFs) of key macro variables over a 15-year horizon, in response to 1% short-term interest rate (left panel), inflation (middle panel) and growth (right panel) shocks. Each panel shows the response to corresponding shock, and the legend of each line refers to the response variable. The IRFs capture the deviation of each response variable relative to the baseline scenario, highlighting the dynamic effects of these shocks on the broader economy.

While the IRFs are useful for understanding the effects of isolated shocks, it is important to note that in practice, multiple shocks often occur simultaneously. Their interactions can lead to macroeconomic dynamics that deviate from the behaviors outlined here.

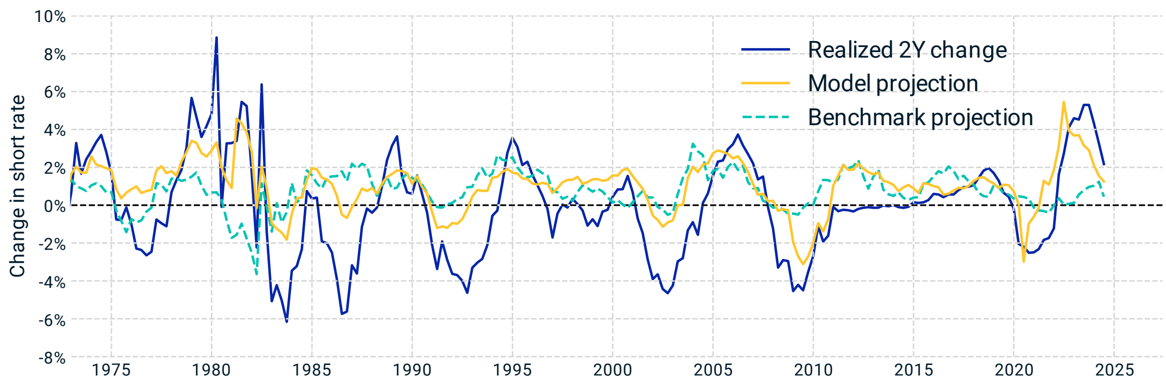
Objective #2: Model the monetary policy response to inflation and growth shocks.

To assess whether the model adequately captures the monetary policy response to inflation and growth shocks, we look at how well it explains interest rate changes based on growth and inflation foresights.

Exhibit 9 compares the model forecast of rate changes, given inflation and growth foresights, with the realized changes. In Exhibit 37 below, we expand the comparison to add a benchmark forecast — where the rate change forecast is solely based on forward rates and the expectation hypothesis. The results show that the Taylor rule embedded in the model significantly improves the explanatory power of the rate forecasts.

This suggests that, with perfect foresight of future growth and inflation, the model more accurately predicts short-term interest rates compared to the expectation hypothesis. In practice, it implies that the model can effectively translate investors' views on future growth and inflation into interest rate expectations.

Exhibit 37: Policy rate model has strong predicting power

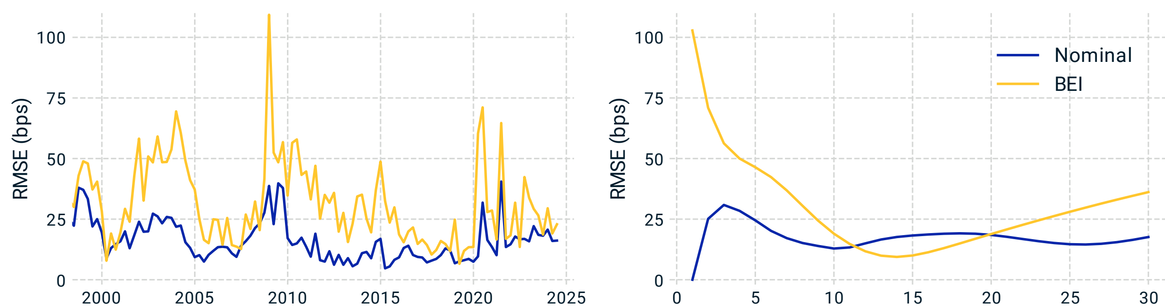


This chart compares realized short-rate (policy rate) changes over two-year periods with two forecasts: one generated by the model incorporating foresight of future inflation and growth, and a benchmark based on yield curves and the expectation hypothesis. The model explains about 40% of variation over the analysis period, significantly outperforming the benchmark forecast, which has essentially zero explanatory power over this period. These results suggest that including policy response to inflation and growth shocks in the model leads to more accurate predictions of interest rate changes.

Objective #3: Estimate market-implied macro expectations across horizons.

As discussed in Appendix C, the model uses market data, such as yield curves, to estimate market-implied expectations for macroeconomic variables that are not directly observable. To assess the goodness of these estimates, we compare the model-estimated yield curves with market curves to ensure the market information are well captured by the model calibration.

Exhibit 38: Discrepancy between model and market yield curves across historical dates and tenors



This figure illustrates the root-mean-square-error (RMSE) for nominal yield (blue) and breakeven inflation curves (yellow), comparing model-estimated curves with market curves. The left panel shows the RMSE averaged across all tenors, over historical analysis dates. The right panel shows the RMSE at different tenors of the curve, averaged across all analysis dates. Low RMSE values occur when the macro--driven yield curves closely align with the market curves. Our model infers yield premia from these discrepancies between the market and model curves, which can expand or contract when bond prices deviate from macro expectations.

Exhibit 38 above displays the discrepancies between the model-estimated and market yield curves, for both the nominal interest rate and the breakeven inflation (BEI), grouped by curve tenor and by analysis date. On average, the discrepancy is about 18 basis points for nominal yields, and about 36 basis points for breakeven inflation. The largest differences are on the short end of the BEI curve, driven by the difficulty in reconciling realized inflation with market-implied inflation: The macro model generates an inflation path starting from the most recent realized one-year inflation, but must also fit the long-term trajectory to the market-based BEI curve. Since short-term BEI often diverges from realized inflation, this contributes to larger disagreement on the short end.

Our model infers yield premia from these discrepancies between the market and model curves, which can expand or contract when bond prices deviate from macro expectations.

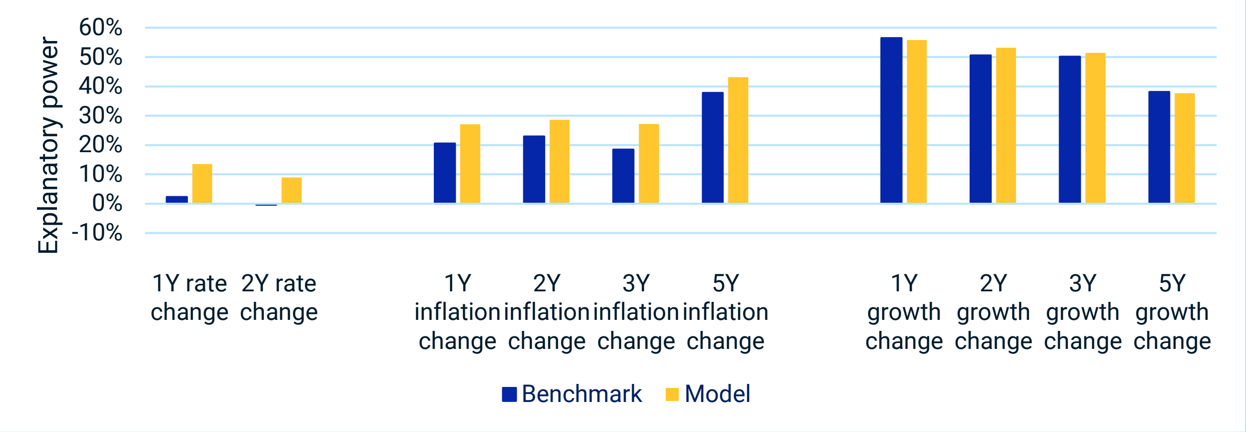
Objective #4: Generate reasonable macro projections given point-in-time levels.

While the primary objective of the model is not to forecast the future of the macroeconomy, reasonable macro projections are a key output, as they are used in asset valuation. Therefore, it's crucial to validate the model's unconditional forecasting ability for short-term macro variables. Exhibit 39 below compares the model's forecasting power for various short-term variables against simple benchmarks.

For interest rate and inflation, the benchmark derives forecasts using the expectation hypothesis and market yield curves. The model performs better than the benchmark across rates and inflation, indicating its capability of capturing macro expectations that yield curves alone would miss.

For real growth forecasts, the model performs similarly well compared to a simple "constant growth" benchmark. While this might initially seem unremarkable, it's important to note that a constant growth forecast lacks sensitivity to growth shocks. Therefore, it's encouraging to see that the model achieves its goal of capturing assets' sensitivity to macroeconomic factors, without compromising its accuracy of the growth forecast.

Exhibit 39: Model explanatory power of key macro variable dynamics



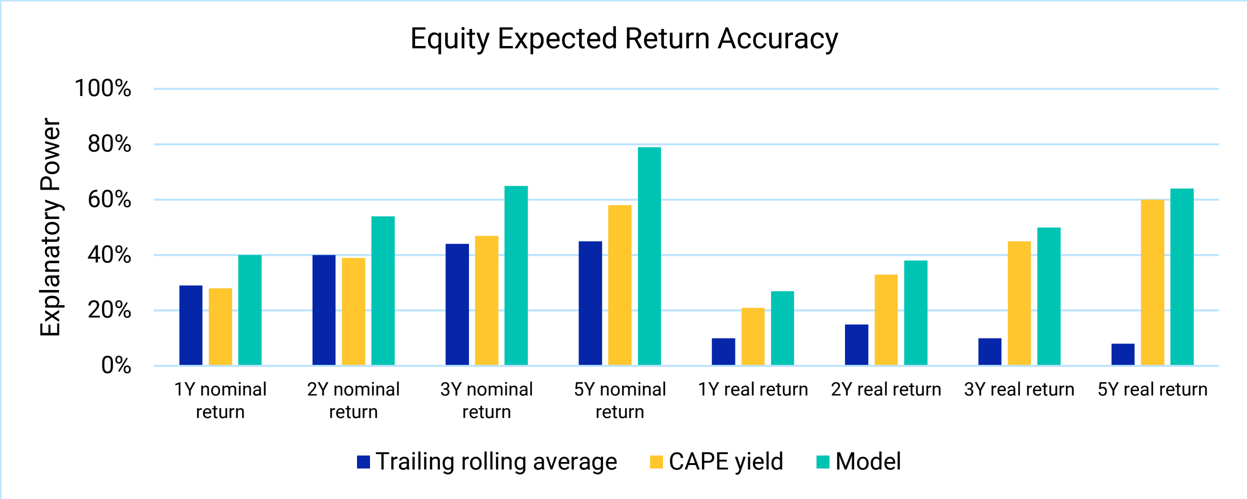
The explanatory power of the model (yellow) and a benchmark (blue) for forecasting N-year (N=1, 2, 3 or 5) changes in key macro variables, including short-term interest rate, inflation and real growth. The benchmark is based on the expectation hypothesis for rate and inflation forecasts, and a constant growth assumption for real growth forecast. All metrics are computed using data from 1998-2023.

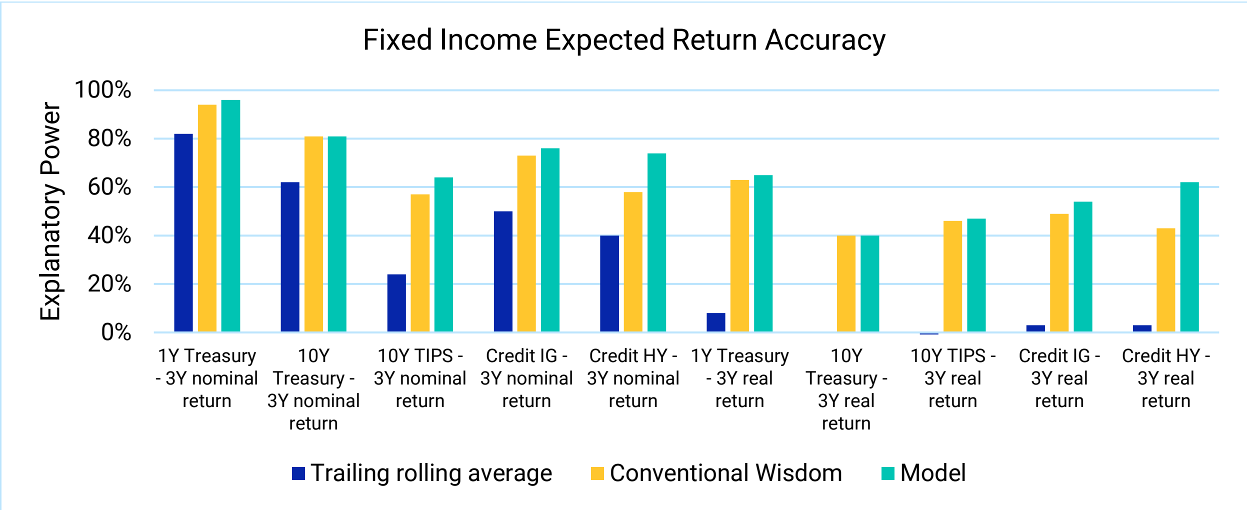
Objective #5: Systematically estimate macro-driven and market-implied capital market assumptions.

A key goal of the model is to estimate capital market assumptions (CMAs) that are informed by the macro conditions and grounded in market valuations. We evaluate the accuracy of our CMA estimates by backtesting them against historical realized returns.

Exhibit 40 compares the model's forecasting accuracy for public equity and fixed-income assets. The model consistently outperforms the trailing average benchmark and improves upon conventional wisdom, particularly for long-horizon forecasts.[62]

Exhibit 40: Model outperforms traditional measures in forecasting equity and bond returns





The explanatory power of model CMAs compared against traditional measures. The top panel highlights expected-return accuracy for U.S. equity, with the model outperforming trailing rolling averages (blue) and CAPE yield (yellow), and increasingly so over longer horizons. The bottom panel shows fixed income, where the model also outperforms trailing rolling average (blue) and conventional wisdom (yellow) for U.S. Treasurys, TIPS and corporate credit. For Treasurys and TIPS, "conventional wisdom" is based on the expectation hypothesis, while for corporate credit, spread changes are split 50/50 between expected default loss and risk premia, combined with the expectation hypothesis for risk-free rates.

Beyond explaining historical returns, the model also offers key advantages for investors. It provides transparency into the underlying macro assumptions, clarifying the economic drivers behind the forecasts — and why they change. Additionally, it supports scenario analysis, allowing adjustments

[62] We also compared the long-horizon expected return forecast generated by the model with a range of CMAs published by various financial institutions. Overall, the model CMAs are well aligned with the industry distribution.

to assumptions such as growth, inflation, or risk premia, helping investors explore various scenarios and assess their impact on CMAs. This transparency and customizability enhance confidence in making informed investment decisions.

Objective #6: Capture asset correlation changes under different macro regimes.

In Section 4.2, we discussed the significance of regime changes and their impact on asset correlations, particularly between rates and equity. Regime shifts, such as transitions from periods of high growth to stagflation or changes in monetary policy, can dramatically alter how assets interact with one another.

Exhibit 5 provides a clear example of how the model accurately captures the correlation change between equity and rates over time. Specifically, during periods of economic expansion, equity and interest rates tend to exhibit positive correlation, as rising growth expectations push both equity prices and rates higher. However, in periods of stagflation, this relationship can reverse, with equities and interest rates showing negative correlation, as central banks raise rates to combat inflation, which dampens growth and, in turn, equity prices.

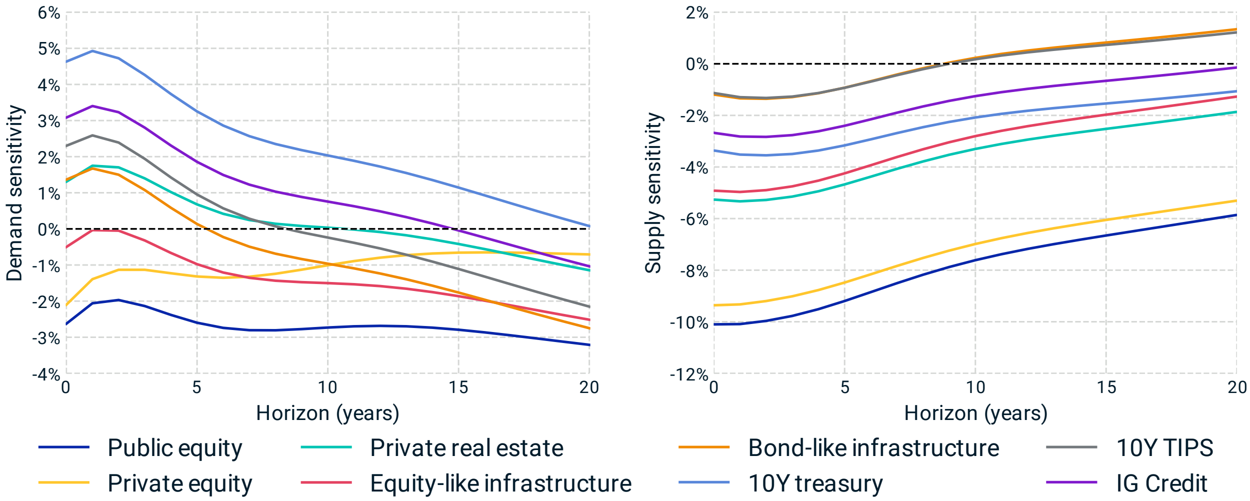
The model's ability to detect these shifts in asset correlation is critical for investors, as it enables them to better anticipate changes in portfolio risk and adjust asset allocations accordingly. By incorporating regime-sensitive dynamics into its framework, the model offers more robust projections that reflect the real-world behavior of asset correlations across different economic environments.

Objective #7: Evaluate long-horizon impact of stress scenarios.

The model adapts to different regimes. Macro shocks *in* different regimes, or shocks that push the macroeconomy *into* a different regime, could have vastly different influences on the asset returns. Exhibit 41 below examines the asset sensitivities to a demand shock and a supply shock scenario.

A demand shock underlies the positive rates-equity correlation that has been the norm of the first two decades of this century, where equity and equity-like assets take the biggest hit, while bond and bond-like assets tend to serve as a hedge. Such hedge between bonds and equity could go away under a supply shock, when higher inflation and slower growth tend to hurt all asset classes.

Exhibit 41: Asset sensitivity in supply shock and demand shock scenarios



The response of U.S. asset valuations to a demand shock (left) and a supply shock (right). Sensitivity is calculated as the real return of each asset in response to an instantaneous shock at t=0. In the demand shock scenario, where growth slows, equity risk premium rises, and inflation declines, equity and equity-like assets show vulnerability, while bonds may serve as a hedge. In contrast, the supply shock scenario, characterized by rising inflation, slowing growth, and a higher equity risk premium, has a negative impact on all assets in the short term, with TIPS and bond-like infrastructure performing the best. These supply and demand shocks should be viewed as a part of a spectrum, rather than two fixed scenarios. Specifically, the demand shock plotted consists of a 1% decline in short-term growth, a 0.3% decline in trend growth, a 0.3% increase in ERP, a 0.75% drop in short-term inflation, and a 0.1% decrease in long-term inflation. The supply shock consists of a 1% rise in short-term inflation, a 0.15% increase in long-term inflation, a 0.9% decline in short-term growth, a 0.2% decline in trend growth, and a 0.2% increase in ERP.

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